



BSI Standards Publication

Geometrical product specifications (GPS) — Geometrical tolerancing — Datums and datum systems

National foreword

This British Standard is the UK implementation of EN ISO 5459:2024. It is identical to ISO 5459:2024. It supersedes BS EN ISO 5459:2011, which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee TPR/1, Technical Product Realization.

A list of organizations represented on this committee can be obtained on request to its committee manager.

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European foreword

This document (EN ISO 5459:2024) has been prepared by Technical Committee ISO/TC 213 "Dimensional and geometrical product specifications and verification" in collaboration with Technical Committee CEN/TC 290 "Dimensional and geometrical product specification and verification" the secretariat of which is held by AFNOR.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by April 2025, and conflicting national standards shall be withdrawn at the latest by April 2025.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

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Endorsement notice

The text of ISO 5459:2024 has been approved by CEN as EN ISO 5459:2024 without any modification.

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Foreword

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The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO document should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

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This document was prepared by Technical Committee ISO/TC 213, *Dimensional and geometrical product specifications and verification*, collaboration with the European Committee for Standardization (CEN) Technical Committee CEN/TC 290, *Dimensional and geometrical product specification and verification*, in accordance with the Agreement on technical cooperation between ISO and CEN (Vienna Agreement).

This third edition cancels and replaces the second edition (ISO 5459:2011), which has been technically revised.

The main changes are as follows:

- update of the Normative references and Bibliography;
- addition of definitions [3.20](#), [3.20.1](#) and [3.20.2](#);
- in [Table 1](#), update of the symbol of the datum feature indicator;
- in [Table 1](#), addition of the symbol of single datum target indicator, moveable datum target indicator, restricted datum feature, indication of a situation feature and datum coordinate system indicator, and addition of a note;
- in [Table 2](#), addition of [SV], [SF] and [SFxx], and addition of a note;
- in [Clause 6](#), addition of a paragraph before the example in [6.1](#), update of [6.2.1](#) and [6.2.2](#), replacement of the first paragraph in [6.2.3](#) and [6.2.4](#), addition of the last paragraph in [6.3.2](#) and table titles added in [6.3.2](#) and subsequent tables renumbered;
- in [Clause 7](#), addition of a note in [7.1](#), update of [7.2.1](#), [7.3](#), [7.4.2.1](#) and [7.4.2.2](#), update of the text and figures in [7.4.2.4](#) up to [Figure 22](#), update of the first paragraph of [7.4.2.6](#), update of [Figure 39](#), addition of a new rule 11 in [7.4.2.11](#) and a new rule 12 in [7.4.2.12](#);
- addition of a new [Clause 8](#);
- in [Annex A](#), update of the text between [Figures A.1](#) and [A.2](#), update of the first paragraph in [A.2.1](#) and of [Figure A.4](#), addition of Notes 1 and 2 in [A.2.2.3](#), and update of the row for the plane in [Table A.2](#);

- addition of a new [Clause D.4](#);
- update of [Annex E](#), addition of new [Annexes G](#) to [J](#), update of [Annex K](#) giving the relation to the GPS matrix model.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

Introduction

This document is a geometrical product specification (GPS) standard and is to be regarded as a general GPS standard (see ISO 14638). It influences chain links A to C of the chains of standards on datums.

The ISO GPS matrix model given in ISO 14638 gives an overview of the ISO GPS system of which this document is a part. The fundamental rules of ISO GPS given in ISO 8015 apply to this document and the default decision rules given in ISO 14253-1 apply to the specifications made in accordance with this document, unless otherwise indicated.

For more detailed information on the relationship of this document to other standards and the GPS matrix model, see [Annex K](#).

For the definitive presentation (proportions and dimensions) of symbols for geometrical tolerancing, see ISO 7083.

The previous version of this document dealt only with planes, cylinders and spheres being used as datums. There is a need to consider all types of surfaces, which are increasingly used in industry. The definitions of classes of surfaces as given in [Annex B](#) are exhaustive and unambiguous.

This document applies new concepts and terms that have not been used in previous ISO GPS standards. These concepts are described in detail in ISO 14638, ISO 17450-1 and ISO 17450-2; therefore, it is recommended to refer to these standards when using this document.

This document provides tools to express location or orientation constraints, or both, for a tolerance zone. It does not provide information about the relationship between datums or datum systems and functional requirements or applications.

Geometrical product specifications (GPS) — Geometrical tolerancing — Datums and datum systems

1 Scope

This document specifies terminology, rules and methodology for the indication and understanding of datums and datum systems in technical product documentation. This document also provides explanations to assist the user in understanding the concepts involved.

This document defines the specification operator (see ISO 17450-2) used to establish a datum or datum system. The verification operator (see ISO 17450-2) can take different forms (physically or mathematically) and is not the subject of this document.

NOTE The detailed rules for maximum and least material requirements for datums are given in ISO 2692.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 128-2:2022, *Technical product documentation (TPD) — General principles of representation — Part 2: Basic conventions for lines*

ISO 1101, *Geometrical product specifications (GPS) — Geometrical tolerancing — Tolerances of form, orientation, location and run-out*

ISO 2692, *Geometrical product specifications (GPS) — Geometrical tolerancing — Maximum material requirement (MMR), least material requirement (LMR) and reciprocity requirement (RPR)*

ISO 4351, *Geometrical product specifications (GPS) — Association*

ISO 17450-1, *Geometrical product specifications (GPS) — General concepts — Part 1: Model for geometrical specification and verification*

ISO 17450-2, *Geometrical product specifications (GPS) — General concepts — Part 2: Basic tenets, specifications, operators, uncertainties and ambiguities*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 1101, ISO 2692, ISO 4351, ISO 17450-1, ISO 17450-2 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <https://www.electropedia.org/>

3.1 situation feature

point, straight line, plane or helix from which the location and orientation of features, or both, can be defined

3.2

datum feature

real (non-ideal) integral feature used for establishing a datum

Note 1 to entry: A datum feature can be a complete surface, a portion of a complete surface, or a feature of size.

Note 2 to entry: An illustration showing the relations between datum feature, associated feature and datum is given in [Figure 4](#).

3.3

associated feature

associated feature for establishing a datum

ideal feature which is fitted to the datum feature with a specific association criterion

Note 1 to entry: The type of the associated feature is by default the same as the type of the nominal integral feature used to establish the datum (for an exception, see [7.4.2.5](#)).

Note 2 to entry: The associated feature for establishing a datum simulates the contact between the real surface of the workpiece and other components.

Note 3 to entry: An illustration showing the relations between datum feature, associated feature and datum is given in [Figure 4](#).

3.4

datum

one or more situation features of one or more features associated with one or more real integral features selected to define the location or orientation, or both, of a tolerance zone or an ideal feature representing for instance a virtual condition

Note 1 to entry: A datum is a theoretically exact reference; it is defined by a plane, a straight line or a point, or a combination thereof.

Note 2 to entry: The concept of datums is inherently reliant upon the invariance class concept (see [Annexes A](#) and [B](#)).

Note 3 to entry: Datums with maximum material condition (MMC) or least material condition (LMC) are not covered in this document (see ISO 2692).

Note 4 to entry: When a datum is established, for example, on a complex surface, the datum consists of a plane, a straight line or a point, or a combination thereof. The modifier [SL], [PL] or [PT], or a combination thereof, can be attached to the datum identifier to limit the situation feature(s) taken into account relative to the surface.

Note 5 to entry: An illustration showing the relation between datum feature, associated feature and datum is given in [Figure 4](#).

3.5

primary datum

datum that is not influenced by constraints from other datums

3.6

secondary datum

datum, in a datum system, that is influenced by an orientation constraint from the primary datum in the datum system

3.7

tertiary datum

datum, in a datum system, that is influenced by constraints from the primary datum and the secondary datum in the datum system

3.8

single datum

datum established from one datum feature taken from a single surface or from one feature of size

Note 1 to entry: The invariance class of a single surface can be complex, prismatic, helical, cylindrical, revolute, planar or spherical. A set of situation features defining the datum (see [Table B.1](#)) corresponds to each type of single surface.

3.9**common datum**

datum established from two or more datum features considered simultaneously

Note 1 to entry: To define a common datum, it is necessary to consider the collection surface created by the considered datum features. The invariance class of a collection surface can be complex, prismatic, helical, cylindrical, revolute, planar or spherical (see [Table B.1](#)).

3.10**datum system**

set of two or more situation features established in a specific order from two or more datum features

Note 1 to entry: To define a datum system, it is necessary to consider the collection surface created by the considered datum features. The invariance class of a collection surface can be complex, prismatic, helical, cylindrical, revolute, planar or spherical (see [Table B.1](#)).

3.11**datum target**

portion of a datum feature which can nominally be a point, a line segment or an area

Note 1 to entry: Where the datum target is a point, a line or an area, it is indicated as a datum target point, a datum target line or a datum target area, respectively.

3.12**moveable datum target**

datum target with a controlled motion

3.13**collection surface**

two or more surfaces considered simultaneously as a single surface

Note 1 to entry: [Table B.1](#) is used to determine the invariance class of a datum or datum systems when using a collection of surfaces.

Note 2 to entry: Two intersecting planes may be considered together or separately. When the two intersecting planes are considered simultaneously as a single surface, that surface is a collection surface.

3.14**objective function**

<association> formula that describes the goal of association from the datum feature and the ideal feature (associated feature)

Note 1 to entry: In this document, the term “objective function” refers to “objective function for association”.

Note 2 to entry: The objective functions are usually named and mathematically described: maximum inscribed, minimum zone, etc.

[SOURCE: ISO 4351:2023, 3.6, modified — “datum feature” replaced “input feature”. Notes to entry added.]

3.15**constraint**

set of restrictions on variability of the mathematical parameters describing an *associated feature* ([3.3](#)) in the optimization process

EXAMPLE Orientation constraint, location constraint, material constraint or intrinsic characteristic constraint are the different types of constraint.

[SOURCE: ISO 4351:2023, 3.9, modified — “constraint” replaced “association constraint” as the term and in the example. “intrinsic characteristic constraint” replaced “size constraint” in the example.]

3.15.1**orientation constraint**

constraint (3.15) related to one or more rotational degrees of freedom of *associated feature* (3.3)

[SOURCE: ISO 4351:2023, 3.10, modified — “constraint” replaced “association constraint”.]

3.15.2**location constraint**

constraint (3.15) related to one or more translational degrees of freedom of *associated feature* (3.3)

[SOURCE: ISO 4351:2023, 3.11, modified — “constraint” replaced “association constraint”.]

3.15.3**material constraint**

constraint (3.15) on the *associated feature* (3.3), in relation to the material boundary of the *datum feature* (3.2)

EXAMPLE The outside material constraint implies that all distances between the associated feature and the datum feature are negative or equal to zero.

[SOURCE: ISO 4351:2023, 3.12, modified — “constraint” replaced “association constraint” and “datum feature” replaced “input feature”.]

3.15.4**intrinsic characteristic constraint**

size constraint

association constraint (3.15) on an intrinsic characteristic of *associated feature* (3.3) whether it is considered fixed or variable

Note 1 to entry: The intrinsic characteristic of a cylinder is its diameter which is a size.

Note 2 to entry: In the collection surface, constituted by two parallel and not coaxial cylinders, there are more than one intrinsic characteristic (two diameters, the distance between their axes and the angle of 0°).

3.16**association criterion**

objective function with or without constraints, defined for an association

Note 1 to entry: Several constraints may be defined for an association.

Note 2 to entry: Association results (associated features) may differ, depending upon the choice of association criterion.

Note 3 to entry: Default association criteria are defined in [Annex A](#).

3.17**integral feature**

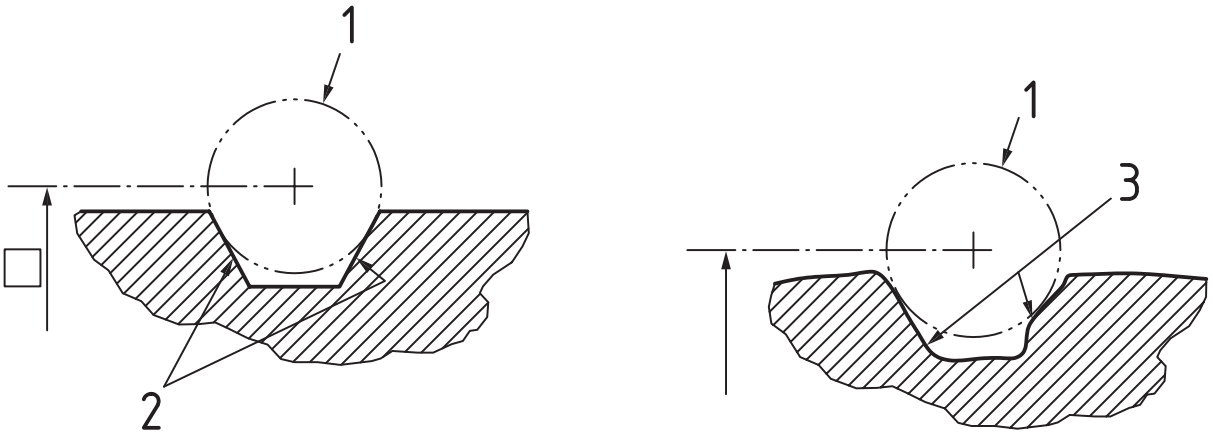
surface or line on a surface

Note 1 to entry: An integral feature is intrinsically defined.

3.18**contacting feature**

ideal feature of any type which is different from the nominal feature under consideration and is associated with the corresponding datum feature

Note 1 to entry: See [Figure 1](#).



a) Contacting feature on nominal model

b) Contacting feature on real workpiece

Key

- 1 contacting feature: ideal sphere in contact with the datum feature or the feature under consideration
- 2 features under consideration: nominal trapezoidal slot (collection of two non-parallel surfaces)
- 3 datum feature: real feature corresponding to the trapezoidal slot (collection of two non-parallel surfaces)

Figure 1 — Example of a contacting feature

**3.19
invariance class**

group of ideal features for which the nominal surface is invariant for the same degrees of freedom

Note 1 to entry: There are seven invariance classes (see [Annex B](#)).

**3.20
datum coordinate system**

coordinate system established from a single datum, a common datum or a datum system

Note 1 to entry: Using a datum coordinate system is optional.

**3.20.1
datum coordinate system indicator**
indication defining a coordinate system

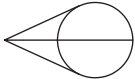
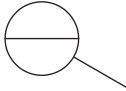
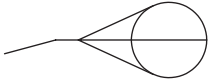
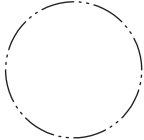
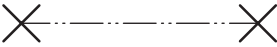
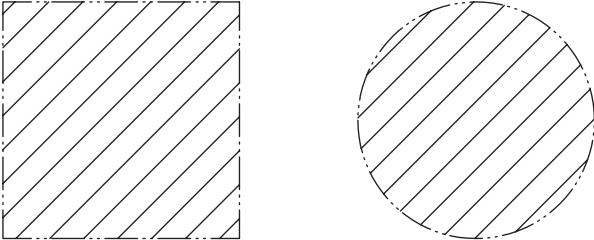
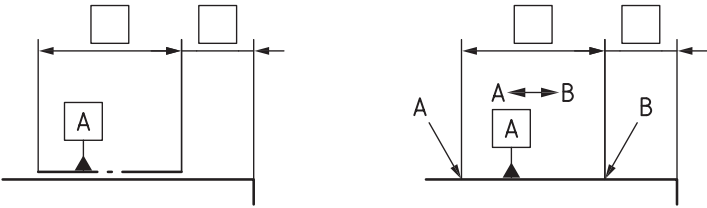
**3.20.2
datum coordinate system identifier**
label identifying a coordinate system

4 Symbols

[Table 1](#) gives symbols to identify the datum feature and datum target used to establish a datum.

[Table 2](#) gives the list of modifier symbols, which can be associated the datum identifier.

Table 1 — Datum features and datum target symbols

Description	Symbol	Sub-clause
Datum feature indicator		7.2.1
Datum feature identifier	Capital letter (A, B, C, AA, etc.)	7.2.2
Single datum target frame		7.2.3.2
Moveable datum target frame		7.2.3.2
Single datum target indicator ^a		7.2.3
Moveable datum target indicator ^a		7.2.3
Datum target point		7.2.3.3
Closed datum target line		7.2.3.3
Non-closed datum target line		7.2.3.3
Datum target area		7.2.3.3
Restricted datum feature		7.4.2.4

^a The terminator of the leader line is dependent of the type of datum target.

Table 1 (continued)

Description	Symbol	Sub-clause
Indication of a situation feature of a single datum, a common datum or a datum system	For single datum For datum system	7.4.2.12
Datum coordinate system indicator		Annex G

^a The terminator of the leader line is dependent of the type of datum target.

Table 2 — Modifier symbols

Symbol	Description	Subclause
[PD]	Pitch diameter	7.4.2.1
[MD]	Major diameter	7.4.2.1
[LD]	Minor diameter	7.4.2.1
[ACS]	Any cross-section	7.4.2.4
[ALS]	Any longitudinal section	7.4.2.4
[CF]	Contacting feature	7.4.2.5
[SV]	Size, variable	7.4.2.2
[SF]	Size, fixed	7.4.2.2
[SFxx]	Size, fixed with specified value xx	7.4.2.2
[DV]	Distance, variable (for common datum)	7.4.2.7
[PT]	(situation feature of type) point	7.4.2.8
[SL]	(situation feature of type) straight line	7.4.2.8
[PL]	(situation feature of type) plane	7.4.2.8
><	For orientation constraint only	7.4.2.8
Ⓟ	Projected (for secondary or tertiary datum)	7.4.2.10
Ⓛ	Least material requirement	See ISO 2692
Ⓜ	Maximum material requirement	See ISO 2692

NOTE Other symbols related to filtration and association are presented in [Tables 8 to 10](#) and in [Table H.1](#).

5 Role of datums

Datums form part of a geometrical specification (see ISO 1101).

Datums are established from real surfaces identified on a workpiece.

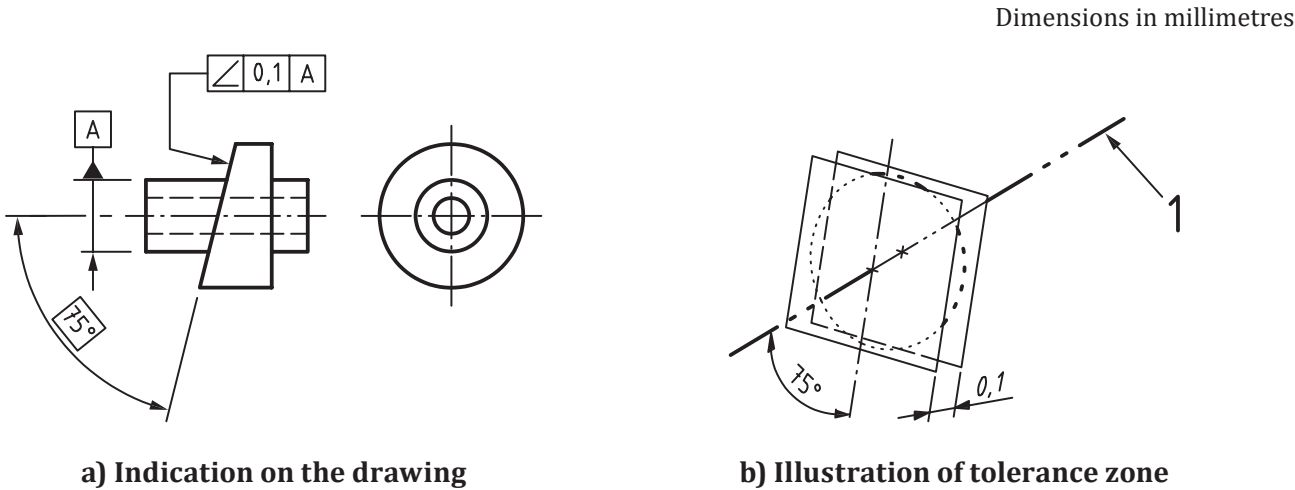
Datums allow tolerance zones to be located or orientated (see Examples 1 and 2) and virtual conditions to be defined (for example maximum material virtual condition $\textcircled{M}$ according to ISO 2692). The datums can be seen as a means to lock degrees of freedom of a tolerance zone. The number of degrees of freedom of the tolerance zone which are locked depends on the nominal shape of the features utilized to establish the datum or datum system; whether the datum is primary, secondary or tertiary; and on the tolerated characteristic indicated in the geometrical tolerance frame.

By default, a datum locks all the degrees of freedom of the tolerance zone that it can lock given its shape and which

- are required by the geometrical characteristic indicated in the tolerance frame, and
- have not already been locked by the preceding datum(s) in the datum system.

When a datum locks only orientation degrees of freedom, this shall be indicated by the modifier $><$.

EXAMPLE 1 The tolerance zone, which is the space between two parallel planes 0,1 mm apart, is constrained in orientation by a 75° theoretically exact angle from the datum. Here, the datum is the situation feature of a cylinder (axis of associated cylinder). See [Figure 2](#).



Key

- 1 datum A constituted by the axis of the associated cylinder

Figure 2 — Example of tolerance zone constrained in orientation from a datum

EXAMPLE 2 The tolerance zone, which is the space between two parallel planes 0,2 mm apart, is constrained in orientation by a 70° angle from a datum, and in location by the distance 20 mm from the gauge plane positioned perpendicular to the axis of 40° cone where its local diameter is 30 mm. Here, the datum consists of the set of situation features of the cone with a fixed angle of 40° , i.e. the cone axis and the point of intersection between the gauge plane and that axis. See [Figure 3](#).



1 datum A constituted by the axis of the associated cone, the point of intersection of the gauging plane and this axis

6 General concepts

Datums and datum systems are theoretically exact geometric features used together with implicit or explicit TEDs to locate or orientate

- A datum consists of a set of situation features for an ideal feature (feature of perfect form). This ideal feature is an associated feature which is established from the identified datum features of a workpiece. Datum features can be complete features, or identified portions thereof (see [Clause 7](#)).

The geometrical type of these associated features belongs to one of the following invariance classes:

- 9

- helical (e.g. a threaded surface)¹⁾;
- revolute (e.g. a cone or a torus);
- prismatic (e.g. a prism);
- complex (e.g. a free-form surface).

Each single or collection feature belongs to one invariance class (for an explanation of invariance classes, invariance degree, and degree of freedom, see [Annex B](#)).

Associated features are established from the real or extracted single features used for the datum. The associated feature can be defined by an operation of association including constraints coming from the feature itself or from one or more other features. The situation features that make up the datum are defined from these associated features. The default association methods are given in [Annex A](#).

One or more single features can be used to establish a datum. If only one single feature is used, it establishes a single datum. If more than one single feature is used, they can either be considered simultaneously to establish a common datum or in a predefined order to establish a datum system (see [6.3](#)).

The datum feature(s) to be used for establishing each datum shall be designated and identified.

The single datums (see [6.3.2](#)), common datums (see [6.3.3](#)) or datum systems (see [6.3.4](#)), as applicable, shall be specified for each geometrical specification.

When applicable, any additional constraints shall be defined for the association.

Datums and datum systems are situation features, not coordinate systems. The set of situation feature is established in a process including associations, but can also include some other geometrical operations (e.g. intersection, translation, rotation). A complete datum system (allowing to lock all six degrees of freedom) is a set of one point, one straight line and one plane (the point on the straight line in the plane). Coordinate systems can be established from a datum or a datum system that locks all six degrees of freedom. In this case, it shall be indicated in accordance with [Annex G](#).

EXAMPLE In [Figure 4](#), the datum is indicated as a single datum derived from a nominal feature, a cylinder, used to orientate or locate a tolerance zone. In order to derive the datum, the following sequence of operations is performed:

- a partition to define the real integral surface corresponding to the nominal feature [see [Figure 4 b](#));
- an extraction to provide the extracted integral feature [see [Figure 4 c](#));
- a filtration (see [Annex A](#));
- an association (see [Annex A](#) for association method) to define the associated feature. (In this case, its type is the same as the nominal feature.) The associated feature [see [Figure 4 d](#))] is established from the non-ideal surface (in the specification process) or from the extracted feature (in the verification process).

The datum is defined as the situation feature (the axis) of the associated cylinder [see [Figure 4 e](#))].

6.2 Intrinsic characteristics of surfaces associated with datum features

6.2.1 General

The default status (variable or fixed) of the intrinsic characteristics is described in [6.2.2](#), [6.2.3](#) or [6.2.4](#). For rules, see also rule 2 (2A to 2D) in [7.4.2.2](#).

For datums with virtual conditions, see ISO 2692.

Issues of orientation and location constraints in datum systems are presented in [Annex I](#).

1) Helical surfaces as such are not considered in this document. They are regarded as cylindrical surfaces because, in most functional cases where helical surfaces (threads, helical slope, endless screw, etc.) are involved, the combined rotation and translation of the helix is not needed for datum purposes. In these cases, the pitch cylinder surface is used to establish the datum. The major or minor cylindrical surface can also be considered and specified.

6.2.2 Single datum established from a single feature

The default status of the intrinsic characteristics for a single feature is:

- variable for the linear size of a feature of size (see [Table 3](#));
- fixed for the angular size of a feature of size (see [Table 3](#));
- fixed for an intrinsic characteristic which is not the linear size of a feature of size.

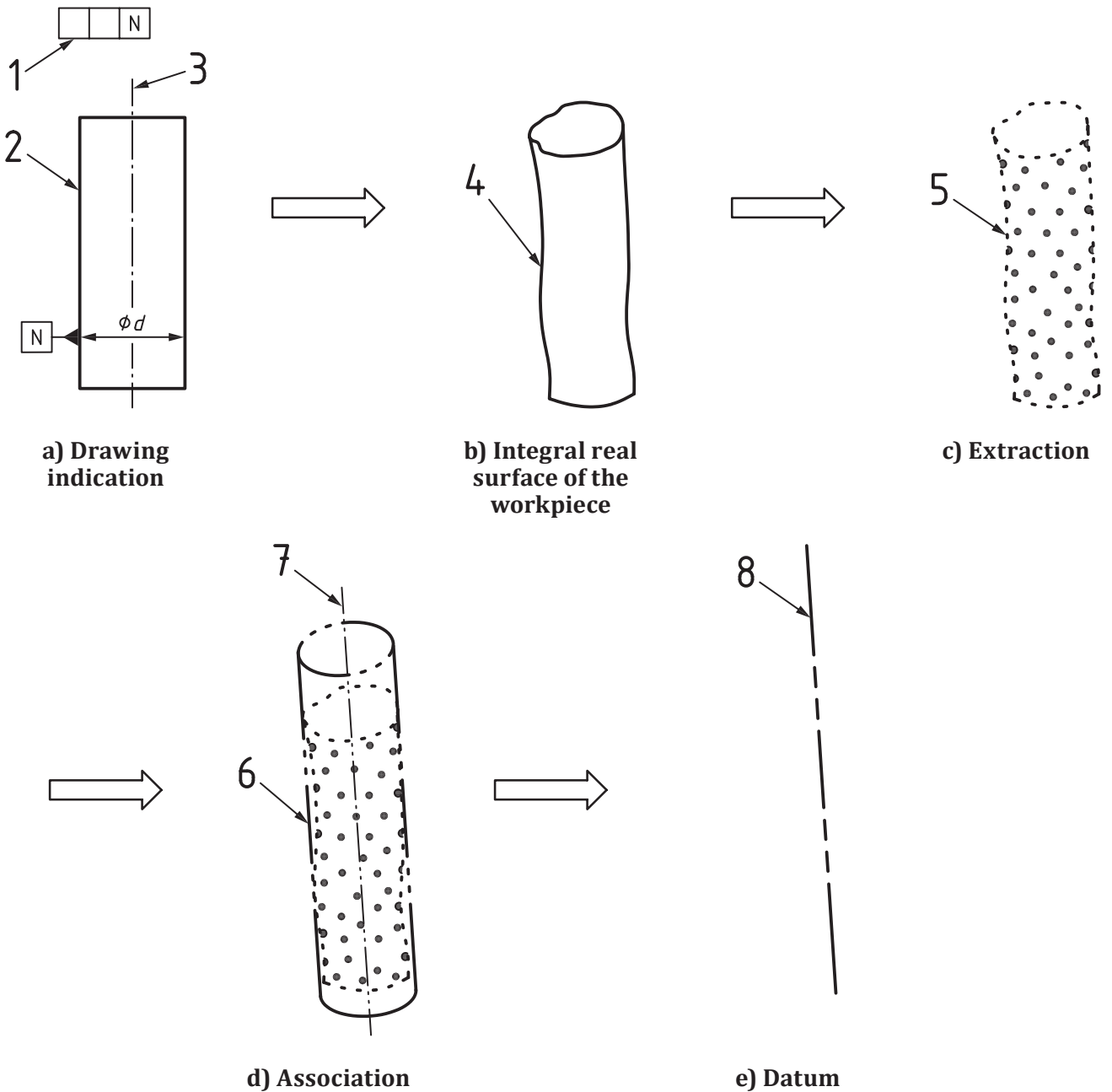
Table 3 — Default status of intrinsic characteristics of features of size

Feature of size	Invariance class	Intrinsic characteristic	Default status
Cylinder	Cylindrical	Diameter	Variable See example in C.2.2
Sphere	Spherical	Diameter	Variable See example in C.2.4
Two parallel opposite planes	Planar	Distance between the two planes	Variable See example in C.2.10
Cone	Revolute	Angle	Fixed See example in C.2.3
Wedge	Prismatic	Angle	Fixed See example in C.2.9

6.2.3 Common datum established from two or more single features simultaneously

The default status (variable or fixed) of the intrinsic characteristics of each associated feature establishing the common datum is as in [6.2.2](#).

Intrinsic characteristics introduced by the collection of features (defining the relation between the associated features) shall by default be considered theoretically exact for both linear and angular dimensions.



Key

- | | | | |
|---|---|---|--|
| 1 | tolerance frame to be linked to a tolerated feature | 5 | extracted integral feature (optional) |
| 2 | nominal integral feature (which is a feature of size) | 6 | associated integral feature |
| 3 | nominal derived feature | 7 | derived feature of associated integral feature |
| 4 | real integral feature (datum feature in this case) | 8 | single datum (situation feature of the associated surface) |

In the example shown in [Figure 4 e](#)), the derived feature (key 7) and the situation feature (key 8) are the same as each other; however, this is not always the case.

Figure 4 — Illustration of features used for establishing a single datum from a cylinder

EXAMPLE A common datum is established from two parallel non-coaxial cylinders (invariance class: prismatic). Each individual cylinder has an intrinsic characteristic, its diameter. The collection of features also has intrinsic characteristics: an angle, defined by the fact that two axes restrict the twist of the datum, and the distance between the two axes (see [C.3.4](#)). The two diameters of the cylinders are considered by default variable and the twist angle and the distance between the two are considered theoretically exact.

NOTE See also [6.3.3](#), [7.4.2.7](#) and [7.4.2.9](#) for the special case where the linear dimensions between features included in the collection can be considered variable by using the modifier [DV].

6.2.4 Datum systems established in a defined sequence from two or more single features

The default status (variable or fixed) of the intrinsic characteristics of each associated feature establishing the datum system is as in [6.2.2](#) for a single datum and as in [6.2.3](#) for a common datum.

The intrinsic characteristics introduced by the collection of features (defining the relation between the associated features) shall be considered variable for linear dimensions but theoretically exact for the angular dimensions.

6.3 Single datums, common datums and datum systems

6.3.1 General

When a single surface or a collection surface is identified as a datum feature, the invariance degrees for which the surface is invariant shall be identified and compared with [Table B.1](#) in order to determine the set of situation features (point, straight line, plane or helix, or a combination thereof) that constitutes the datum.

[Annex C](#) gives examples of single datums, common datums and datum systems.

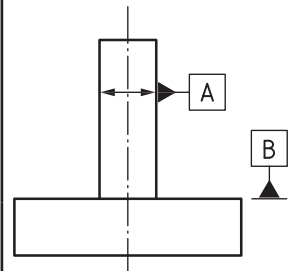
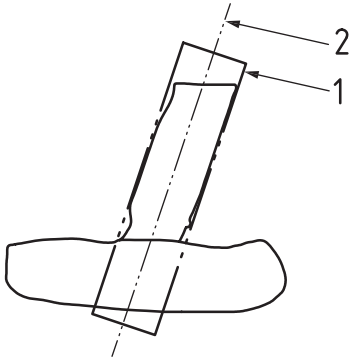
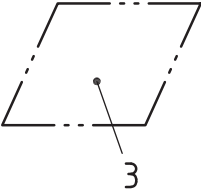
6.3.2 Single datums

A single datum consists of one or more situation features based on one single feature or a portion thereof.

NOTE A single datum taken from a cone has two situation features: its axis and a point on that axis.

For an example, see [Table 4](#).

Table 4 — Example single datum taken from a cylinder or a plane

Indication of datum feature	Indication of datum in tolerance frame	Illustration of the meaning	Invariance class and situation feature (see Annex B)	Datum
			Cylindrical Axis of associated cylinder	
			Planar Associated plane	
Key 1 associated feature (without orientation constraint) 2 straight line which is the situation feature of the associated cylinder (its axis) 3 plane which is the situation feature of the associated plane (the associated plane itself) NOTE Association for single datums is described in Annex A.				

If a single datum is used as the only datum in a tolerance frame, or if it is the primary datum in a datum system, the associated feature to the real integral feature (or to the portions of it) used for establishing the datum is obtained without external orientation constraints or location constraints.

For the constraints that are applied, when a single datum is used as a “secondary datum” or a “tertiary datum”, see 6.3.4.

Additional examples of single datums are given in Clause C.2.

For a feature of size made up of two opposite parallel planes, the collection surface associated with the surfaces (or to the portions of the surfaces) used for establishing the datum is obtained with an internal orientation constraint; the associated surfaces (constituting the collection surface) are individually defined with an internal parallelism constraint and a variable intrinsic characteristic constraint (see C.2.10).

A secondary datum shall not be specified when it does not constrain more degrees of freedom of the tolerance zone than the primary datum.

A tertiary datum shall not be specified when it does not constrain more degrees of freedom of the tolerance zone than the primary and the secondary datums.

6.3.3 Common datums

A common datum consists of one or more situation features established by taking into account the collection surface.

If the common datum is used as the only datum in a tolerance frame, or if it is the primary datum in a datum system, the collection of associated features used for establishing the datums, is established without

external orientation constraints or location constraints; therefore, the surfaces (constituting the collection surface) are associated together, simultaneously. For the constraints that are applied when a common datum is used as a “secondary datum” or a “tertiary datum”, see [6.3.4](#).

A secondary datum shall not be specified when it does not constrain more degrees of freedom of the tolerance zone than the primary datum.

A tertiary datum shall not be specified when it does not constrain more degrees of freedom of the tolerance zone than the primary and the secondary datums.

Examples of common datums are given in [6.3.4](#) (Examples 1 and 2 illustrate the difference between a datum system and a common datum) and [Clause C.3](#).

The complementary indication [DV] (meaning “distance variable”) following the letters indicating a common datum in a tolerance frame means that the linear dimensions between the situation features are variable. See also [7.4.2.9](#) and [Clause E.4](#).

6.3.4 Datum systems

A datum system is constituted by an ordered sequence of two or three single or common datums. A datum system consists of two or three situation features resulting from the collection of the considered surfaces.

The associated features used to establish the datum system are derived sequentially, in the order defined by the geometrical specification. The relative orientation of the associated surfaces is theoretically exact but their relative location is variable.

This order defines the orientation constraints for the association operation: the primary datum imposes orientation constraints on the secondary datum and tertiary datum; the secondary datum imposes orientation constraints on the tertiary datum.

A secondary datum shall be specified when it is necessary to constrain more degrees of freedom of the tolerance zone than are constrained by the primary datum.

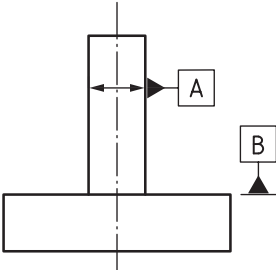
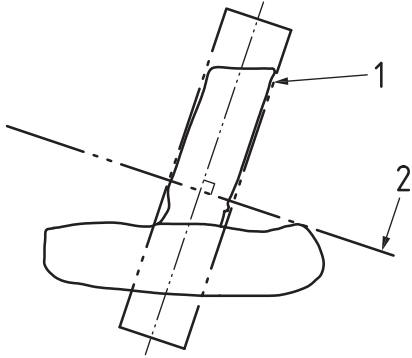
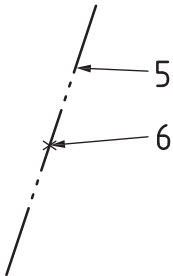
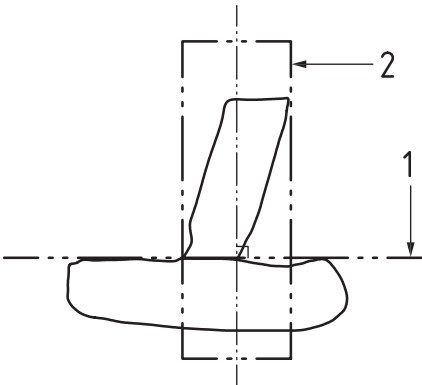
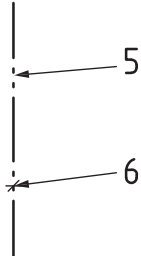
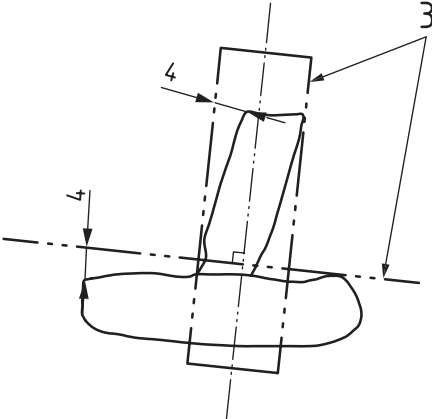
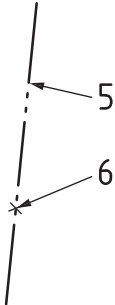
A tertiary datum shall be specified when it is necessary to constrain more degrees of freedom of the tolerance zone than are constrained by the primary and the secondary datums.

A secondary or tertiary datum shall not be specified when it does not constrain more degrees of freedom of the tolerance zone than are constrained by the primary datum and in the case of a tertiary datum also by the secondary datums.

Additional examples of datum systems are given in [Clause C.4](#).

For examples, see [Tables 5, 6 and 7](#).

Table 5 — Example common datum or datum system taken from a cylinder and a plane

Indication of datum feature	Indication of datum in tolerance frame	Meaning on workpiece	Resulting common datum or datum system
			
			
			

Key

- 1 first associated feature without orientation constraint
- 2 second associated feature with orientation constraint from the first associated feature
- 3 simultaneously associated features with orientation constraint and location constraint
- 4 maximum distance balanced between the associated features and datum features
- 5 straight line which is the situation feature of the associated cylinder (its axis)
- 6 point of intersection between the straight line and the plane

NOTE 1 The orientation and location of datums are different depending on the datum indications in the tolerance frame.

NOTE 2 The association for datum systems is described in [A.2.4](#).

Table 6 — Example common datum or datum system taken from two cylinders

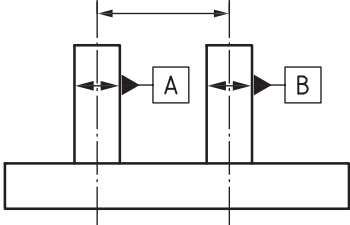
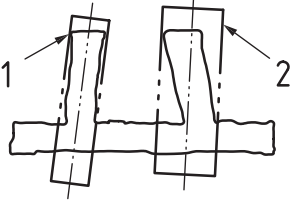
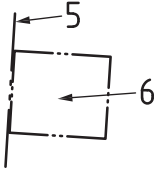
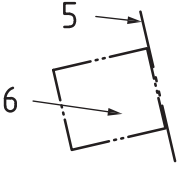
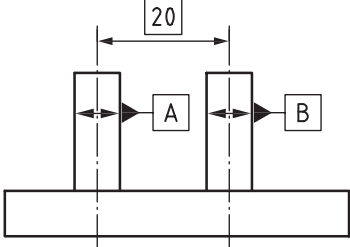
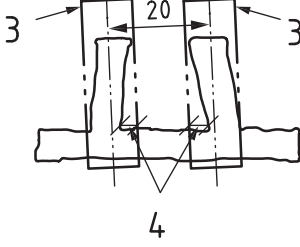
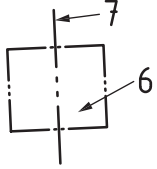
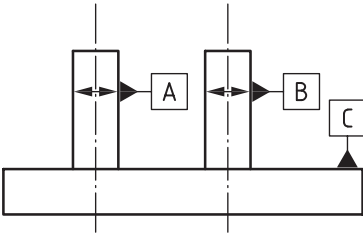
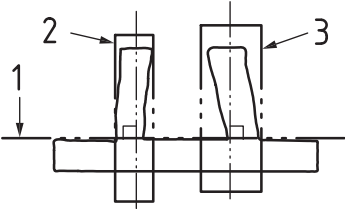
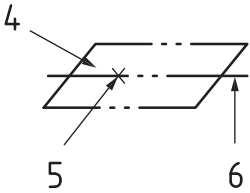
Indication of datum feature	Indication of datums in tolerance frame	Meaning on workpiece	Resulting datum or datum system
			
			
			
Key 1 first associated cylinder without constraint 2 second associated cylinder with parallelism constraint from the first associated feature 3 simultaneously associated cylinders with parallelism constraint and location constraint 4 maximum distance balanced between the associated cylinders and datum features 5 straight line which is the axis of the first associated cylinder 6 plane including the axes of the two associated cylinders 7 median straight line of the axes of the two simultaneously associated cylinders NOTE 1 The orientation and location of datums are different depending on the datum indication in the tolerance frame. Not all possibilities for establishing the datums are covered. NOTE 2 For the association method, see Annex A.			

Table 7 — Example datum system taken from two cylinders and a plane

Indication of datum feature	Indication of datums in tolerance frame	Meaning on workpiece	Resulting datum or datum system
			
Key 1 first associated feature without a constraint 2 second associated feature with a perpendicularity constraint from the first associated feature 3 third associated feature with a perpendicularity constraint from the first associated feature (and parallelism constraint from the second one) 4 plane which is the first associated feature 5 point of intersection between the plane and the axis of the second associated feature 6 straight line which is the intersection between the associated plane and the plane containing the two axes NOTE For the association method, see Annex A.			

7 Graphical language

7.1 General

The expression of geometrical tolerances with datums in geometrical specifications on drawings includes the following steps:

- Indicate the integral surfaces of the workpiece which are to be used as datum features. When a complete integral surface is not required for a datum feature, the considered portion(s) [area(s), line(s) or point(s)] and the corresponding dimensions and locations shall be indicated (see [7.2](#)).
- Indicate individual single datums, individual common datums or datum systems (see [7.3](#)).
- Indicate the orientation constraints and/or location constraints for the tolerance zones in relation to the indicated datums (see [7.4](#)).
- Indicate the association criteria. This document defines the default association criteria in [Annex A](#). If a different association criterion is required, it shall be indicated (see [Clause 8](#)).
- Indicate, where necessary, the use of the maximum or least material requirement (see ISO 2692) or projected datum (see [7.4.2.10](#)).

NOTE [Annex D](#) presents former practices in relation to ISO 5459:1981²⁾ and ISO 5459:2011³⁾.

7.2 Indication of datum features

7.2.1 Datum feature indicator

A single integral feature to be used as a datum feature shall be indicated by a datum feature indicator. The datum feature indicator consists of a rectangular frame, which contains the datum feature identifier, connected to a leader line terminating with a filled or open datum feature triangle. The datum feature

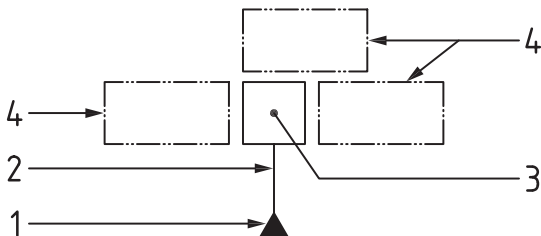
2) Cancelled and replaced by ISO 5459:2011.
3) Cancelled and replaced by this document.

triangle shall be attached to an integral feature directly or indirectly via a reference line and a leader line, see 7.4.2.1. Optional complementary indications shall be stated in the adjacent indication area, see Figure 5.

NOTE 1 If a datum feature indicator is repeated in several views or sheets of drawing, it can introduce specification ambiguity/interpretation, for instance if some complementary indications are not repeated.

NOTE 2 There is no difference in meaning between a filled and an open datum triangle.

NOTE 3 The width of the rectangular frame is dependent of the length of the datum feature identifier.



Key

- 1 datum feature triangle
- 2 leader line
- 3 area where the datum feature identifier is indicated
- 4 adjacent indication area

Figure 5 — Datum feature indicator without datum feature identifier

7.2.2 Datum feature identifier

Single features used for establishing datum features shall be identified by a datum feature identifier placed in the datum feature indicator. A datum feature identifier consists of one or more capital letters which are not to be separated by hyphens.

The letters I, O, Q and X (which can be misinterpreted) should not be used.

If a drawing has exhausted the alphabet, or if it is useful for the comprehension of the drawing, it is recommended that an unambiguous coding system be defined; for example by repeating the same letter, e.g. two times, three times, etc. (e.g. BB, CCC, etc.). In order to facilitate the reading of this document, only one letter is used in the rest of the document.

7.2.3 Datum targets

7.2.3.1 General

When it is not desirable to use a complete integral feature to establish a datum feature, it is possible to indicate portions of the single feature (areas, lines or points) and their dimensions and locations. These portions are called datum targets. They usually simulate the interface between the considered single feature of the workpiece and one or more contacting ideal features (assembly interface features or fixture features).

A datum target is indicated by a datum target indicator. This indicator is constructed from a datum target frame, a datum target symbol and a leader line linking the two symbols (directly, or through a reference line).

It can be necessary to indicate the same datum target on several appropriate views in order to have an unambiguous definition (see Figures 27, 28 and 29).

7.2.3.2 Datum target frame

The datum target frame is a circle, divided into two compartments by a horizontal line (see [Figure 6](#)). The lower compartment is reserved for the datum feature identifier followed by a digit (from 1 to *n*), corresponding to the datum target number.

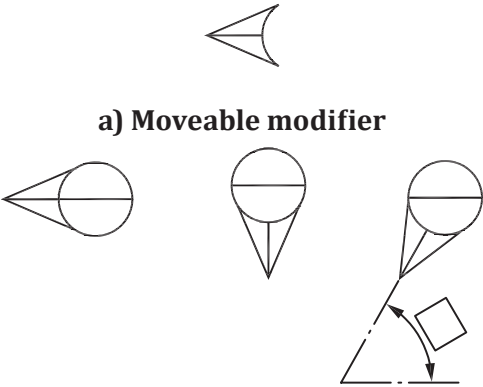
The upper compartment is reserved for additional information, such as dimensions of the target area.



Figure 6 — Single datum target frame

The moveable modifier is constituted by two tangential lines from the circle of the datum target frame to the median segment (see [Figure 7](#)). The direction of the median segment of the moveable modifier is important. It specifies the movement direction where the distance between the moveable datum target and other datums or datum targets is not specified.

The moveable modifier is used to indicate the direction of movement of a physical feature or component simulating the datum or the datum system.



b) Moveable datum target (horizontal, vertical or inclined motion)

Figure 7 — Moveable modifier and examples of datum target frame with the moveable modifier

7.2.3.3 Datum target symbol

The datum target symbol indicates the type of datum target: point, line, or area, identifying a datum target point, a datum target line or a datum target area, respectively:

- a cross (see [Figure 8](#));
- a long-dashed double-dotted narrow line (type 05.1 of ISO 128-2:2022), which, when this line is not closed, is terminated by two crosses (see [Figures 9](#) and [10](#)). This line may be straight, circular or a line of any shape;
- a hatched area surrounded by a long-dashed double-dotted narrow line (type 05.1 of ISO 128-2:2022) (see [Figure 11](#)).



Figure 8 — Datum target point

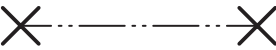


Figure 9 — Non-closed datum target line

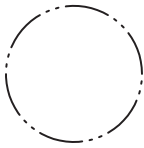


Figure 10 — Closed datum target line

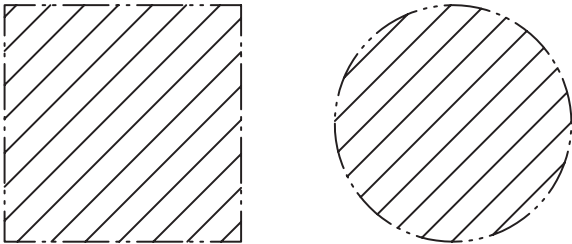


Figure 11 — Datum target area

7.2.3.4 Leader line

The datum target frame is connected directly, or through a reference line, to the datum target symbol by a leader line terminated with or without an arrow or a dot (see [Figures 12, 13](#) and [14](#)).

When the datum target point or the datum target line is not hidden, the leader line shall be continuous and shall be terminated with or without an arrow (see [Figures 12](#) and [13](#)).

When the datum target area is shown as an area in the relevant view and is not hidden, the leader line shall be continuous and shall be terminated with a dot [see [Figure 15 b](#)].

When the considered surface is hidden, the leader line shall be dashed and terminated by an open circle for a datum target area [see [Figure 15 a](#)] or without terminator for a datum target point or a datum target line.

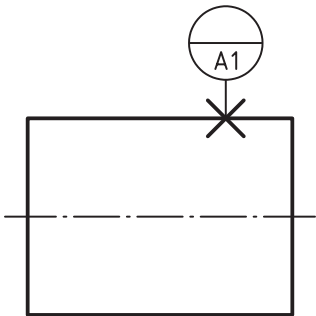


Figure 12 — Indicator for single datum target point

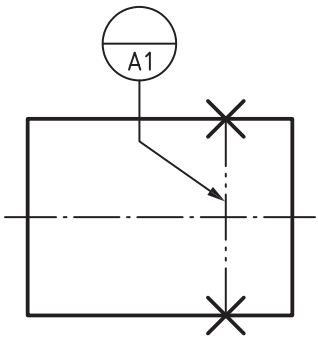


Figure 13 — Indicator for single datum target line

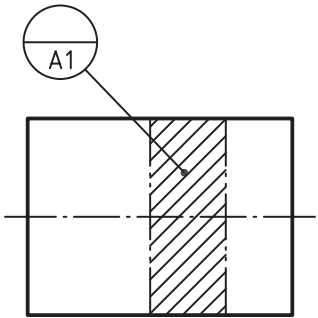


Figure 14 — Indicator for single datum target surface

The orientation of the leader line connecting the frame with the datum target symbol is unimportant. A datum target should be indicated on a view where it is not hidden if possible (see [Figure 15](#)).

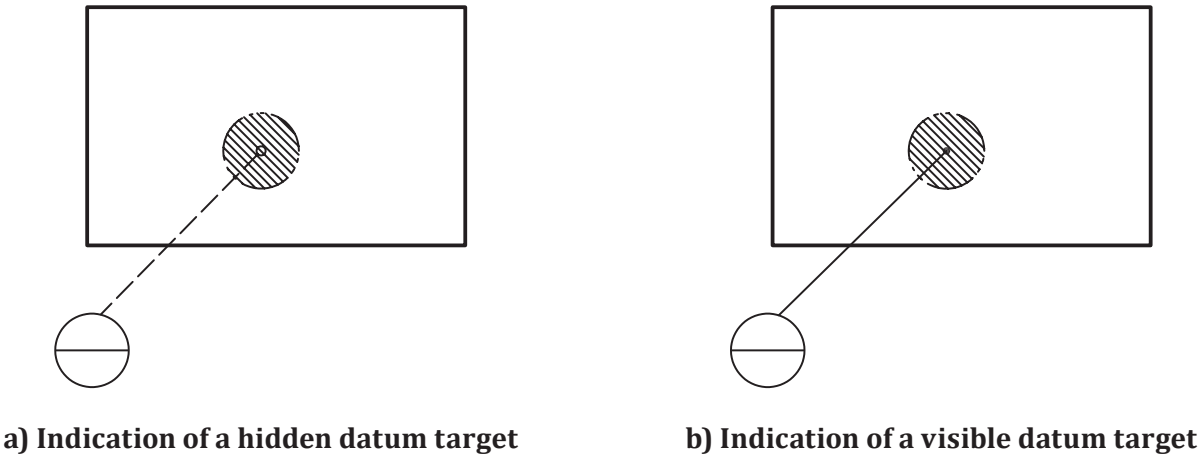
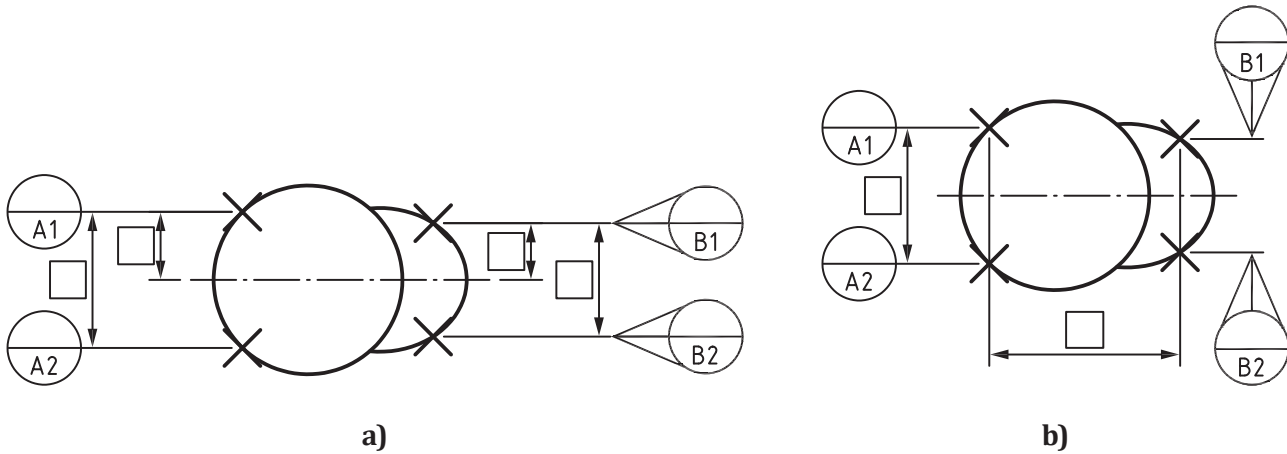


Figure 15 — Example of indication of datum target

The moveable datum target modifier is used to define the direction in which the location of the datum target is not fixed.

The direction of the motion is indicated by the direction given by the moveable modifier and not by the leader line (see [Figure 16](#)).



NOTE 1 For a), the motion of the moveable datum target, which is given by the direction of the moveable modifier, is perpendicular to a line going through the datum targets A1 and A2.

NOTE 2 For b), the motion of moveable datum target, which is given by the direction of the modifier moveable, is parallel to a line going through the datum targets A1 and A2.

Figure 16 — Example of moveable datum targets

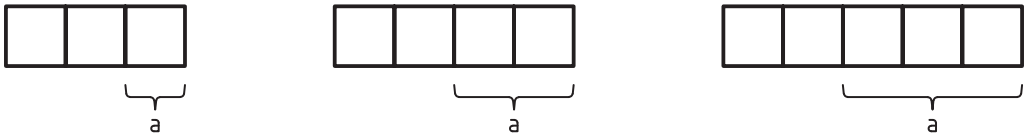
7.3 Specification of datums and datum systems

A datum (single or common) or the datum system is specified in the datum section of:

- a tolerance indicator (see [Figure 17](#) and ISO 1101:2017, 8.2);
- an intersection plane indicator (see ISO 1101:2017, Clause 13);
- an orientation plane indicator (see ISO 1101:2017, Clause 14);
- a direction feature indicator (see ISO 1101:2017, Clause 15);
- a collection plane indicator (see ISO 1101:2017, Clause 16).

A datum section is specified by one, two or three datum indicators.

For more details, see rule 6 in [7.4.2.6](#).



a Datum section, indicated by one, two or three datum indicators.

Figure 17 — Datum section in a tolerance indicator

7.4 Indication and meaning of rules

7.4.1 General

7.4.1.1 Humans interact with technical product documentation in two ways:

- designers encode the datum requirements into graphical language;
- other users decode the datum requirements from the graphical language.

The necessary rules are given in [7.4.2](#).

7.4.1.2 To encode the location constraints or orientation constraints, or both, for the tolerance zone from features on the workpiece, the following steps shall be performed.

- Indicate and identify all datum features used to establish the datum features with datum feature indicators (see rule 1) and, if necessary, define datum target(s) (see rules 1, 3 and 4).
- Consider how the datums are established from the datum features by defining the manner of association, relevant constraints and types of features (see rules 2, 5, 6, 7, 8, 9 and 10, and [Clause 8](#)).

7.4.1.3 If more than one associated feature is used to establish a datum or datum system, the associations shall be realized sequentially or simultaneously (see rules 6 and 7).

7.4.1.4 For the datum or each datum comprising the datum system, see the applicable rule as indicated below.

- If the type of associated feature is not the same as the datum feature (contacting feature), see rule 5.
- If the size of a feature of size is considered fixed or variable, see rule 2.
- If the constraint given by a datum needs to be reduced, see rules 8 and 9.
- If a geometrical modifier ($\textcircled{M}$, $\textcircled{L}$ or $\textcircled{P}$) is required for a datum, see rule 10.

7.4.1.5 To decode the meaning of a datum or datum system, the following steps shall be performed.

- Read the tolerance frame to determine whether datums are used. If the tolerance frame contains more than two compartments, then at least one datum is used (see rule 6).
- Read all datum feature identifiers corresponding to single datums or common datum components (see rule 7).
- Identify the datum feature corresponding to these datum feature identifiers (see rule 1).
- Determine the datum feature is the complete integral feature (see rule 3). If not, it is necessary to read the datum target(s) and its (their) definition(s) (consider TEDs: see rule 4).
- Consider how the associations shall be processed to establish the datum (see rules 2, 5, 6, 7, 8, 9 and 10).

7.4.2 Rules

7.4.2.1 Rule 1 — Datum features (established from a single feature)

Where the single feature used for establishing a datum is a feature of size, the surface shall be designated by a datum feature indicator placed either:

- as an extension of a dimension line [see [Figure 18 a\)](#) and [Figure 18 b\)](#)];
- on the reference line of a dimension [see [Figure 18 c\)](#)];

- on a tolerance frame linked to a reference line with a dimension and pointing to the surface [see [Figure 18 d\)](#)];
- on a tolerance indicator pointing to an extension of a dimension line [see [Figure 18 e\)](#) and [Figure 18 f\)](#)].

If the single feature is a helical surface (e.g. a screw) or a complex surface (e.g. a cylindrical gear), it is considered a cylindrical surface. To establish the datum, the datum feature indicator shall be designated as indicated in [7.2.1](#). If there is no complementary indicator, the datum is established from a pitch cylinder having the pitch diameter of the single feature and the modifier [PD] can be omitted. When the datum is established from a major cylinder (having the major diameter of the single feature) or a minor cylinder (having the minor diameter of the single feature), the symbol [MD] or [LD], respectively, is placed in the adjacent indication area of the datum feature indicator.

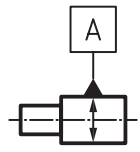
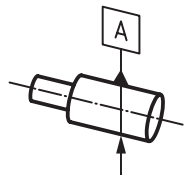
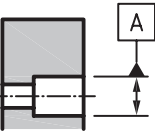
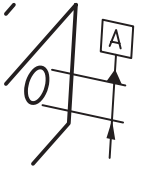
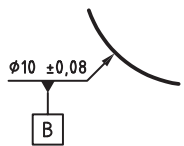
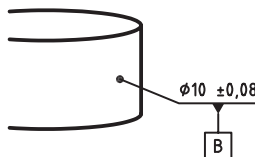
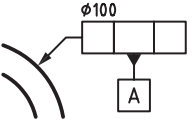
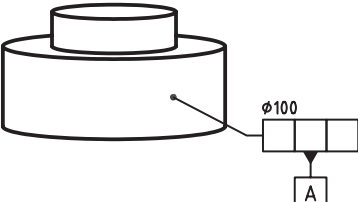
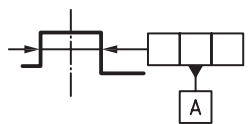
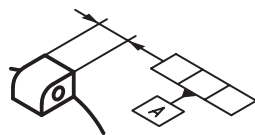
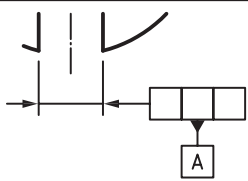
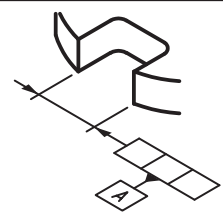
a)		
b)		
c)		
d)		
e)		
f)		
	For 2D representation	For 3D representation

Figure 18 — Attachment of a datum feature indicator for a single datum feature that is treated as a feature of size (linear or angular)

Where the single feature used for establishing a datum is not a feature of size, the surface shall be designated by a datum feature indicator placed either

- on the outline of the surface [see [Figure 19 a\)](#)];

- on an extension line of the surface [see [Figure 19 b](#)];
- on a reference line with a leader line having as terminator a dot [see [Figure 19 c](#)] for hidden surface and [Figure 19 d](#)] for visible surface];
- on a tolerance indicator pointing directly to the surface [see [Figure 19 e](#)];
- on a tolerance indicator pointing to an extension line of the surface [see [Figure 19 f](#)];
- on a tolerance indicator pointing to a reference line with a leader line having as terminator a dot [see [Figure 19 g](#)]].

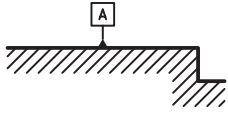
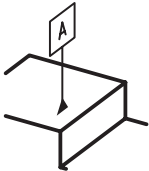
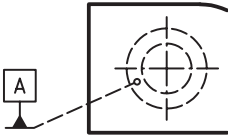
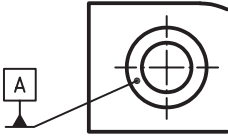
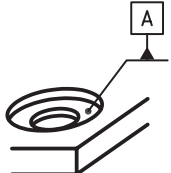
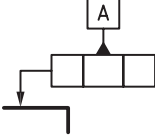
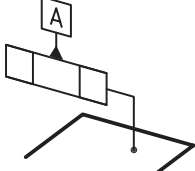
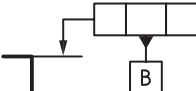
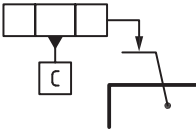
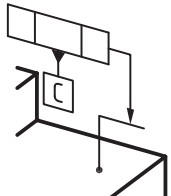
a)		
b)		Non-applicable
c)		Non-applicable
d)		
e)		
f)		Non-applicable
g)		
	For 2D representation	For 3D representation

Figure 19 — Attachment of a datum feature indicator for a single datum feature that is not treated as a feature of size

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It is recommended to indicate the datum feature indicator on a view where the surface is not hidden.

NOTE 1 A datum feature indicator can be placed on the top [see [Figures 19 e\)](#) and [19 g\)](#)] or on the bottom of a tolerance indicator [see [Figure 19 g\)](#)].

When the surface is hidden, the leader line or reference line shall be dashed and terminated by an open circle [see [Figure 19 c\)](#)].

NOTE 2 The meaning of a datum feature indicator attached to a tolerance frame for a single feature that is a feature of size is identical to that of a datum feature indicator identifying a feature of size placed in any other permitted manner (see [Figure 18\)](#).

NOTE 3 See example [C.2.1](#) for a feature which is not a feature of size and example [C.2.2](#) for a feature which is a feature of size.

7.4.2.2 Rule 2 — Size status for associated features of size

7.4.2.2.1 General

The size of an associated feature of size can be variable or fixed. This status can be defined explicitly or implicitly, in accordance with rules 2A, 2B, 2C and 2D. Examples of their applications are given for feature of linear size in [Table 8](#) and for feature of angular size in [Table 9](#). An explicit indication of size shall always take precedence over an implicit size status.

NOTE Implicit rules are retained in this document, as in many cases they correspond to established practice. Explicit indications reduce the risk of misinterpretation.

Table 8 — Examples illustrating the status of size of the associated feature of linear size, in link with the indications

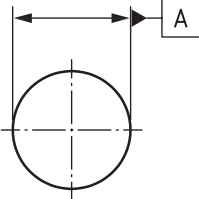
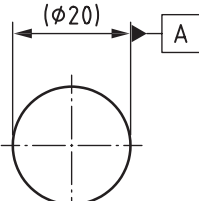
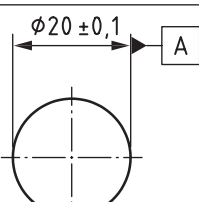
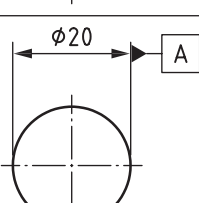
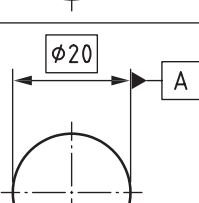
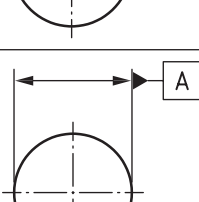
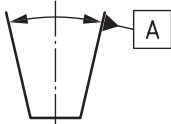
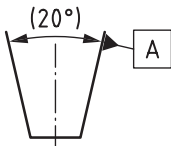
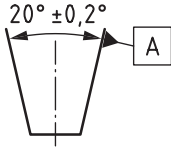
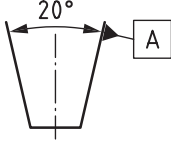
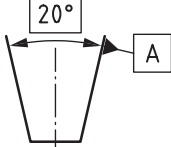
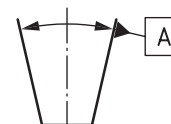
		Datum indicator			
		without [SV] or [SF] A	with [SV] A[SV]	with [SFxx] A[SF20]	with [SF] A[SF]
Indication of size value in line with the datum feature indicator (see rule 1)		Variable	Variable	Fixed with size value equal to 20 mm	Incomplete specification (fixed, however with an undefined size value)
					
					
					
		Fixed with size value equal to 20 mm			Fixed with size value equal to 20 mm
		Fixed with size value equal to CAD value			Fixed with size value equal to CAD value
TEDs according to CAD model 007a					

Table 9 — Examples illustrating the status of size of the associated feature of angular size, in link with the indications

		Datum indicator						
		without [SV] or [SF] A	with [SV] A[SV]	with [SFxx] A[SF20]	with [SF] A[SF]			
Indication of size value in line with the datum feature indicator (see rule 1)		Incomplete specification (fixed, however with an undefined size value)	Variable	Fixed with size value equal to 20°	Incomplete specification (fixed, however with an undefined size value)			
								
		Variable						
								
		Fixed with size value equal to 20°			Fixed with size value equal to 20°			
	 TEDs according to CAD model 007a	Fixed with size value equal to CAD value			Fixed with size value equal to CAD value			

7.4.2.2.2 Associated feature of size with variable size

7.4.2.2.2.1 General

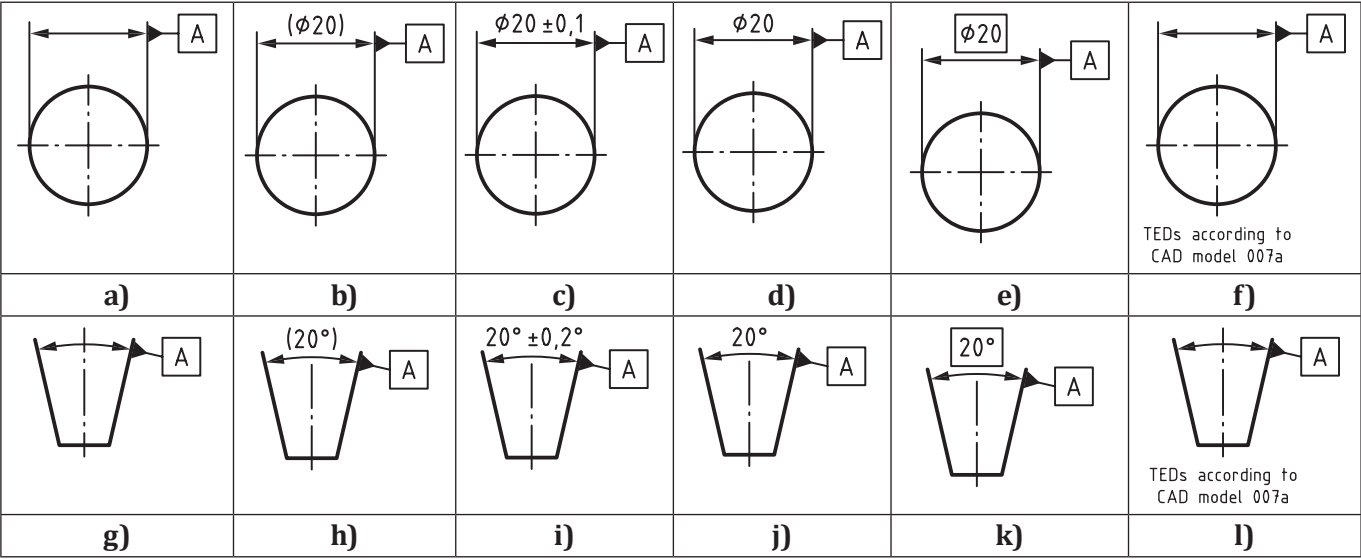
The associated feature of size shall have a variable size (linear or angular) when this is defined explicitly or implicitly, in accordance with rules 2A and 2B.

7.4.2.2.2.2 Rule 2A — Implicit definition of variable size

The associated feature shall have a variable size when:

- the datum feature indicator is attached to a dimension with a tolerance value, applied directly or indirectly as a general tolerance value [see Figure 20 c), d), i) and j)]; or
- the datum feature indicator is attached to an auxiliary dimension for a feature of linear size [see Figure 20 b)]; or

- the datum feature indicator is attached to a dimension without a dimension value for a feature of linear size [see [Figure 20 a\)](#)], and there is no indirect indication that the dimension value is a TED (e.g. with a reference to the CAD model with a statement such as “TEDs in accordance with CAD model 007a”).



NOTE Other indications of size in line with datum feature indicator are available, see rule 1.

Figure 20 — Illustration of datum feature indicator attached to a dimension line, of size (linear or angular)

7.4.2.2.2.3 Rule 2B — Explicit indication of variable size

The associated feature shall have a variable size when the modifier [SV] is placed immediately after the datum identifier in a datum indicator of the datum section. There shall be no space between the datum identifier and the modifier [SV].

NOTE The [SV] modifier can be used with features of linear size and features of angular size.

7.4.2.2.3 Associated feature of size with fixed size

7.4.2.2.3.1 General

The associated feature of size shall have a fixed size when this is defined explicitly or implicitly, in accordance with rules 2C and 2D.

7.4.2.2.3.2 Rule 2C — Implicit definition of fixed size

The associated feature shall have a fixed size when the datum feature identifier is attached to a dimension line:

- with a TED indicated directly, or defined indirectly by a reference to a CAD model, with a statement such as ‘TEDs in accordance with CAD model 007a’ [see [Figure 20 e\), f\), k\) and l\)](#)]; or
- with an auxiliary dimension for a feature of angular size [see [Figure 20 h\)](#)]; or
- without a dimension value for a feature of angular size [see [Figure 20 g\)](#)].

7.4.2.2.3.3 Rule 2D — Explicit indication of fixed size

The associated feature shall have a fixed size when the modifier [SF] or [SFxx] (where “xx” is the fixed size value) is placed immediately after the datum identifier in a datum indicator of the datum section. There

shall be no space between the datum identifier and the modifier. The size value “xx” shall be indicated after the letter sequence “SF”, followed by the angular unit when it is an angular size, for example [SF10], [SF30°]. There shall be no space between “SF” and the value.

The [SFxx] modifier defines the fixed size of the associated feature, and the value indicated takes precedence over any other value given in a dimension or the CAD model.

When the modifier [SF] is used, the fixed size is equal to the TED value indicated directly on the dimension line to which the datum feature identifier is attached, or indirectly by an explicit reference to a CAD model, with a statement such as “TEDs in accordance with CAD model 007a” placed nearby the title block.

NOTE Explicit indication of the fixed size, i.e. [SFxx], reduces the risk of misinterpretation compared to the use of [SF].

7.4.2.3 Rule 3 — Datum features established from a complete feature

If a datum is established from a complete integral feature, it shall be indicated with a datum feature indicator only.

NOTE See [Figures 18](#) and [19](#) and [C.2.1](#).

7.4.2.4 Rule 4 — Datum features established from one or more datum targets

If a single datum feature is established from one or more portions of a single surface, by using datum targets, one of the following indications shall be placed in the adjacent indication area of the datum feature indicator:

- the datum feature identifier followed by the list of numbers, separated by a comma and a space, identifying each datum target to be used, see [Figure 21 a\)](#);
- the datum feature identifier followed by the first and last numbers separated by a hyphen without space, identifying the set of datum targets to be used (all datum targets between the first and the last number are used), see [Figure 21 b\)](#).



Figure 21 — Indication of a datum feature defined from datum targets

Each individual datum target shall be identified by a datum target indicator, indicating the datum feature identifier, the number of the datum target and, if applicable, the dimensions of the datum target.

NOTE 1 See [C.2.11](#).

When there is one datum target only, the datum feature defined as a restricted area can be defined as a complement of rule 1 of the datum feature indication in one of the following ways:

- by attaching the datum feature indicator on:
 - the long-dashed dotted wide line (in 2D) (type 04.2 of ISO 128-2:2022) [see [Figure 22 a\)](#) for non-feature of size and [Figure 22 g\)](#) for feature of size];
 - the restricted area indication [see [Figures 22 d\)](#) and [e\)](#)];
 - the reference line connected to a leader line pointing directly to the restricted area [see [Figure 22 b\)](#)];
 - anywhere inside of the considered restricted area of the surface, completed in its adjacent indication area by the description of it, using the between symbol and the labels of its boundaries or corners [for 2D, see [Figure 22 c\)](#); for 3D, see [Figure 22 f\)](#)];

- the tolerance indicator for which its leader line shall point directly or indirectly to the restricted area;
- by indicating:
 - ACS above the tolerance indicator [see Figure 23 a)] or [ACS] after the datum identifier in the tolerance indicator [see Figure 23 c)], allowing the datum feature to be defined as any cross-section of the integral feature (concurrently with the tolerated feature);
 - ALS above the tolerance indicator [see Figure 23 b)] or [ALS] after the datum identifier in the tolerance indicator [see Figure 23 c)], allowing the datum feature to be defined as any longitudinal section of the integral feature (concurrently with the tolerated feature).

When ACS or ALS is indicated above the tolerance indicator, the tolerated feature and the datum feature are established in the same cross-section.

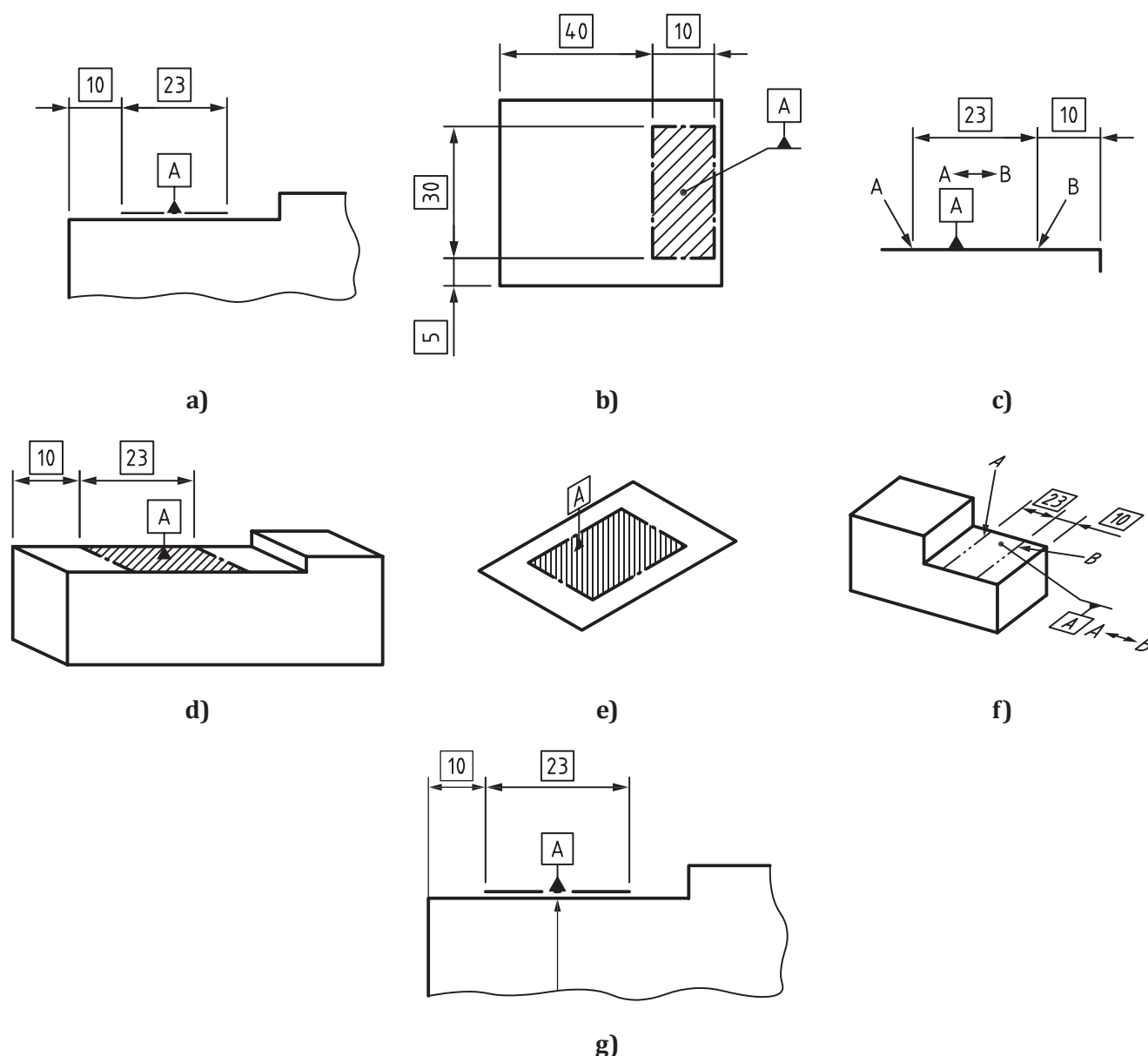


Figure 22 — Simplification of drawing indication when a datum feature is defined by only one datum target area

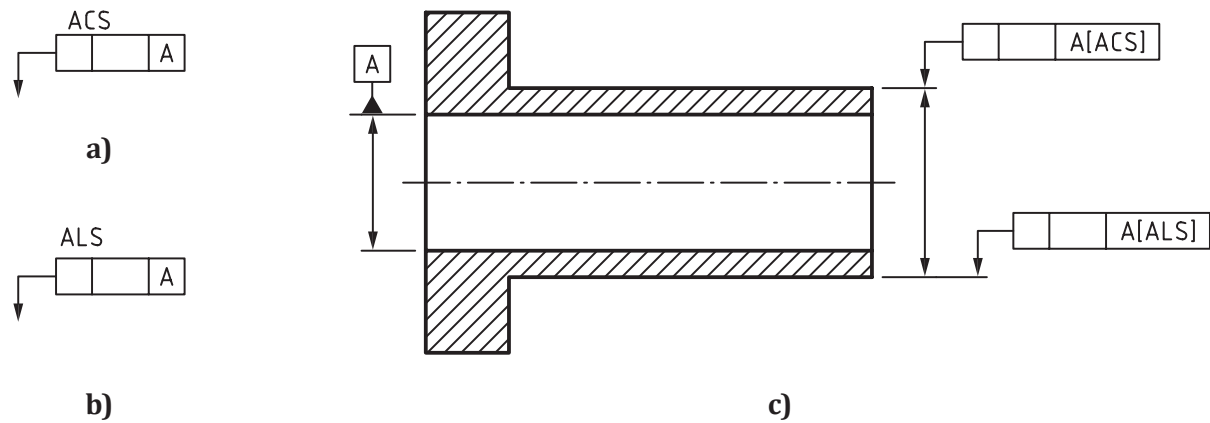
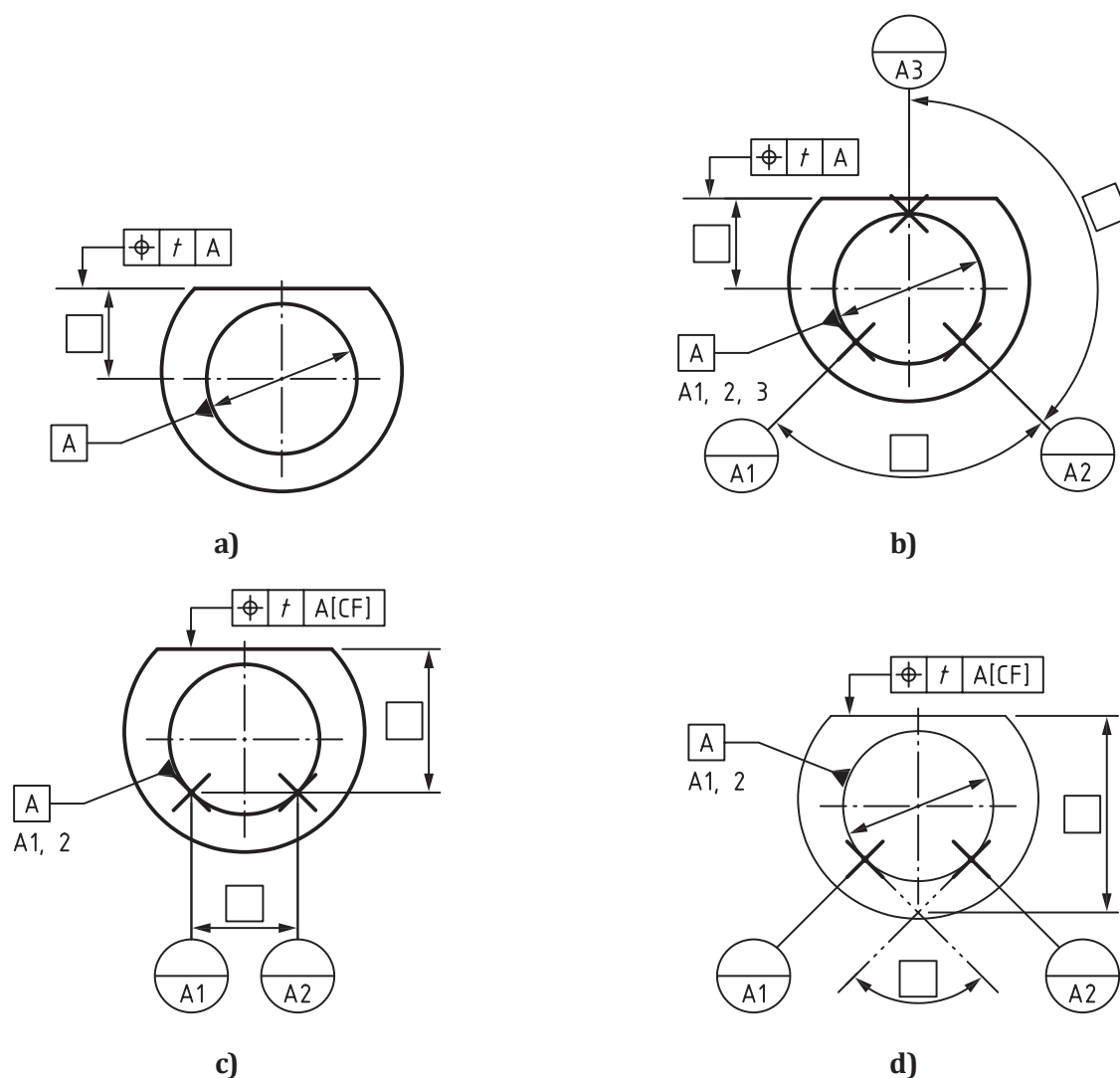


Figure 23 — Indication of [ACS] and [ALS] modifier

The relative location of datum targets on one single feature shall be defined by angular and/or linear TED(s) when it shall be considered fixed [see [Figure 24 c\)](#)].

When a set of datum targets is used to simulate a feature of size, the number and the location of the datum targets shall be adequate to simulate its size and the relative location shall be defined by a linear or angular TED [see [Figure 24 b\)](#)]. Where datum targets are located from another feature, linear TED are required [see [Figure 27\)](#)].



NOTE 1 For a), the associated feature used to establish the datum from the complete integral feature is a cylinder and the datum is its axis.

NOTE 2 For b), the angular locations between the datum target lines A1, A2 and A3 are indicated by TEDs and are considered as fixed. The associated feature used to establish the datum from 3 datum target lines is a cylinder and the datum is its axis. (To fully define datum target lines A1, A2 and A3 more than one drawing view is required.)

NOTE 3 For c), the distance between datum targets A1 and A2 is indicated by a TED and is fixed. The associated feature used to establish the datum is not a cylinder and the datum is not its axis. The datum consists of two planes: one plane through two datum target lines and a second perpendicular median plane.

NOTE 4 For d), the datum targets A1 and A2 are defined by the interface between the cylinder A and a contacting feature. The distance between datum targets A1 and A2 is variable. It depends of the actual diameter of the cylinder and the contacting feature defined in this case by a "V" block of angle α . The associated feature used to establish the datum is the "V" block, not a cylinder.

Figure 24 — Examples of single datums established from a complete cylinder, a portion of a cylinder or with a contacting feature

When the location of a datum target is not fixed, the datum target shall be specified by using

- one or more contacting features, when the contact zone cannot be predetermined [see rule 5 and [Figure 24 d\)](#)] (implicit datum targets can exist when datum targets are defined without ambiguity by the definition of a contacting feature and the interface of the datum feature), or
- a moveable datum target (see [Clause E.3](#)) when the distance between that datum target and other datum targets on the same surface is not fixed but their location relative to other datum features is defined.

When two or more moveable datum targets are used for one datum, they move synchronously.

For example, when three distinctive points on a cross-section of a cylinder are used as the datum target point to establish a datum, only the angular relation between these three points on the cylinder needs to be defined. The moveable modifier, pointing in the direction normal to the cylinder, can be added or omitted in this case (no distances from other features are implicated in the relation).

When only two points on a cross-section of a cylinder with fixed distance between them are used as datum target points, a TED between these two points needs to be indicated. The associated feature in this case is not a cylinder but two perpendicular planes.

The location of a datum target relative to one or more other features shall be considered fixed and shall be defined by a TED. Exceptions exist, for example when moveable modifiers are not used.

When the location of a datum target on one feature relative to another feature is not fixed in a particular direction (other than normal to the surface profile), and when the [ACS] modifier is not used, then the datum target shall be defined as a moveable datum target (indicated with the “moveable” modifier in the datum target indicator) to indicate the direction in which the distance between the features is variable.

When the location of a datum target is defined in a cross-section with a variable location on one feature, is not fixed in a particular location relative to another feature, and is available in any cross-section, then the [ACS] modifier shall be used.

The beginning and the end of the location of a datum target line shall be considered theoretically exact locations and defined by TEDs.

The extent of a datum target area shall be considered theoretically exact. The dimensions of the area shall be indicated

- either in the upper compartment of the datum target frame when the area is circular, square (see [Figure 25](#)), or rectangular, or placed outside and connected to the appropriate compartment by either a leader line or by a leader line and a reference line if the space within the compartment is limited [see [Figure 26 a\)](#)], or
- directly on the drawing by TEDs when the area is neither square nor circular [see [Figure 26 b\)](#)].

NOTE 2 In the case of a point or a line, it can be necessary to indicate the datum target on several views in order to have an unambiguous definition.

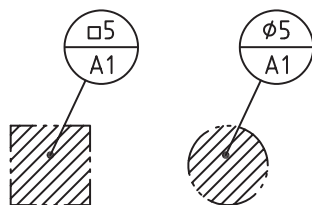


Figure 25 — Indication of dimension of a circular/square area

Dimensions in millimetres

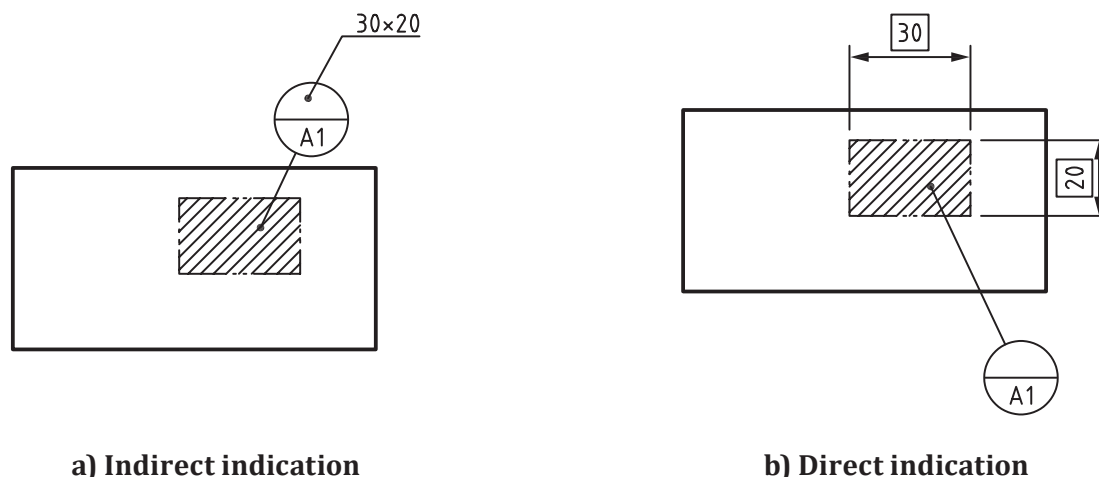


Figure 26 — Indication of dimensions of a rectangular area

7.4.2.5 Rule 5 — Associated feature with a different type than the nominal datum feature

If the associated feature establishing the datum is not of the same type as the nominal feature, datum targets shall be used and the indication [CF] shall be added in the tolerance frame after the datum feature identifier (see [Figures 27, 28, 30 and 31](#)).

The dimensions of the contacting feature shall be considered fixed and shall be indicated on the drawing, when it is not implicit, by drawing the contacting feature with a long-dashed double-dotted narrow line (type 05.1 of ISO 128-2:2022) in contact with the datum feature.

NOTE 1 The modifier [CF] implies that some portions of the workpiece are used to establish the datum and that the location of the contact between the contacting feature and the workpiece cannot be determined exactly (it depends on the dimensions and the geometry of the real workpiece); see examples in [Annex E](#). The modifier [CF] allows some dimensions between the datum targets on a single feature to become variable (see [Figure 28](#)). The datum targets can be omitted when they do not reduce the potential contact between the contacting feature and the workpiece (see [Figure 30](#), compared to [Figure 31](#)).

NOTE 2 Datum targets are used to express the nominal contact between the contacting features and the surface of the workpiece used to establish the datum. In some cases, the datum target can be omitted.

EXAMPLE In [Figure 24 d](#)), the datum A with modifier [CF] is a set of two perpendicular planes and not the axis of a cylinder. The first plane contains the datum target points; the second plane is the median plane of the datum target points, perpendicular to the first one. This median plane contains the set of axes of possible associated cylinders (with variable diameters) from the datum target points. If the modifier [CF] had not been used, the datum A would have been the axis of the associated cylinder (with a variable diameter or with a fixed diameter depending on the application of rule 2).

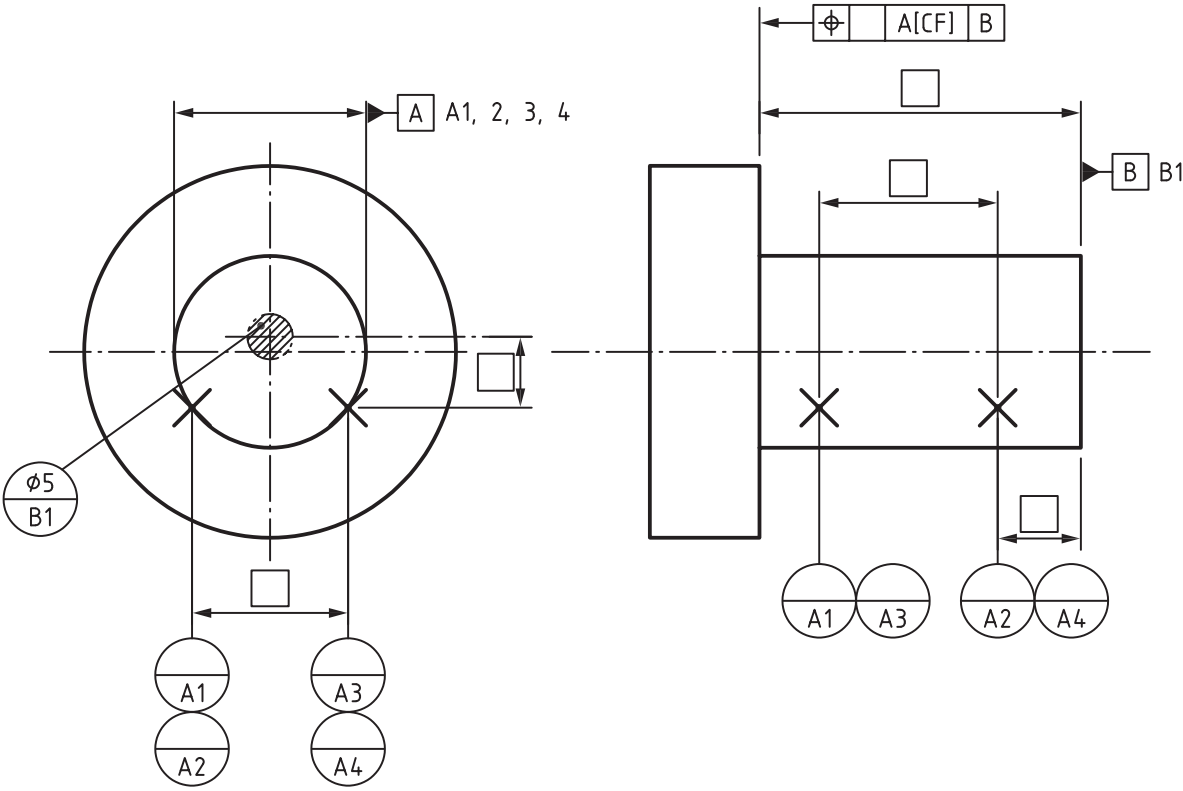


Figure 27 — Example of indication of datum with modifier [CF]

Dimensions in millimetres

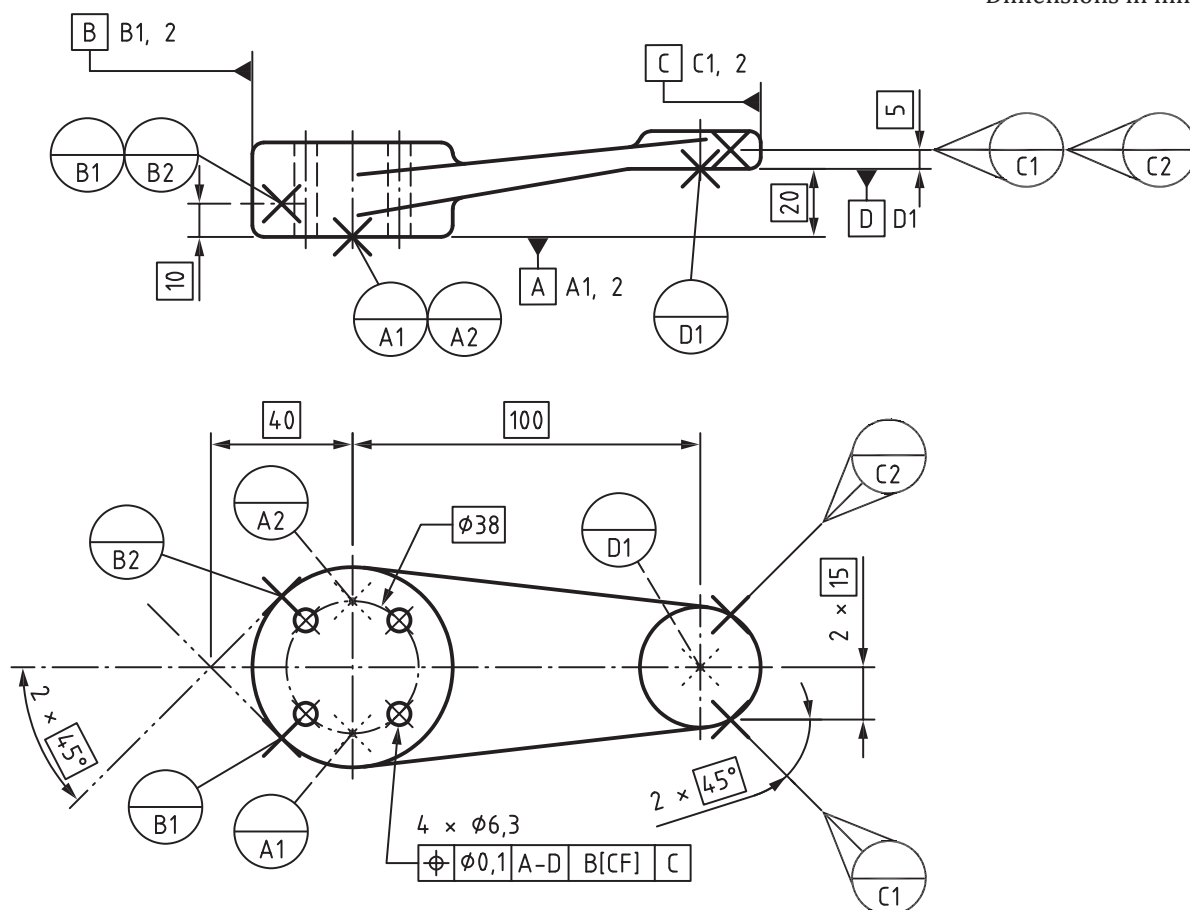


Figure 28 — Example of datum system established from datum targets

NOTE 3 In [Figure 28](#), the distance between the datum targets B1, B2 and C1, C2 is not predefined. Therefore, C1 and C2 are defined as moveable datum targets in relation with datum targets B1 and B2. C1 and C2 move synchronously.

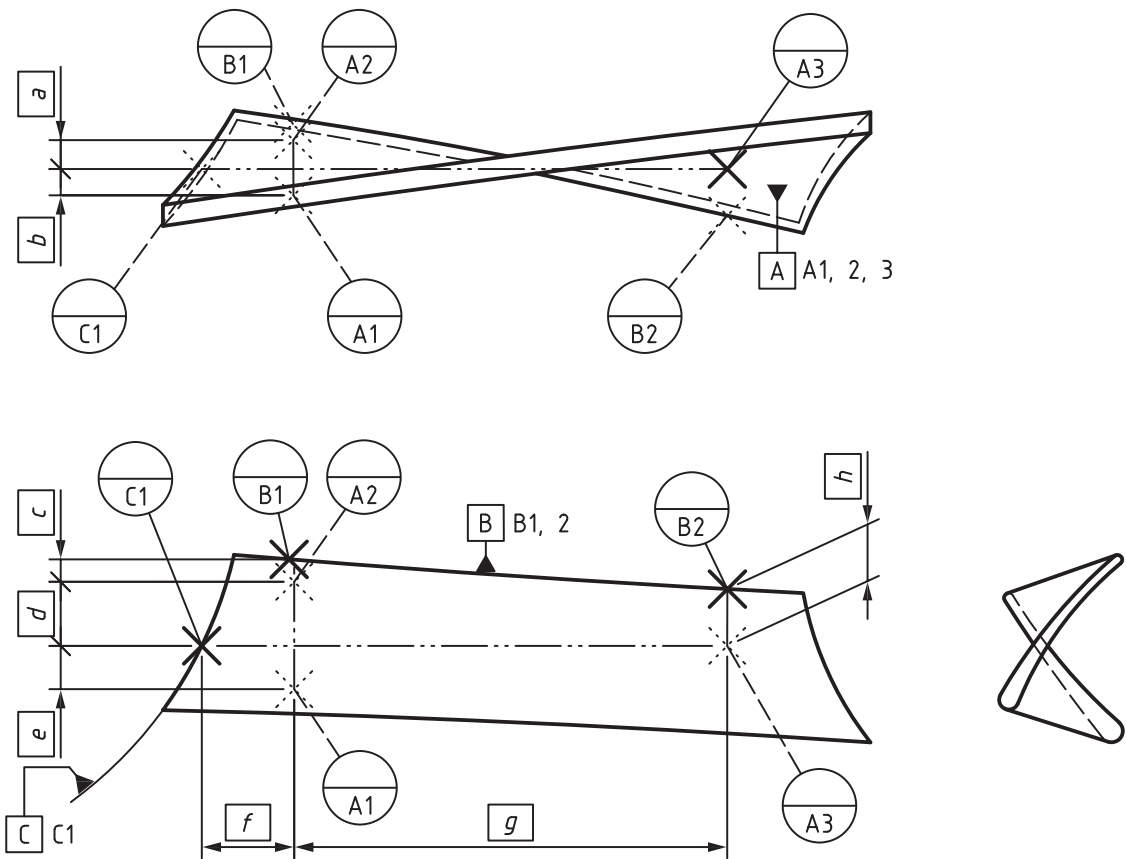


Figure 29 — Example of indications of datum targets on complex surfaces

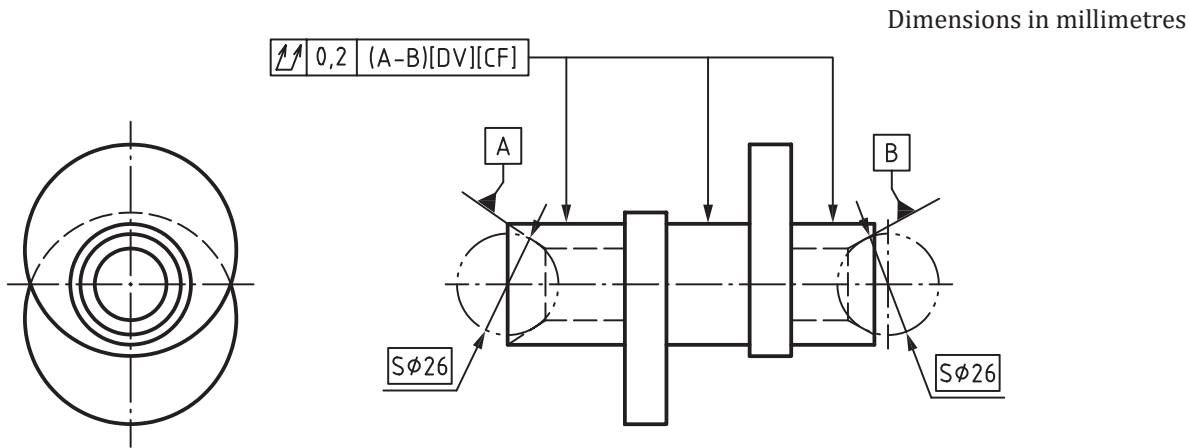


Figure 30 — Example of datum system in which the datum target indication is omitted

Dimensions in millimetres

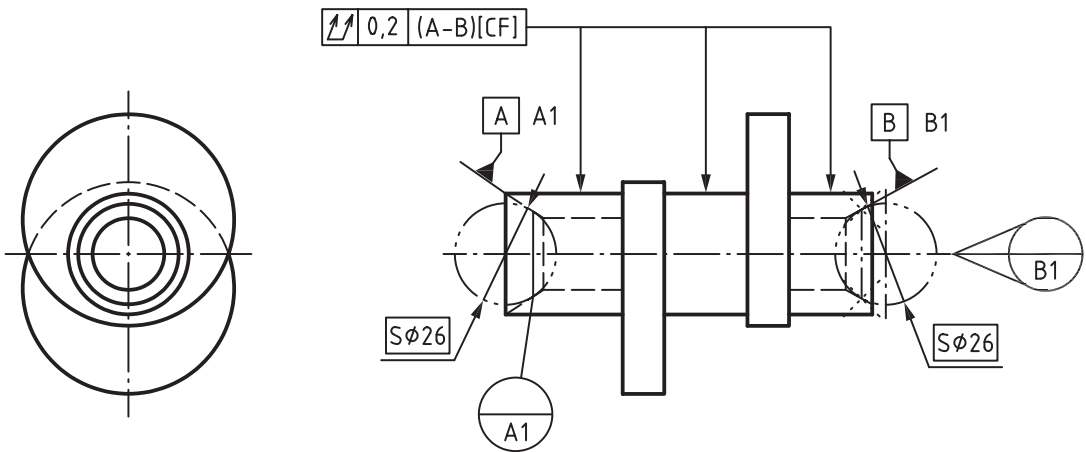


Figure 31 — Example of datum system in which the datum target is indicated

7.4.2.6 Rule 6 — Tolerance frame layout with a datum or a datum system

If the datum section has only one datum indicator, this may contain:

- only a single datum [see Figure 32 a)];
- a common datum used as the only datum [see Figure 32 b)];
- an indirect indication of a datum section [see Figure 42 b) and rule 11].

If the tolerance zone is oriented or located from more than one datum established in a specific order, then the tolerance frame shall have more than three compartments and each compartment after the second compartment shall indicate a single or common datum [see Figures 32 c) to 32 e)]. The specified order defines the orientation constraints between the primary, secondary and tertiary datums (single or common). The value of the constraints shall be specified by TEDs. TED values of 0°, 90°, 180° and 270° are implicit and not indicated.

NOTE 1 In a datum system, the primary datum is identified in the third compartment of the tolerance frame; the secondary datum is identified in the fourth compartment of the tolerance frame; the tertiary datum is identified in the fifth compartment of the tolerance frame.

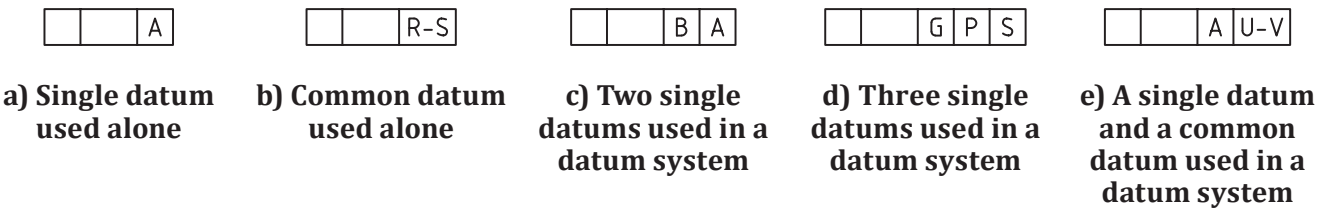


Figure 32 — Examples of indication of datums in a tolerance frame

NOTE 2 See Clauses C.2 and C.3 for examples where there are only three compartments in the tolerance frame and Clause C.4 for examples where there are more than three compartments in the tolerance frame.

7.4.2.7 Rule 7 — Indication of a single or common datum in a datum indicator of a datum section

When the datum is a single datum, it shall be indicated by one datum identifier which corresponds to one datum feature identifier in a datum indicator (compartment) of the datum section [see Figures 32 a), 32 c), 32 d) and 32 e)].

When the datum is a common datum, it shall be indicated by one datum identifier, which is a sequence of more than one datum feature identifier separated by hyphens, in a datum indicator of the datum section [see [Figures 32 b\)](#) and [32 e\)](#)].

In the case of a common datum, the following points apply:

- The associated features that establish a common datum are by default constrained in location and orientation to each other. If the modifier [DV] is placed in the tolerance frame after the letter(s) identifying a common datum, then the linear distance between the members of the collection of features that make up this common datum shall be considered variable.
- The orientation and location constraints correspond to the intrinsic characteristics introduced by the collection of features and are specified by TEDs. The values of 0 mm, 0°, 90°, 180° and 270°, and equally divided linear or angular dimensions may be implicit TEDs and not indicated.
- There are as many datum feature identifiers in the tolerance frame as single features used for establishing the common datum, unless a simplified indication is used.
- The sequence of the letters identifying the common datum has no significance.
- It is possible to simplify the drawing indication by:
 - using only one datum feature indicator;
 - using only one doubled letter separated by a hyphen in the datum indicator of the datum section [see [Figure 33 d\)](#)];
 - adding the complementary indication “n×” giving the number, *n*, of surfaces in the collection in the adjacent indication area of a datum feature indicator attached to one of the surfaces [see [Figure 33 a\)](#)]. When the datum feature indicator points to the tolerance frame, the indication “n×” is not written on the adjacent indication area of a datum feature indicator but above the tolerance frame [see [Figure 33 b\)](#)], or, when the datum feature indicator points to the tolerance frame, by using *n* leader lines indicating each surface included in the common datum [see [Figure 33 c\)](#)].

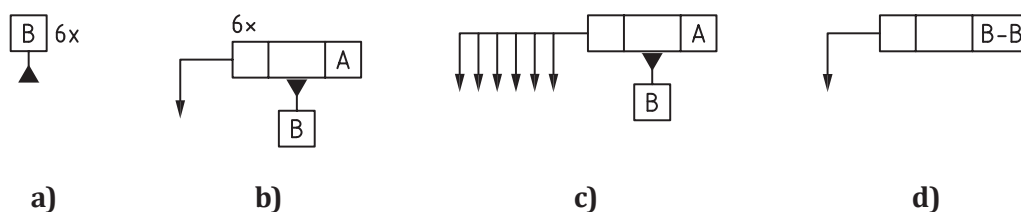


Figure 33 — Examples of complementary indication

NOTE See the examples given in [Clause C.2](#) for single datums and in [Clause C.3](#) for common datums. See the example in [C.3.5](#) for the simplified drawing indication.

7.4.2.8 Rule 8 — Locked or released degrees of freedom from a datum

If all situation features of a single or common datum are required to lock all possible degrees of freedom of the tolerance zone in relation to the geometrical characteristic, no further indication (PL, SL, PT, ><) shall be added after the datum identifier in its datum indicator (see [C.2.7](#)).

If all the situation features of a single or common datum and/or the location from a datum are not required, a complementary indication (PL, SL, PT, ><) shall be added after the datum identifier in its datum indicator, except when it is obvious from the specification which situation feature is to be used:

- the complementary indication is [PL] if the plane (situation feature) is needed (see [Figures 34](#) and [37](#));
- the complementary indication is [SL] if the straight line (situation feature) is needed (see [Figures 35](#) and [37](#));

- the complementary indication is [PT] if the point (situation feature) is needed (see [Figure 36](#));
- the complementary indication is >< if the datum is only used to lock the orientation degrees of freedom and not the location [see [Figure 38 a](#)]. The >< symbol shall be omitted when the geometrical characteristic only controls the orientation of the feature (e.g. a perpendicular specification). [Figure 38 b](#) illustrates the meaning of the specification including the >< symbol.

NOTE 1 By default, when a datum is established on a complex surface, for example, the datum consists of a plane, a straight line and a point. The modifier [SL], [PL] or [PT], or a combination thereof, can be attached to the datum identifier to limit the situation feature(s) obtained from the surface.



Figure 34 — Indication when only the plane is needed in the set of situation features



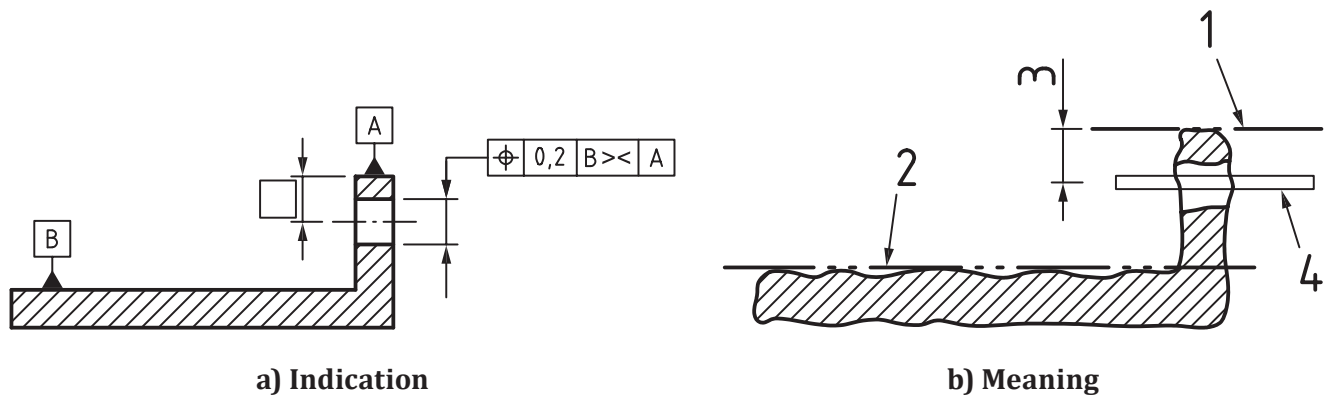
Figure 35 — Indication when only the straight line is needed in the set of situation features



Figure 36 — Indication when only the point is needed in the set of situation features



Figure 37 — Indication when only the plane and the straight line are needed in the set of situation features



Key

- 1 associated plane with outside material constraint and with orientation constraint from datum B
- 2 associated plane with outside material constraint (datum B)
- 3 distance relative to the location
- 4 tolerance zone with orientation constraint from datum B and location constraint from datum A

NOTE The modifier >< allows only the orientation degrees of freedom of the tolerance zone and not its translation degree of freedom (managed by the datum A, in this case) to be locked from the datum B.

Figure 38 — Example of datum with the >< modifier

NOTE 2 See the examples in [C.2.5](#) where all the situation features are needed, example [C.2.6](#) for obvious situation feature and example [C.2.7](#) where only one situation feature is needed.

7.4.2.9 Rule 9 — Special indications for common datum

When a complementary indication ([CF], [SF], [SV], [DV], [PT], [SL] or [PL]) applies to all elements of the collection of surfaces of a common datum, the sequence of letters identifying the common datum shall be indicated within parentheses [see [Figures 39 a\)](#) and [39 c\)](#)].

When a complementary indication ([CF], [SF] or [SV]) applies only to one element of the collection of surfaces of a common datum, the sequence of letters identifying the common datum shall not be indicated within brackets, and the complementary indication applies only to the feature identified by the letter just before the indication(s) [see [Figures 39 b\)](#) and [39 d\)](#)].

If the linear distance between the members of the collection of features that make up a common datum shall be considered variable, then the modifier [DV] shall be placed in the datum indicator after the letter(s) identifying the common datum [see [Figure 39 c\)](#)].

NOTE See example [Clause E.4](#).

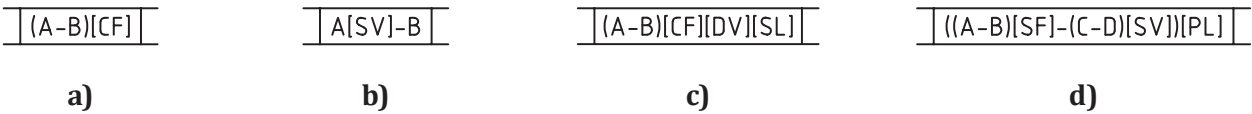


Figure 39 — Examples of complementary indications for common datums

7.4.2.10 Rule 10 — Application of geometrical modifiers in a tolerance frame

If the modifier $\textcircled{M}$, $\textcircled{L}$ or $\textcircled{P}$ is placed in the tolerance frame after the letter, then the default meaning changes.

If one of the modifiers $\textcircled{M}$ or $\textcircled{L}$ is placed in the tolerance frame after the letter indicating a datum, the datum shall be established in accordance with ISO 2692.

When the modifier $\textcircled{P}$ is placed in the tolerance frame after the letter indicating a datum established from a feature of size, then the datum feature shall be established by fitting an associated feature of the projected length to the extension of the real feature by considering the criteria defined in [Annex A](#) and not the real integral feature itself.

When using the modifier $\textcircled{P}$, the extension of the feature shall be indicated, directly on the drawing (see [Figure 40](#)) or after the modifier $\textcircled{P}$ in the tolerance frame. The dimension(s) of this extension shall be seen as a TED.

NOTE The modifier $\textcircled{P}$ can be applied to a secondary or tertiary datum. In [Figures 40](#) and [41](#), the effect of this modifier is shown when the datum is a secondary datum in a datum system. The modifier has no effect when it is applied to a primary datum, and the same association criteria are used to define the projected datum feature and to determine the datum.

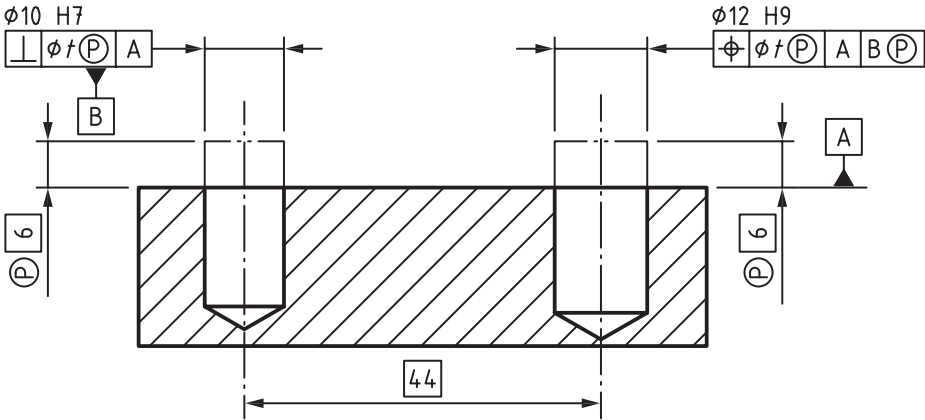
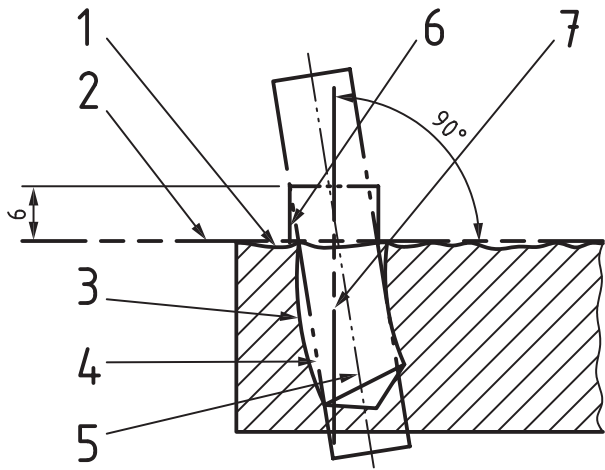


Figure 40 — Example of application of modifier $\textcircled{P}$ on the secondary datum

Dimensions in millimetres



Key

- 1 real integral feature of the planar surface
- 2 datum A: an associated integral feature to 1
- 3 real integral feature of the cylindrical surface
- 4 associated integral feature to 3
- 5 derived feature of 4
- 6 associated integral feature to the portion of 4, with the constraint perpendicular to 2
- 7 datum B: the derived feature of 6 (as a secondary datum)

Figure 41 — Meaning of the specification given in [Figure 40](#)

7.4.2.11 Rule 11 — Indirect indication of a datum section

A datum section indicating one or more datums (single datum or common datum) may be indirectly indicated by using a datum system identifier in only one datum indicator of a datum section.

The datum system identifier consists of the symbol DS, optionally followed, without space, by a positive integer *n*.

The datum system, DS*n*, shall be defined in or near the title block as follows: a datum section, consisting in one datum indicator (one compartment), containing the datum system identifier, followed by an equal sign, followed by a datum section (see example in [Figure 42](#)).

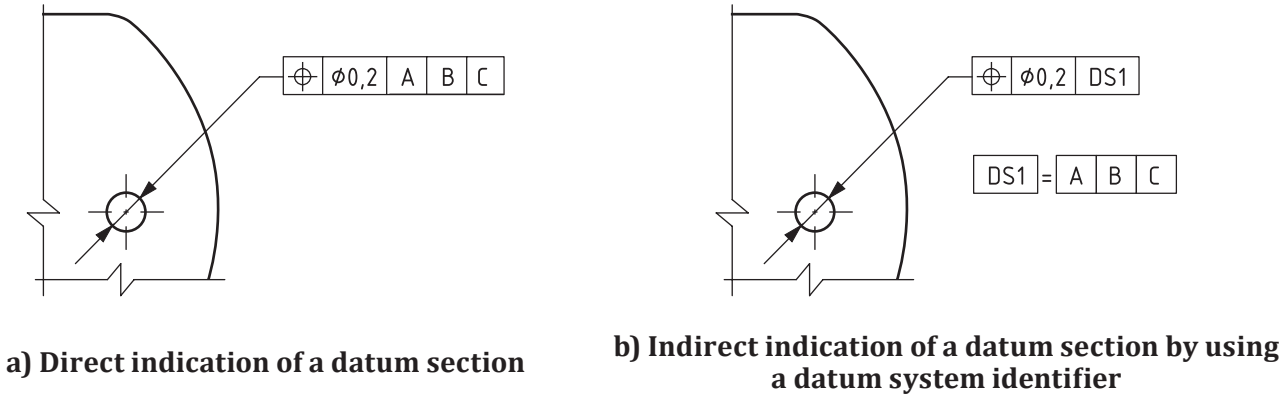


Figure 42 — Example of direct and indirect indications of a datum section

7.4.2.12 Rule 12 — Indication of a situation feature

If a situation feature of a datum (single or common) or a datum system needs to be indicated explicitly, then it shall be represented and identified.

NOTE 1 When a datum consists of only one situation feature [e.g. axis of an associated cylinder, median plane of two associated parallel opposite planes, median point (centre) of an associated sphere], it is not necessary to include any additional indication to identify the situation feature.

The representation of the situation feature on the drawing shall be made in accordance with [Table 10](#).

Table 10 — Representation of situation features on a drawing

Situation feature	Representation on the drawing
Point	A wide line cross (according to Figure F.2).
Straight line	A long-dashed double-short-dashed wide line (type 09.2 of ISO 128-2:2022); or, if viewed end view, a wide line cross.
Plane	Four straight lines with a long-dashed double-short-dashed wide line (type 09.2 of ISO 128-2:2022) forming a rectangle, representing the borders of a plane; or, if viewed end view, a long-dashed double-short-dashed wide line (type 09.2 of ISO 128-2:2022).

NOTE 2 The situation features are, for instance, the origins from which the angular and linear theoretical exact dimensions (TEDs) allows to orientate and locate the theoretically exact features (TEF).

The representation of the situation feature shall be connected with the situation feature identifier by a leader line. Its terminator depends of the representation of the geometrical feature, which is pointed:

- When the leader line points on a point (projection of the situation straight line or the situation point), there is no terminator.
- When the leader line points on a straight line (a border of the representation of the situation plane or the situation straight line), the terminator is an arrow.
- When the leader line points on a planar surface (situation plane), the terminator is a full dot.

The situation feature identifier consists of a datum identifier or a datum system identifier followed, without a space, by a situation feature type indicator. For a common datum, its identifier is placed between parenthesis. For a datum system, it recommended to use an indirect indication of a datum e.g. DS1.

The situation feature type indicator is [PT], [SL] or [PL], for a situation feature of type point, straight line or plane, respectively.

EXAMPLE

- A[PT], A[SL], A[PL];
- (A-B)[PT], (A-B)[SL], (A-B)[PL];
- DS1[PT], DS1[SL], DS1[PL].

See [Figures 43](#) and [44](#).

NOTE 3 In a 2D drawing, it can be necessary to show the same situation feature in several views to provide a complete definition [see [Figure 43 a](#))].

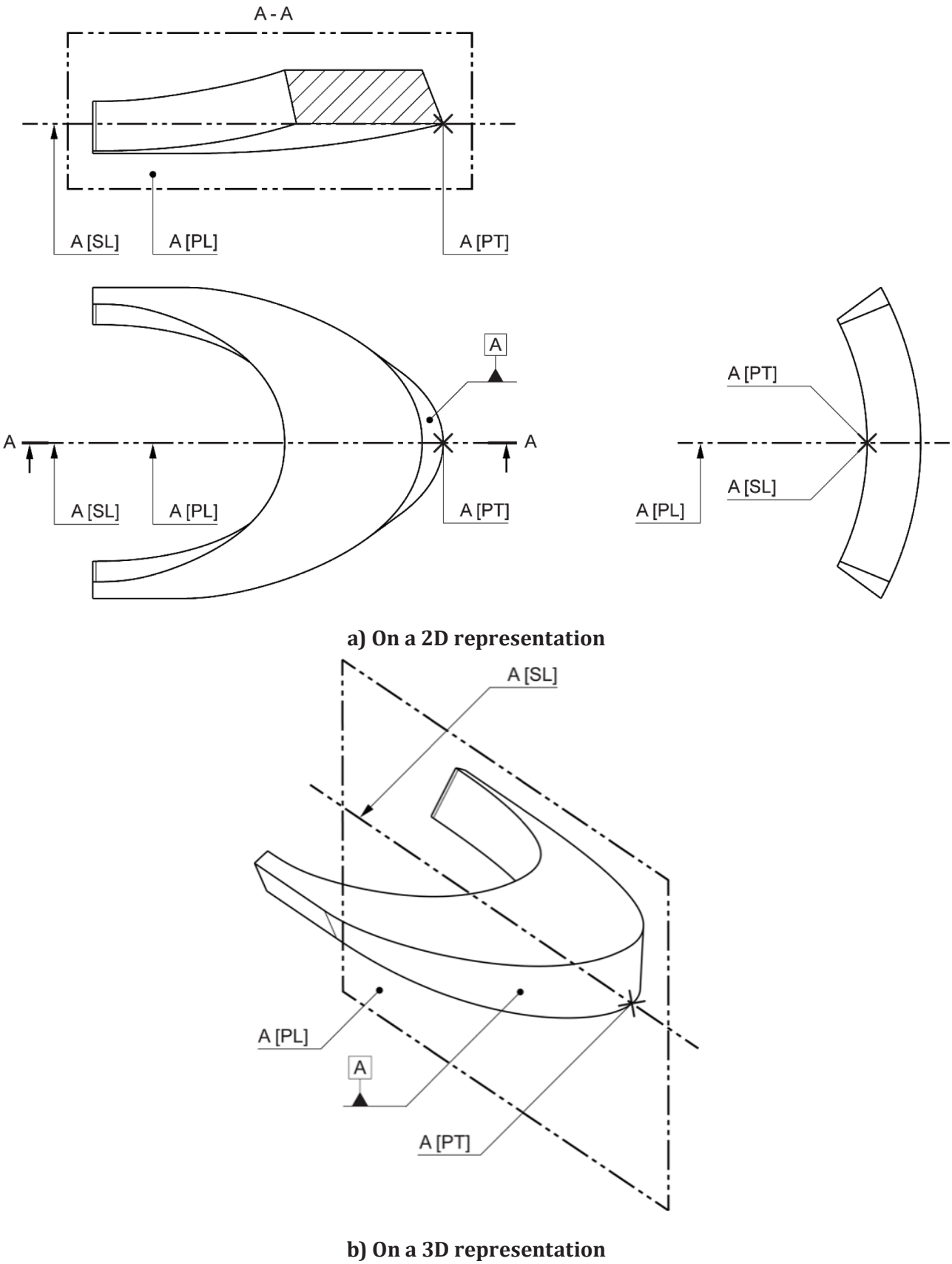


Figure 43 — Example of indication of the situation features constituting a datum established from a complex integral surface

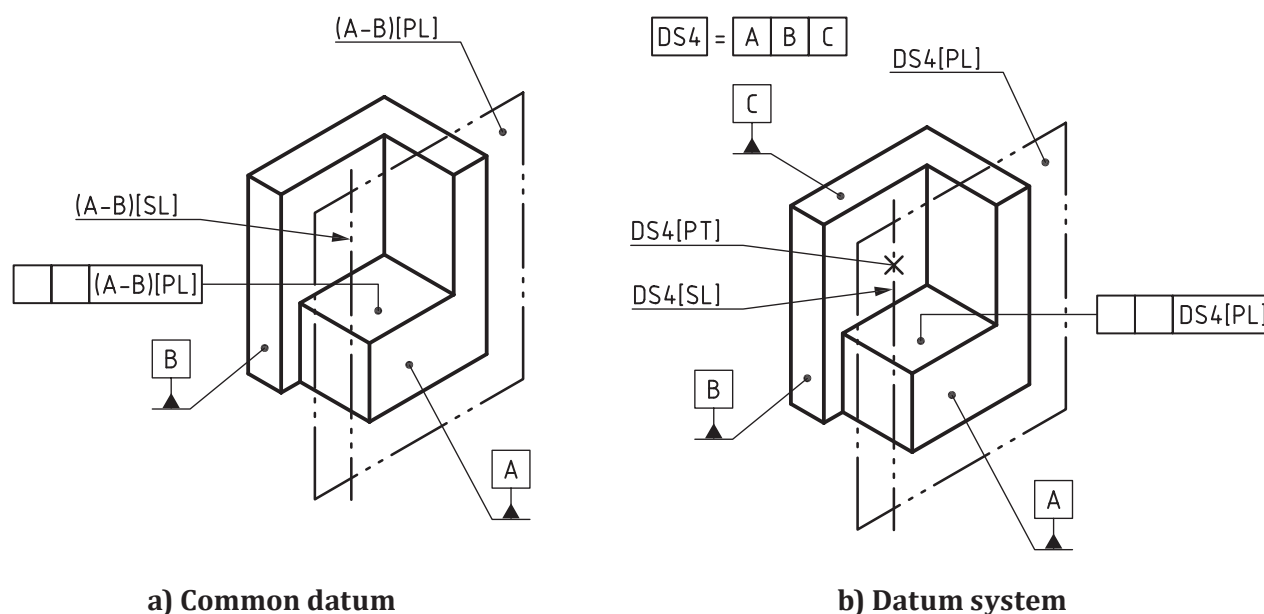


Figure 44 — Example of indication of the situation features constituting a common datum or a datum system

8 Specification operators for datum

8.1 ISO default specification operator for datum

[Annex A](#) defines, for the specification operator, the default association criteria and a basic approach for filtration applicable for a datum or datum system.

Defined association criteria shall be in accordance with [Annex A](#).

8.2 Special specification operator for datum

8.2.1 General

When a special specification operator applies to establish a datum, it shall be indicated in the datum indicator of the datum section using the following ordered sequence of specification elements:

- the datum feature identifier, see [7.2.2](#);
- if necessary, the filtration indication, see [8.2.2](#);
- if necessary, the [PD], [MD] or [LD] modifier, see [7.4.2.1](#);
- if necessary, the [SV], [SF] or [SFxx] modifier, see [7.4.2.2](#);
- if necessary, the [ACS] or [ALS] modifier, see [7.4.2.4](#);
- if necessary, the modifier [CF], see [7.4.2.5](#);
- if necessary, the association specification elements, see [8.2.3](#);
- if necessary, the modifier [DV], see [7.4.2.7](#);
- if necessary, the list of situation features to consider, see [7.4.2.8](#);
- if necessary, the orientation only modifier, see [7.4.2.8](#);

- if necessary, the modifier $\textcircled{P}$, see 7.4.2.10;
- if necessary, the modifier $\textcircled{M}$ or $\textcircled{L}$, see 7.4.2.10.

An overview of the indications is given in Table 11. There shall be no space between specification elements from different columns. Any default indication in the sequence can be omitted.

Specification elements for datums can also be used in an indirect indication of a datum section, see 7.4.2.11.

Table 11 — Examples of the specification elements used in a datum section

Datum feature identifier	Filtration	Datum feature and associated feature					Locked or unlocked degrees of freedom			Geometrical modifiers	
	Filtration specification elements	Threads and gears	Size	Section	Contacting feature	Association specification elements	Variable distance	Situation features	Orientation only modifier	Projection	Material requirement
A	[W0,25×0,25] [CB1,5;W0,25×0,25]	[PD] [MD] [LD]	[SV] [SF] [SF14,2]	[ACS] [ALS]	[CF]	[CM] [C-] [GE]	[DV]	[PL] [SL] [PT]	><	$\textcircled{P}$	$\textcircled{L}$ $\textcircled{M}$

8.2.2 Filtration specification elements for datum

The filtration specification elements consist of the filter symbol, followed by the nesting index (see the ISO 16610 series). There shall be no space between the symbol and the nesting index.

When a morphological filter (e.g. a closing ball filter) is shown with a nesting index which corresponds to the size of the structuring element, this is a long-wave pass filter. To be consistent with ISO 1101, and to facilitate the understanding of long-wave pass filters, the nesting index value shall be followed by a hyphen with no space.

The filtration specification elements shall be placed between square brackets, see Figure 45.

If a sequence of filters is used, then the complete sequence shall be placed between square brackets and separated by semi-colons.

The symbols for the standardized filters are given in Annex H.

NOTE The rules for indicating filters for datum features are the same as the rules for indicating filters for toleranced features, see ISO 1101. The use of indirect indication of the datum system can be useful, e.g. where the sequence of indications in a datum identifier with a filter is long and is repeated in several places on the technical product documentation.

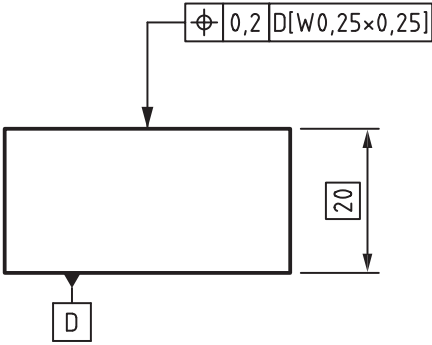


Figure 45 — Example of a special specification operator for datum with filtration specification elements

8.2.3 Association specification elements for datum

When a non-default association criterion needs to be specified, the objective function and the material constraint shall be specified. They shall be placed between square brackets, by indicating without a space first the objective function symbol (as defined in Table 12), and second the material constraint symbol (as defined in Table 13) or the material offset symbol (as defined in Table 14), e.g. [CM], [C-], [GI], see Figure 46.

The objective function and material constraint and material offset modifiers are listed in Tables 12, 13 and 14, respectively. The definitions of objective functions, material constraints and material offset shall be in accordance with ISO 4351.

Table 12 — Objective function symbols

Objective function	
Symbol	Designation
G	Least squares (Gaussian)
C	Minimax (Chebyshev)
K	Minimum volume
X	Maximum inscribed
N	Minimum circumscribed
L_p	Minimizing the p -norm distance (p is the parameter; the value of p shall be a positive integer different from 1, 2 and ∞)
NOTE The objective functions maximum inscribed and minimum circumscribed are only applicable to features of linear size.	

Table 13 — Material constraint

Material constraint or material offset	
Symbol	Designation
E	External material
I	Internal material
M	Without material constraint

Table 14 — Material offset

Material constraint or material offset	
Symbol	Designation
+	Shifted tangent outside material
-	Shifted tangent inside material
$x\%$	Material ratio percentage, with x the value of the percentage.

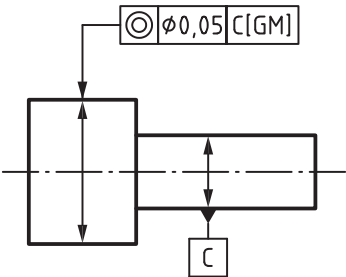


Figure 46 — Example of a non-default specification operator for datum with association specification elements

8.3 Drawing-default specification operator for datums

When a drawing-default specification operator applies for datums, it shall be indicated in or near the title block with the text “DS ISO 5459”, followed by the filtration specification elements between square brackets, followed by the association specification elements between square brackets.

The drawing-default specification operator for datums changes both the default filtration and the default association. It shall apply unless otherwise specified in the datum indicator of the datum section.

EXAMPLE DS ISO 5459[CB1,5][GM] means that the drawing-default specification operators are as follows:

- the filtration is a closing ball filter with a 1,5 mm radius structuring element;
- the association criterion is a least squares objective function without material constraint.

Annex A
(normative)

Association for datums

A.1 Basic concepts

Association methods for datums relate the real features to the datums and to an unambiguous set of constraints that lead to the identification of unique datums or datum systems.

To establish an associated feature, it is necessary to perform a partition, an extraction, a filtration and finally an association (see [Figure A.1](#)).

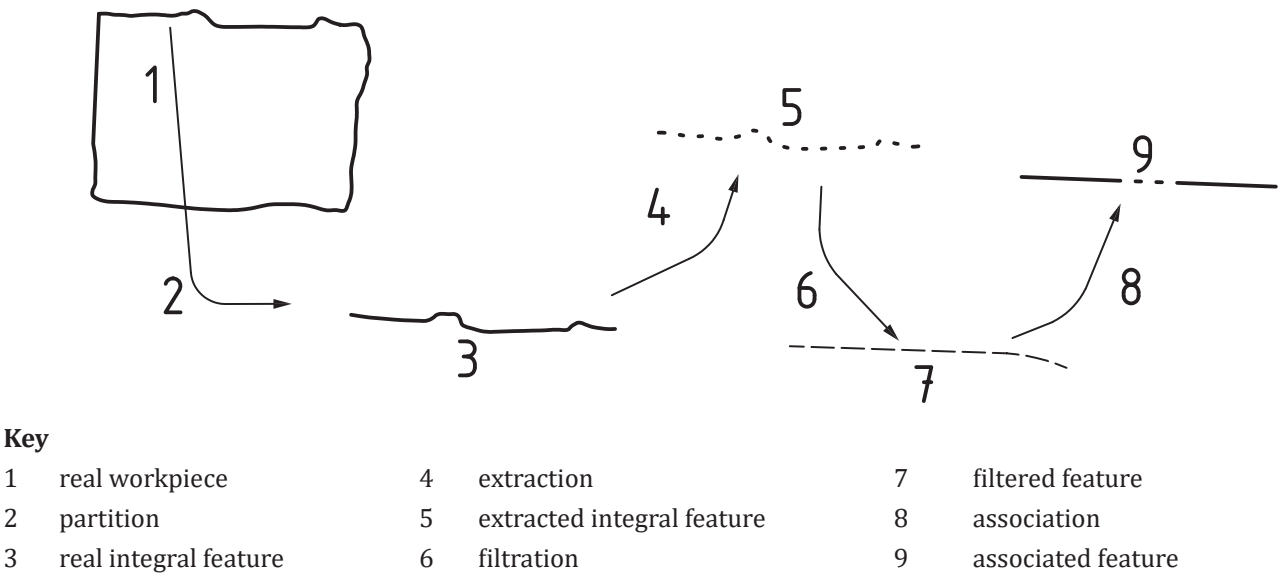
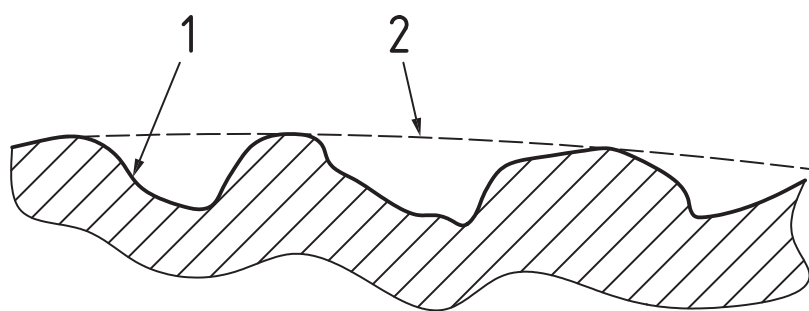


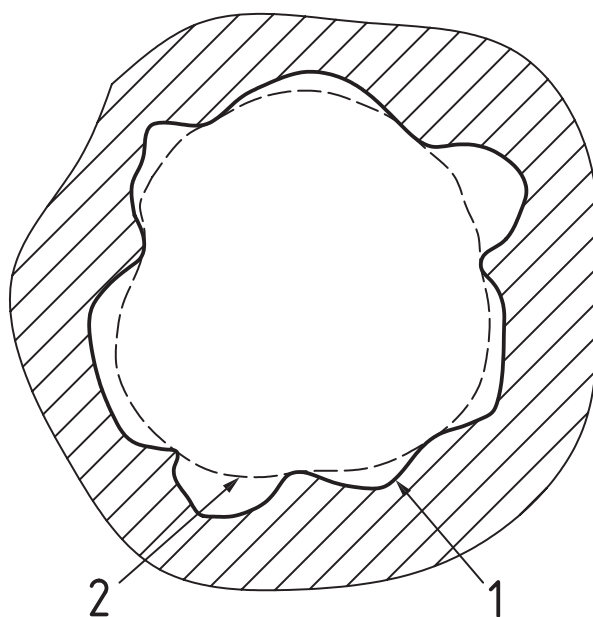
Figure A.1 — Example of procedure to establish an associated feature

Association methods for datums are defined independently of partition, extraction and filtration. The relevant methods for partition, extraction and filtration are not defined in this document.

The default filtration shall retain the highest points in the real integral feature. For a flat or convex nominal feature, such as a shaft, the default filtration shall result in a convex feature (see [Figure A.2](#)). For other types of nominal feature, such as a hole, voids in the surface shall similarly be removed (see [Figure A.3](#)). The default filtration is not mathematically completely defined in this document, except for the case of a single plane. Filtration defined in [Annex J](#) shall be applied as the default filtration for a single plane.

**Key**

- 1 real integral feature
- 2 filtered feature

Figure A.2 — Illustration of filtration applied to a nominally flat surface**Key**

- 1 real integral feature
- 2 filtered feature

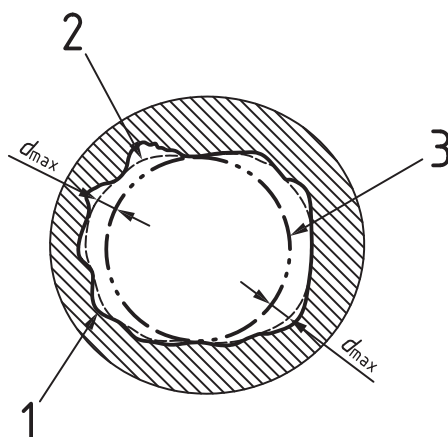
Figure A.3 — Illustration of filtration applied to a nominally cylindrical surface

The default association criterion (objective function with or without constraints), used for the association of datums, is meant to simulate the contact between a surface having a perfect shape and the non-ideal surface. This perfect contact surface is by default of the same type as the nominal surface. There are cases where the associated surface, used to establish the datum, is not of the same type as the nominal datum feature (e.g. when simulating contacting features, see rule 5).

A.2 Association methods

A.2.1 General

For any geometry type, the associated features, used to establish the datums or datum systems, simulate by default the contact with the real integral features in a way that ensures that the associated feature is external material to the non-ideal feature. When the result of this process is not unique, then the associated feature to be used is the one that minimizes the maximum distance normal to the associated feature between the associated feature and the filtered feature representing the real feature (see [Figure A.4](#)).



Key

- 1 real integral feature
- 2 filtered feature
- 3 maximum inscribed cylinder which minimizes the maximum distance ($d_{\max}$) with the filtered feature (2)

Figure A.4 — Examples of associated feature (for a cylindrical surface)

When the datum is established from a feature of size whose intrinsic characteristic (size) is linear, this intrinsic characteristic shall be considered:

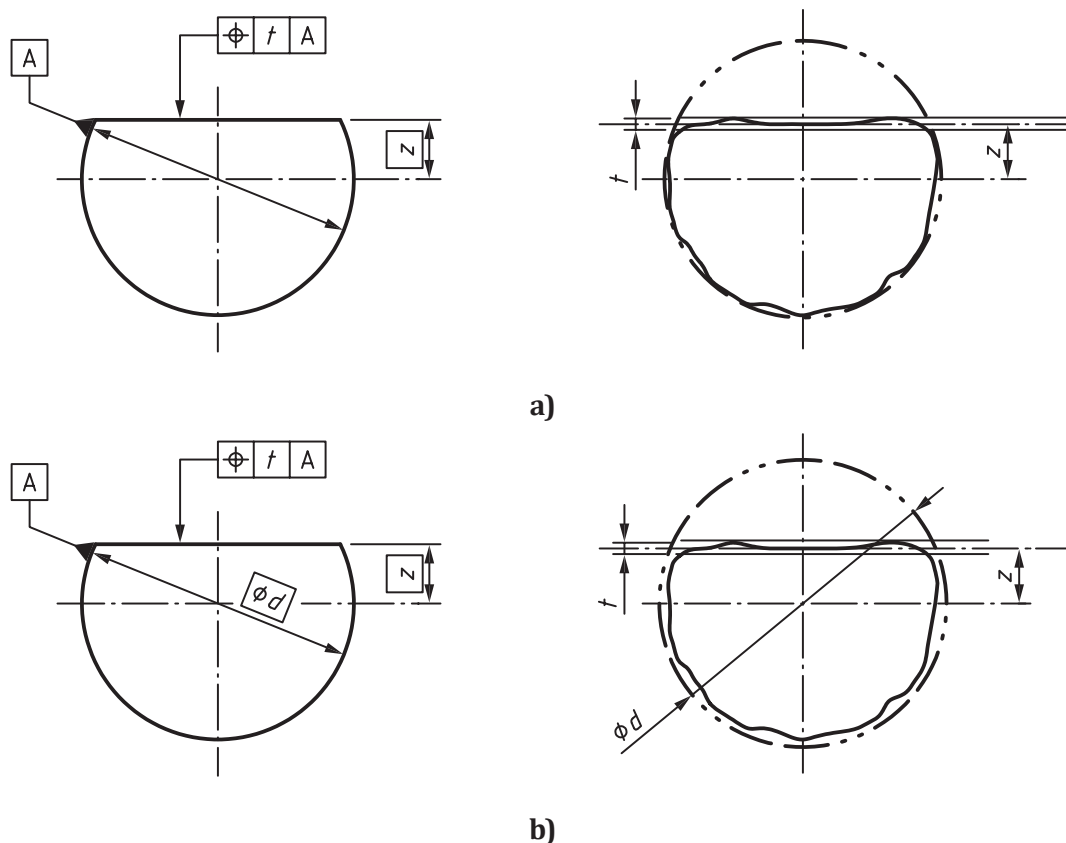
- variable (rule 2), if contact between the associated feature and the real feature is required [see [Figure A.5 a](#)], or
- fixed (rule 2), if contact between the associated feature and the real feature is not required [see [Figure A.5 b](#)].

When the datum is established from a feature of size whose intrinsic characteristic (size) is angular, contact between the associated feature and the real feature is required.

A.2.2 Association for single datums

A.2.2.1 Association for single datums without the [CF] modifier

Only one ideal feature or feature of size (of the same type as the nominal feature) shall be fitted to the indicated non-ideal surface when the modifier [CF] is not specified after the datum identifier in the tolerance frame [see [Figures A.5 a](#)) and [A.5 b](#))].



NOTE When associating a cylinder of fixed size, the maximum distance between the associated cylinder and the real (extracted) cylinder is minimized, without orientation, location and material constraints.

Figure A.5 — Examples of single datums established on a nominally cylindrical surface, without [CF] (contacting feature) modifier, used in a geometrical specification

A.2.2.2 Association for single datums with the modifier [CF]

When the modifier [CF] is specified after the datum identifier in the tolerance frame, it is necessary:

- to identify the contacting features used to define the interface (1) between these contacting features and the datum feature;
- to identify the ideal feature type (2) of the contacting features;
- to associate an ideal feature to the interface [defined above as (1)]. The type of the ideal feature is a plane, a straight line, a point or a collection of these [depending on the type defined above as (2)] (see [Figure A.6](#)).

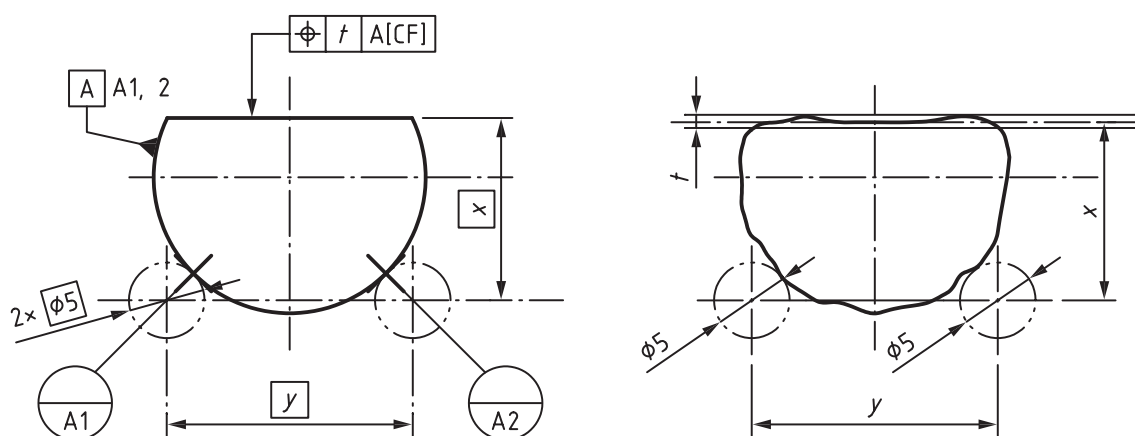


Figure A.6 — Examples of single datums established on a nominally cylindrical surface, with [CF] (contacting feature) modifier, used in a geometrical specification

A.2.2.3 Default association criteria

When a single datum is established from a feature of size and its size is considered variable for the association (e.g. a cylinder, a sphere, two parallel planes or a torus), the default association criteria is given in [Table A.1](#).

When a single datum is established from a feature of size and its size is considered fixed for the association (e.g. a cone or a wedge), the default association criterion is to minimize the distance between the datum feature and the associated feature (in this particular case, the material constraint is not applied between the associated feature and the datum feature).

The distance to be minimized is in all cases normal to the associated feature.

When a single datum is established from a plane or a complex surface (which are not features of size) or from a cone or a wedge (which are features of size, with angular size), the default association criterion is given in [Table A.2](#).

When a single datum established from a conical surface is considered variable for the association, the default association criterion is to minimize the distance between the datum feature and the associated feature with the outside material constraint and fixed intrinsic characteristic constraint.

NOTE If a non-default association criterion is needed, modifiers are available to define an explicit association criterion (see [8.2.3](#)).

Table A.1 — Default association criteria for a feature of size with variable intrinsic characteristics

Drawing indication referencing a datum established from this nominal type of geometrical feature	Internal/ external	Default association criteria for single datums established from a feature of size with variable intrinsic characteristic constraint		Situation feature
		Objective function	Material constraint	
Sphere	Internal	Maximum inscribed: Maximize the diameter of the associated inscribed sphere in the datum feature ^a .	Outside the material	The centre of the associated sphere (point)
	External	Minimum circumscribed: Minimize the diameter of the associated circumscribed sphere in the datum feature ^a .		
Cylinder	Internal	Maximum inscribed: Maximize the diameter of the associated inscribed cylinder in the datum feature ^a .	Outside the material	The axis of the associated cylinder (straight line)
	External	Minimum circumscribed: Minimize the diameter of the associated circumscribed cylinder in the datum feature ^a .		
Two parallel planes	Internal	Maximum inscribed: Maximize the distance between the two planes associated simultaneously to the two datum features. These two planes are constrained to be parallel to each other ^a .	Outside the material	Median plane of the two associated planes (plane)
	External	Minimum circumscribed: Minimize the distance between the two planes associated simultaneously to the two datum features. These two planes are constrained to be parallel to each other ^a .		
Torus	Internal	Maximum inscribed: Maximize the diameter of the cross-section of the inscribed torus (torus with variable diameter of directrix and fixed diameter of generator in internal contact) associated with the datum feature.	Outside the material	Plane and centre of the associated torus (plane and point)
	External	Minimum circumscribed: Minimize the diameter of the cross-section of the circumscribed torus (torus with variable diameter of directrix and fixed diameter of generator in external contact) associated with the datum feature.		
^a In cases where the linear size (of the feature of size) is considered variable, the result of the association can lead to several solutions with the same datum feature ("unstable association"). In this case, the following alternative association criterion shall be used: minimize the maximum distance normal to the associated feature between the associated feature and the datum feature or between the two associated features and the two datum features (in the case of two parallel planes).				

Table A.2 — Default association criteria for a feature which is not a feature of size or for a feature of size with fixed intrinsic characteristics

Drawing indication referencing a datum established from this nominal type of geometrical feature	Internal/ external	Default association criteria for a single datum		Situation feature
		Objective function	Material constraint	
Cone	Internal	Minimax: Minimize the maximum distance between the associated cone and the datum feature with fixed intrinsic characteristic constraint (fixed angle).	Outside the material	Situation features of the associated cone (straight line and point)
	External			
Wedge	Internal	Minimax: Minimize the maximum distance between the associated wedge and the datum feature with fixed intrinsic characteristic constraint (fixed angle).	Outside the material	Situation features of the associated wedge (plane and straight line)
	External			
Complex surface	Non-applicable	Minimax: Minimize the maximum distance between the associated complex surface with fixed parameters and the datum feature.	Outside the material	Situation features of the associated complex surface (plane, straight line and point)
Plane	Non-applicable	For a single datum, the objective function is the total least squares (Gaussian) objective: minimize the sum of the square distances from the points on the datum feature to the associated feature.	External material ^a	Situation features of the associated plane (plane)
		For a common datum, the objective function is the minimax: minimize the maximum distance from the points on a datum feature of the collection to the associated plane of the collection.		
^a Implies [GE] (see Tables 12 and 13).				
NOTE The combination of the Gaussian objective function with the external material constraint forms the [GE] association criterion (not the [G+] association criterion).				

A.2.3 Association for common datums

A.2.3.1 General

The association method for common datums requires that a collection of ideal single surfaces be fitted simultaneously (in one step) to several non-ideal surfaces.

The process of association for common datums includes location and orientation constraints between the different associated features. These constraints are the new intrinsic characteristics defined by the collection of the features. These constraints are either defined explicitly by TEDs or implicitly (implicit orientation constraint: 0°, 90°, 180°, 270° and implicit location constraint: 0 mm). The internal constraints for association described for single datums are also applicable for common datums, but complementary constraints (e.g. coplanarity, coaxiality) between the associated features shall be added.

A.2.3.2 Default association criteria

The default association criterion is defined by constraints and an objective function.

The following constraints for establishing a common datum apply to each associated feature included in the collection defined by the common datum indication:

- be outside the material of its corresponding filtered feature;

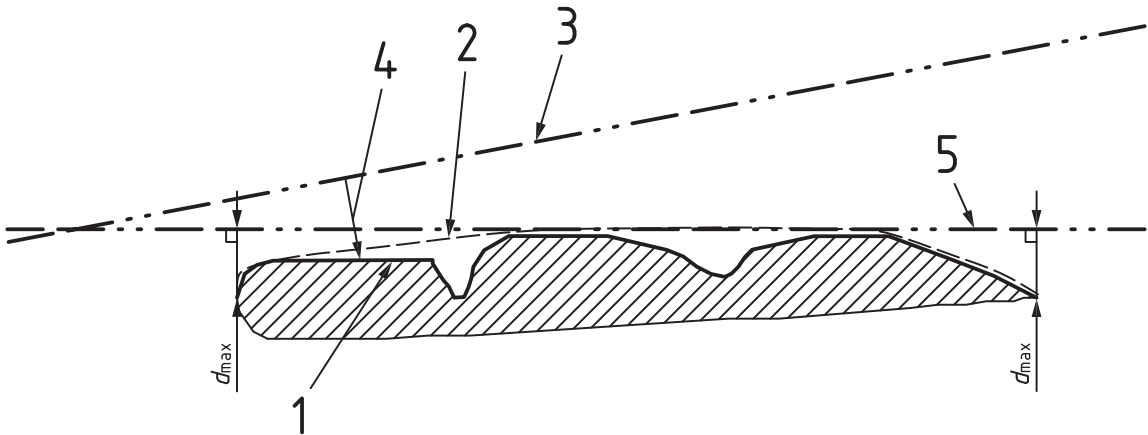
- respect the orientation and location constraints defining the relationship between the nominal features in the collection (indicated by an explicit or implicit TED), while taking into account any modifiers (e.g. [DV]).

The objective function is to simultaneously minimize the maximum distance normal to the associated feature between each associated feature and its filtered feature, as illustrated by the following formula, while respecting the default constraints (see [Figure A.7](#)).

$$\text{minimize} \left[\max_{i=1, \dots, N} d(A_i, F_i) \right]$$

where

- $d(A_i, F_i)$ is the distance between the features A_i and F_i ;
- i is the index of a single feature member of the collection surface for the common datum;
- N is the number of single features constituting the collection surface for the common datum;
- A_i is the associated feature of the filtered feature;
- F_i is the filtered feature of the real integral feature.

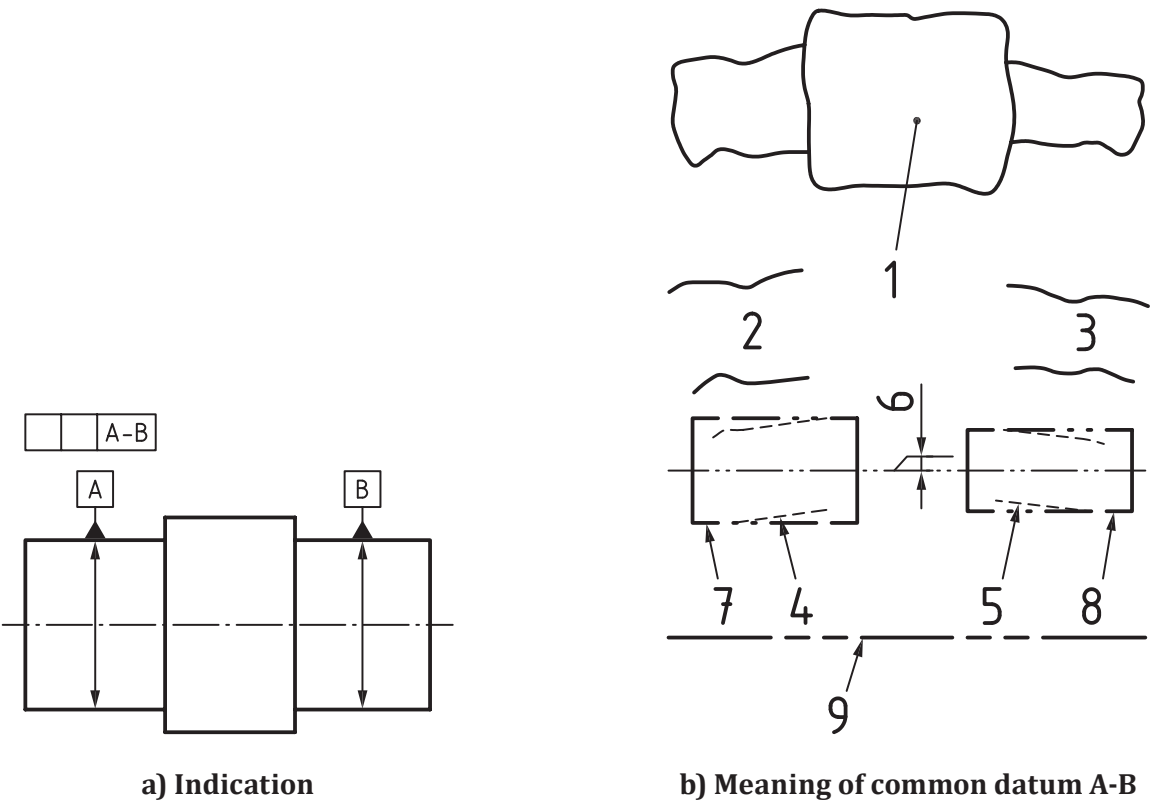


Key

- 1 real integral feature
- 2 filtered feature
- 3 ideal feature
- 4 local distance defined from the ideal feature to the filtered feature
- 5 associated feature with the objective function "minimax" and the association constraint tangent outside the material ($d_{\max}$ is minimized)

Figure A.7 — Illustration of association process with minimax objective function and outside material constraint

[Figure A.8](#) illustrates the process used to establish a common datum from two surfaces nominally cylindrical and coaxial.



Key

- 1 real workpiece
- 2 datum feature arbitrarily numbered N° 1 of the collection surface for the common datum
- 3 datum feature numbered N° 2 of the collection surface for the common datum
- 4 filtered feature of datum feature N° 1
- 5 filtered feature of datum feature N° 2
- 6 orientation constraint and location constraint between the associated features of the collection feature (coaxiality)
- 7 associated feature of datum feature N° 1 respecting constraint 6), variable intrinsic characteristic and outside material constraint with objective function “minimax” to minimize, simultaneously and globally, the maximum distances between the associated feature of N°1 datum feature and its filtered feature and between the associated feature of N°2 datum feature and its filtered feature
- 8 associated feature of datum feature N° 2 respecting constraint 6), variable intrinsic characteristic and outside material constraint with objective function “minimax” to minimize, simultaneously and globally, the maximum distances between the associated feature of N°1 datum feature and its filtered feature and between the associated feature of N°2 datum feature and its filtered feature
- 9 common datum (in this case, the axis of the collection surface – the two coaxial associated cylinders)

Figure A.8 — Common datum established from two coaxial cylinders

A.2.4 Association for datum systems

A.2.4.1 General

The association method for datum systems requires that a collection of ideal single surfaces be fitted in a specified order (in several steps) to several non-ideal surfaces.

A datum system is constituted by an ordered list of two or three datums. These datums (primary, secondary, tertiary) can each be a single or a common datum. The association of the surfaces to each datum feature is performed one after the other in the order defined by the system. The association of the secondary and tertiary datums respects the constraints created by the associations already carried out.

Moreover, the following supplementary constraints are required.

- The primary datum imposes orientation constraints on the secondary datum, defined by the theoretically exact relative orientation between the primary and the secondary datums.
- If a tertiary datum exists,
 - the primary datum imposes orientation constraints on this tertiary datum, defined by the theoretically exact relative orientation between the primary and the tertiary datums, and
 - the secondary datum imposes orientation constraints on the tertiary datum, defined by the theoretically exact relative orientation between the secondary and tertiary datums.

A.2.4.2 Default association criteria

The default association criterion for the primary datum is the default association criterion for a single datum or a common datum, respectively, if the primary datum is a single datum or a common datum without additional constraint.

The default association criterion for the secondary datum is the default association criterion for a single datum or a common datum, respectively, if the secondary datum is a single datum or a common datum with the additional orientation constraints from the primary datum (defined explicitly by TED angles and/or implied angles of 0°, 90°, 180° or 270°).

The default association criterion for the tertiary datum is the default association criterion for a single datum or a common datum, respectively, if the tertiary datum is a single datum or a common datum with the additional orientation constraints from the primary datum and the secondary datum (defined explicitly by TED angles and/or implied angles of 0°, 90°, 180° or 270°).

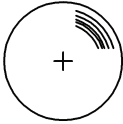
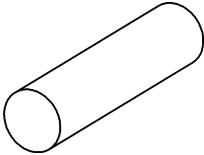
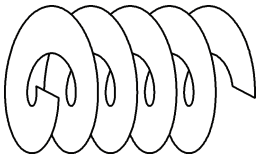
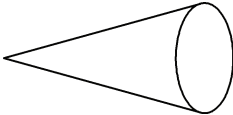
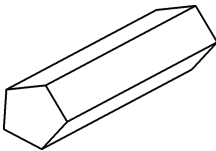
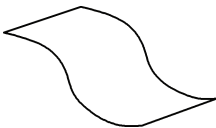
Annex B
(informative)

Invariance classes

All surfaces can be classified into seven classes based on the degrees of freedom for which the surface is invariant (a collection of two or more surfaces also belongs to one of these classes).

For each class of surface, situation features (point, straight line, plane or helix) are defined.

Table B.1 — Table of invariance classes

Invariance class	Unconstrained degrees of freedom	Illustration	Situation features	Example of types of surfaces
Spherical	3 rotations around a point		Point	Sphere
Planar	1 rotation perpendicular to the plane and 2 translations along 2 lines of the plane		Plane	Plane
Cylindrical	1 translation and 1 rotation around a straight line		Straight line	Cylinder
Helical	Combination of 1 translation and 1 rotation around a single straight line		Straight line ^a	Helical surface with a basis of involute to a circle
Revolute	1 rotation around a straight line		Straight line Point	Cone Torus
Prismatic	1 translation along a line of a plane		Plane Straight line	Pentagonal prism
Complex	None		Plane Straight line Point	Bezier surface based on an unstructured cloud of points in space

^a Helical surfaces as such are not considered in this document. They are regarded as cylindrical surfaces because, in most functional cases where helical surfaces (threads, helical slopes, endless screws, etc.) are involved, the combined rotation and translation of the helix is not needed for datum purposes. In these cases, the pitch cylindrical surface is used for the datum; the major or minor cylindrical surface can also be considered and specified. Natively, the situation feature of a feature belonging to a helical invariance class is a helix, but in this document we consider only its axis.

When a feature is used to establish a datum, it shall constrain, or lock, some degrees of freedom for the tolerance zone. The maximum number of degrees of freedom that can be constrained is equal to, or less than, six minus the invariance degree of the feature (see [Table B.1](#)).

When a datum or datum system is indicated in a tolerance frame, the number of unlocked degrees of freedom of the tolerance zone that are left free is equal to, or greater than, the invariance degree of that datum or datum system.

NOTE The term “invariance degree”, used in geometry, is the correct term for “degree of freedom” used in kinematics. The way in which these terms are used in this document is such that the number of invariance degrees is equal to the number of unconstrained degrees of freedom for a given geometrical feature.

EXAMPLE 1 For a nominal cylindrical surface, used to establish a single datum, the nominal surface is invariant in two directions (1 translation and 1 rotation), so it belongs to the “cylindrical” invariance class, and has an invariance degree of 2 (see [Table B.1](#)). The situation feature used to establish the datum from this feature is a straight line (axis of cylinder). Indicating the corresponding datum alone in a tolerance frame can lock up to four degrees of freedom of the tolerance zone, but will leave at least two degrees of freedom (a translation and a rotation) unlocked.

EXAMPLE 2 For a nominal conical surface, used to establish a single datum, the nominal surface is invariant only in one direction (1 rotation), so it belongs to the “revolute” invariance class and has an invariance degree of 1 (see [Table B.1](#)). The situation features used to establish the datum from this feature are a straight line (the axis of the cone) and a point (one point on the axis). Indicating the corresponding datum alone in a tolerance frame can lock up to five degrees of freedom of the tolerance zone, but will leave at least one degree of freedom (a rotation) unlocked.

EXAMPLE 3 For two nominally cylindrical, non-coaxial surfaces with parallel axes, used to establish a common datum, the nominal collection surface is invariant only in one direction (1 translation), so it belongs to the “prismatic” invariance class and has an invariance degree of 1 (see [Table B.1](#)). The situation features used together to establish the datum from these features are a straight line (the median line of the two axes of the associated cylinders) and a plane (the plane containing the two axis of the associated cylinders). Indicating the corresponding datum alone in a tolerance frame can lock up to five degrees of freedom of the tolerance zone, but will leave at least one degree of freedom (a translation) unlocked.

Annex C
(informative)

Examples

C.1 General

In the examples given in this annex, the default association criterion used is defined in [Annex A](#).

C.2 Examples of single datums

C.2.1 Plane

[Figure C.1](#) illustrates the drawing indication of the design intent.

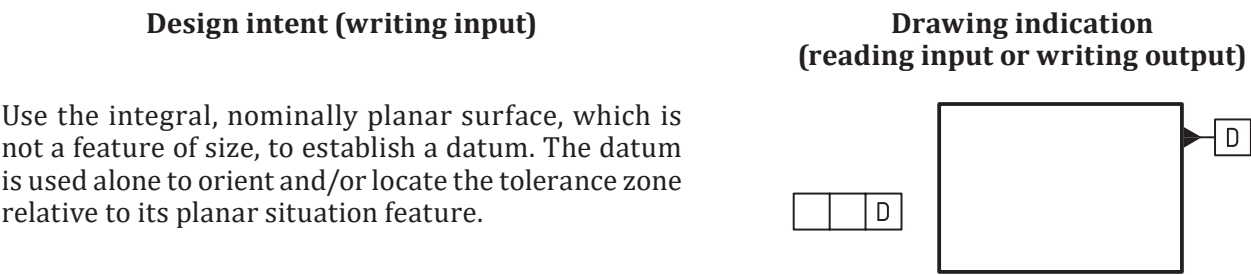
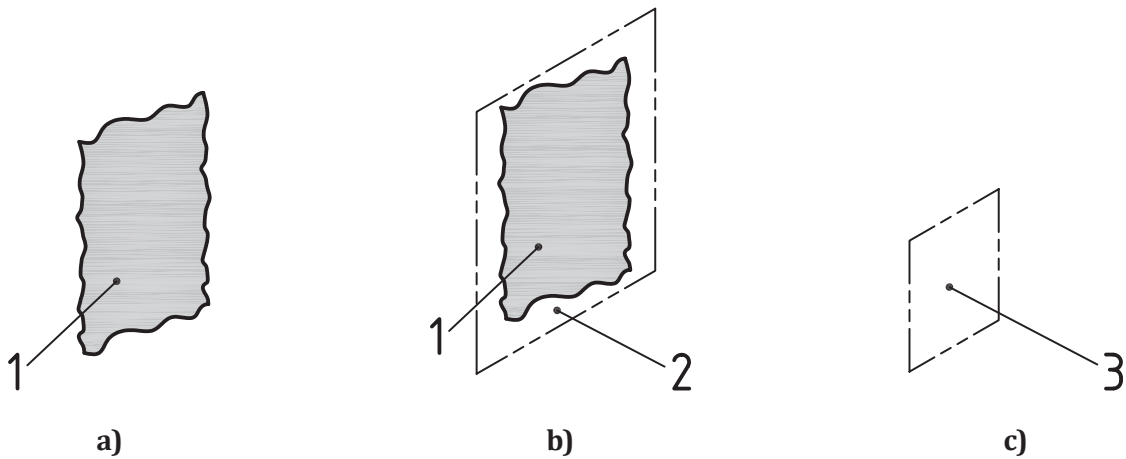


Figure C.1 — Plane — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surface is obtained after partition/extraction [see [Figure C.2 a](#)].
- According to [6.3.2](#), the single datum is characterized by the situation feature of the plane associated with the real integral feature without external constraints [see [Figure C.2 b](#)]. The invariance class of the nominal surface is planar and the situation feature is a plane [see [Figure C.2 c](#)].



Key

- 1 datum feature: real integral feature
- 2 associated feature constraint: tangent outside of material
- 3 single datum – situation feature of the associated feature: plane

NOTE Illustrations are given in 2D view.

Figure C.2 — Establishing a single datum from a planar surface

C.2.2 Cylinder

[Figure C.3](#) illustrates the drawing indication of the design intent.

Design intent (writing input)

Use the integral, nominally cylindrical surface, which is a feature of size, to establish a datum by considering its size variable. The datum is used alone to orient and/or locate the tolerance zone relative to its situation feature, which is the axis of the associated cylinder.

**Drawing indication
(reading input or writing input)**

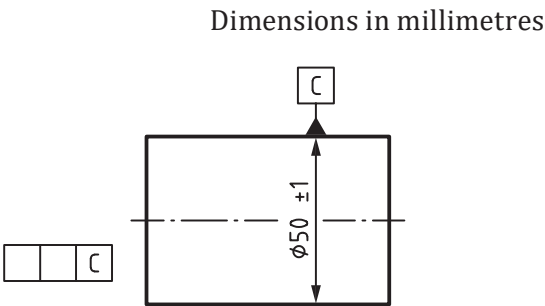
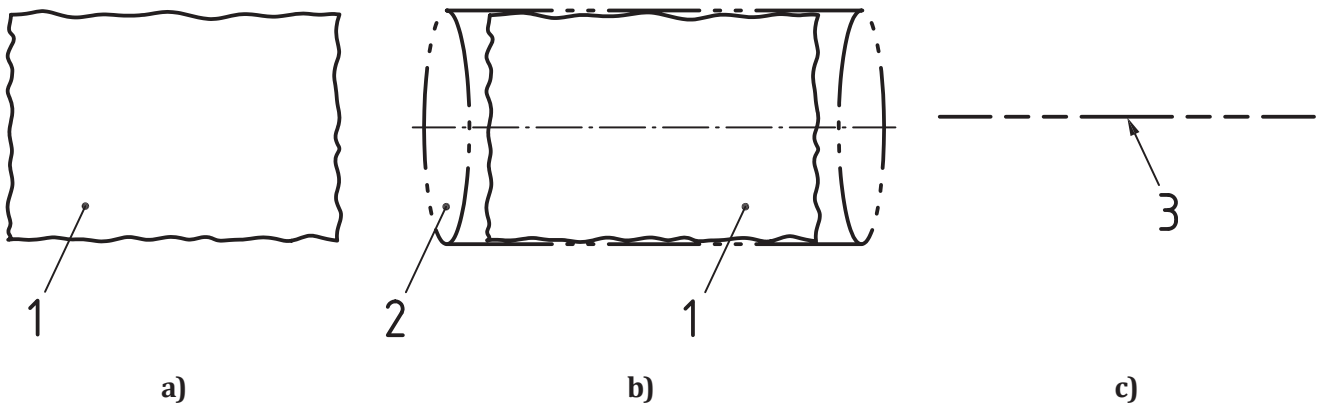


Figure C.3 — Cylinder — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surface is obtained after partition/extraction [see [Figure C.4 a\)](#)].
- According to [6.3.2](#), the single datum is characterized by the situation feature of the cylinder associated with the real integral feature without external constraints [see [Figure C.4 b\)](#)]. The invariance class of the nominal surface is cylindrical and the situation feature is the axis of the cylinder [see [Figure C.4 c\)](#)].



Key

- 1 datum feature: real integral feature
- 2 associated feature with variable diameter
- 3 single datum – situation feature of the associated feature: straight line (axis of the cylinder)

Figure C.4 — Establishing a single datum from a cylindrical surface

C.2.3 Cone

[Figure C.5](#) illustrates the drawing indication of the design intent.

Design intent (writing input)

Use the integral, nominally conical surface, which is a feature of size, to establish a datum by considering its size fixed. The datum is used alone to orient and/or locate the tolerance zone relative to its situation features, which are the axis of the associated cone and a point along this axis.

Drawing indication (reading input or writing input)

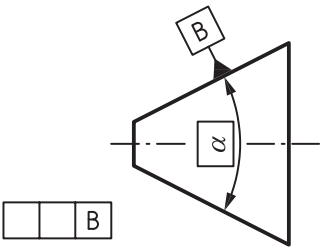
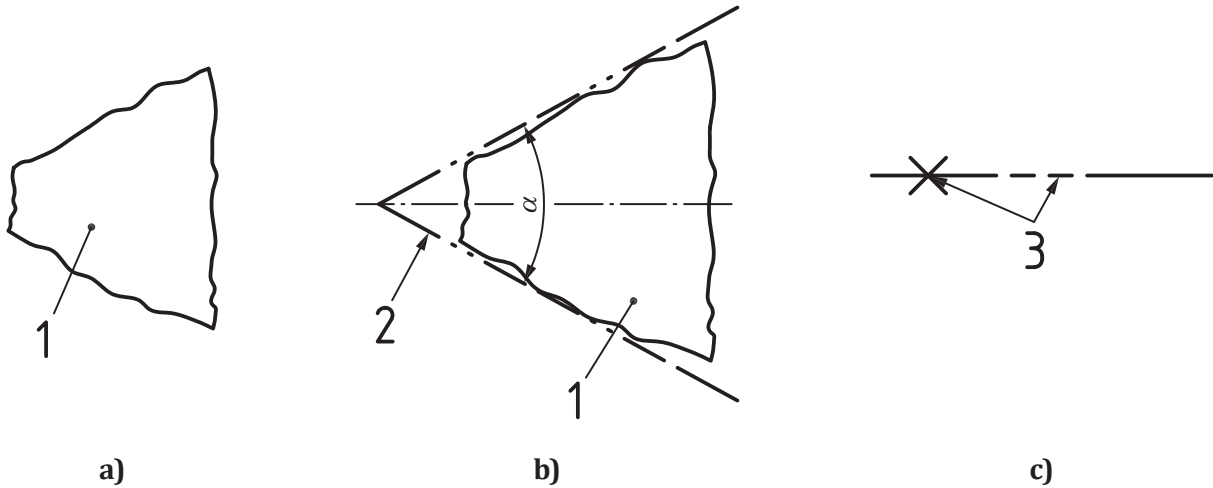


Figure C.5 — Cone — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surface is obtained after partition/extraction [see [Figure C.6 a](#)].
- According to [6.3.2](#), the single datum is characterized by the situation features of the cone associated with the real integral feature without external constraints [see [Figure C.6 b](#)]. The invariance class of the nominal surface is revolute and the situation features are the axis of the cone and a point along this axis [see [Figure C.6 c](#)].



Key

- 1 datum feature: real integral feature
- 2 associated feature with fixed intrinsic characteristic constraint (angle α)
- 3 single datum – situation features of the associated feature: straight line (axis of the cone) and point (point on the axis)

Figure C.6 — Establishing a single datum from a conical surface

C.2.4 Sphere

Figure C.7 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use the integral, nominally spherical surface, which is a feature of size, to establish a datum by considering its size variable. The datum is used alone to orient and/or locate the tolerance zone relative to its situation feature, which is the centre of the associated sphere.

Drawing indication (reading input or writing input)

Dimensions in millimetres

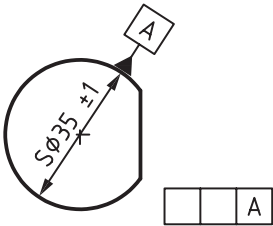
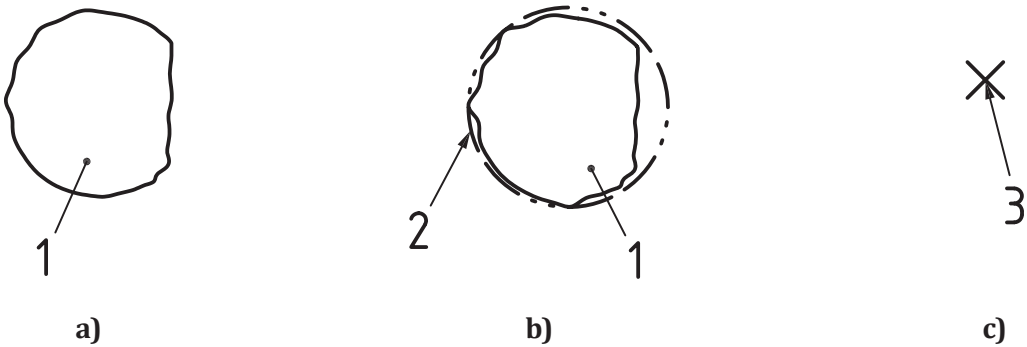


Figure C.7 — Sphere — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surface is obtained after partition/extraction [see Figure C.8 a)].
- According to 6.3.2, the single datum is characterized by the situation features of the sphere associated with the real integral feature without external constraints [see Figure C.8 b)]. The invariance class of the nominal surface is spherical and the situation feature is the centre of the sphere [see Figure C.8 c)].



Key

- 1 datum feature: real integral feature
- 2 associated feature with variable diameter
- 3 single datum — situation feature of the associated feature: point (centre of the sphere)

NOTE The fourth contacting point is not visible in 2D view

Figure C.8 — Establishing a single datum from a spherical surface

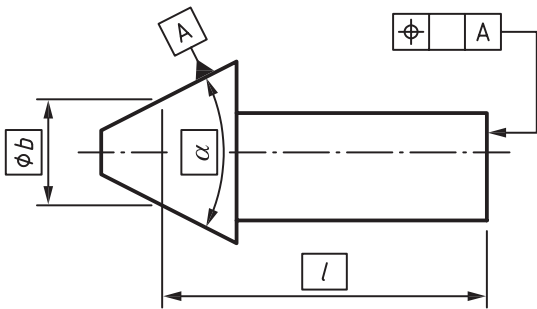
C.2.5 Particular situation feature

Figure C.9 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use the integral, nominally conical surface, which is a feature of size, to establish a datum by considering its size fixed. The datum is used alone to orient and locate the tolerance zone relative to its situation features, which are the axis of the associated cone and a particular point along the axis, defined by the location where the section diameter is specified.

Drawing indication (reading input or writing input)

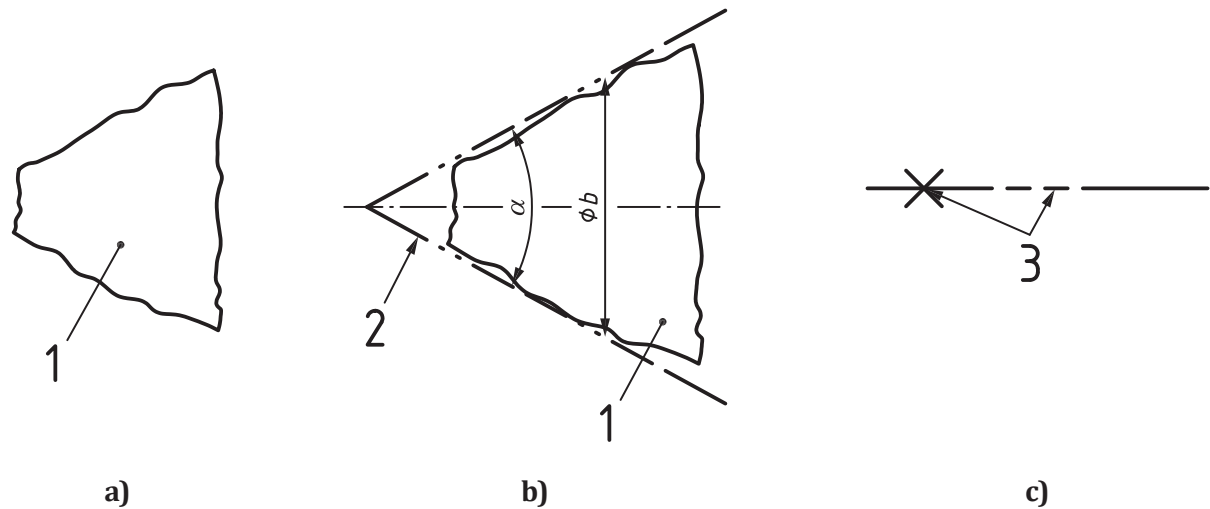


NOTE This particular point is defined by the intersection between a plane and the axis of the cone. This plane is perpendicular to the axis of the cone and its location defined by the TED ϕb that defines the diameter of a circle. This circle is the intersection between the associated cone and the plane.

Figure C.9 — Particular situation feature — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surface is obtained after partition/extraction [see Figure C.10 a)].
- According to 6.3.2, the single datum is characterized by the situation features of the cone associated with the real integral feature without external constraints [see Figure C.10 b)]. The invariance class of the nominal surface is revolute and the situation features are the axis of the cone and a particular point along this axis [see Figure C.10 c)].



Key

- 1 datum feature: real integral feature
- 2 associated feature with fixed intrinsic characteristic constraint (angle α)
- 3 single datum – situation feature of the associated feature: straight line (axis of the cone) and point (point on the axis of the cone where the section diameter is ϕb)

Figure C.10 — Establishing a single datum from a conical surface, where the specific relationship between the tolerated feature and the datum is given

C.2.6 Obvious situation feature

Figure C.11 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use the integral, nominally conical surface, which is a feature of size, to establish a datum by considering its size fixed. The datum is used alone to orient and locate the tolerance zone relative to an axis. This axis is the situation feature of the associated cone.

Drawing indication (reading input or writing input)

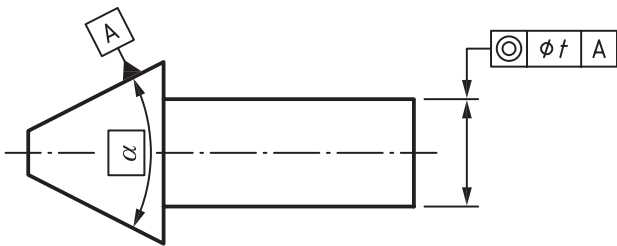
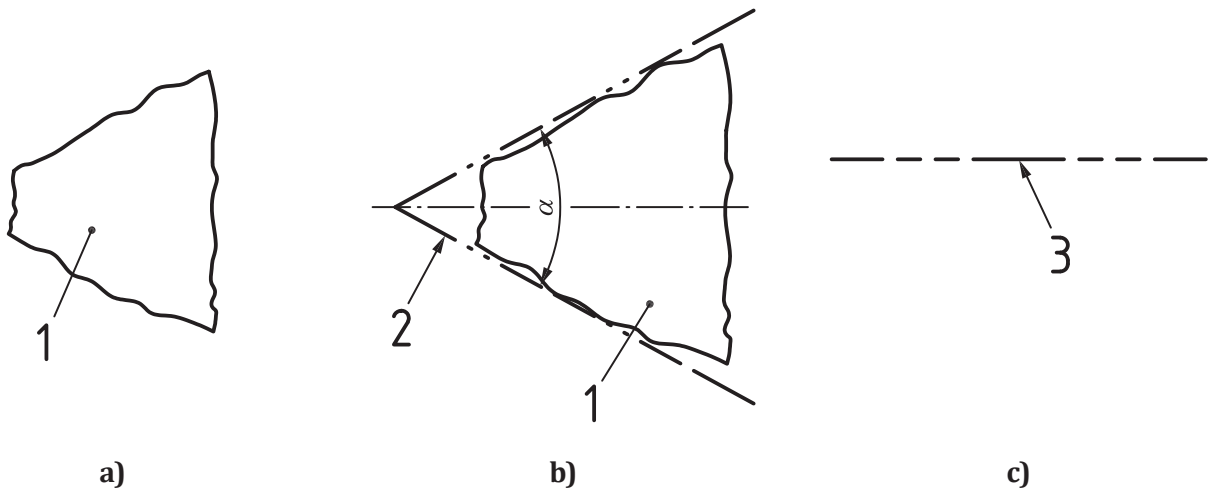


Figure C.11 — Obvious situation feature — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surface is obtained after partition/extraction [see Figure C.12 a)].
- According to 6.3.2, the single datum is characterized by the situation feature of the cone associated with the real integral feature without external constraints [see Figure C.12 b)]. The invariance class of the nominal surface is revolute and the situation features are the axis of the cone and a particular point along this axis. In this case, the point is not involved in the location of the tolerance zone [see Figure C.12 c)].



Key

- 1 datum feature: real integral feature
- 2 associated feature with fixed intrinsic characteristic constraint (angle α)
- 3 single datum – only the straight line of the situation feature of the associated feature: straight line

Figure C.12 — Establishing a single datum where the situation feature is obvious

C.2.7 Only one situation feature needed

[Figure C.13](#) illustrates the drawing indication of the design intent.

Design intent (writing input)

Use the integral, nominally conical surface, which is a feature of size, to establish a datum by considering its size fixed. The datum is used alone to orient and locate the tolerance zone relative to one of the situation features of the associated cone (the situation feature is a straight line: the axis of the associated cone).

**Drawing indication
(reading input or writing input)**

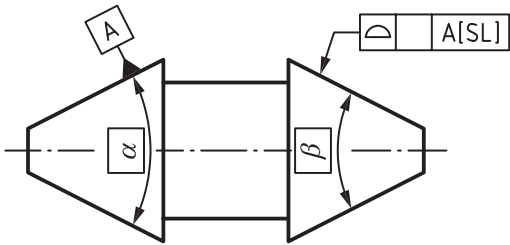
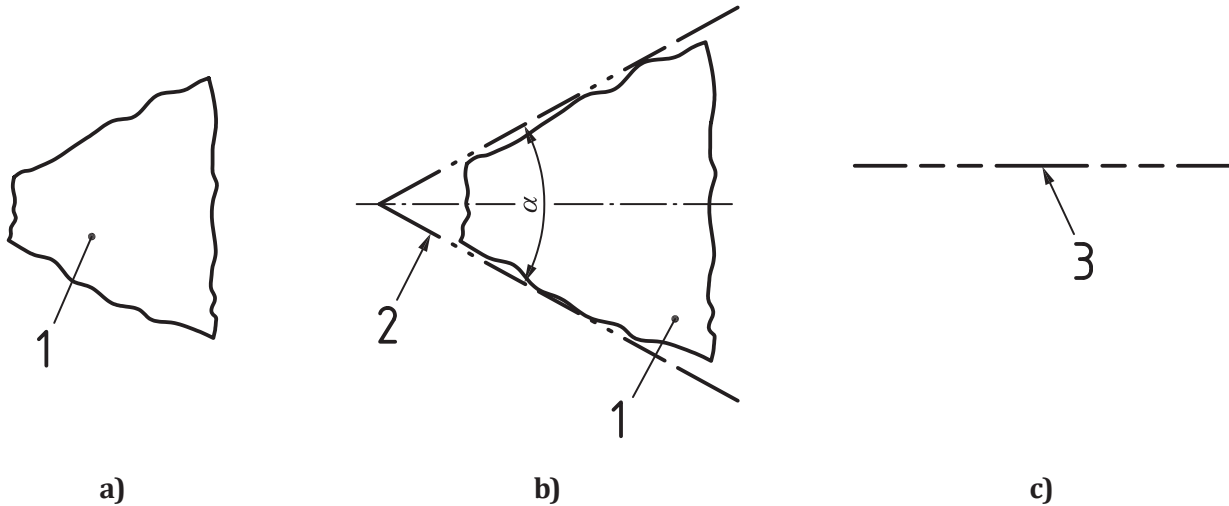


Figure C.13 — Only one situation feature needed — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surface is obtained after partition/extraction [see [Figure C.14 a](#)].
- According to [6.3.2](#), the single datum is characterized by the situation features of the cone associated with the real integral feature without external constraints [see [Figure C.14 b](#)]. The invariance class of the nominal surface is revolute and the situation features are the axis and the cone and a particular point along this axis. Because a complementary indication [SL] is given, only the axis of the cone is considered [see [Figure C.14 c](#)].



Key

- 1 datum feature: real integral feature
- 2 associated feature with fixed intrinsic characteristic constraint (angle α)
- 3 single datum – only the straight line of the situation features of the associated feature (straight line and point)

Figure C.14 — Establishing a single datum where only one situation feature is needed

C.2.8 Complex surface

Figure C.15 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use the integral, nominally complex surface, which is not a feature of size, to establish a datum. The datum is used alone to orient and/or locate the tolerance zone relative to the situation features of the associated surface which are a plane, a straight line and a point.

Drawing indication (reading input or writing input)

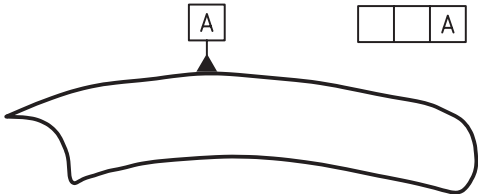
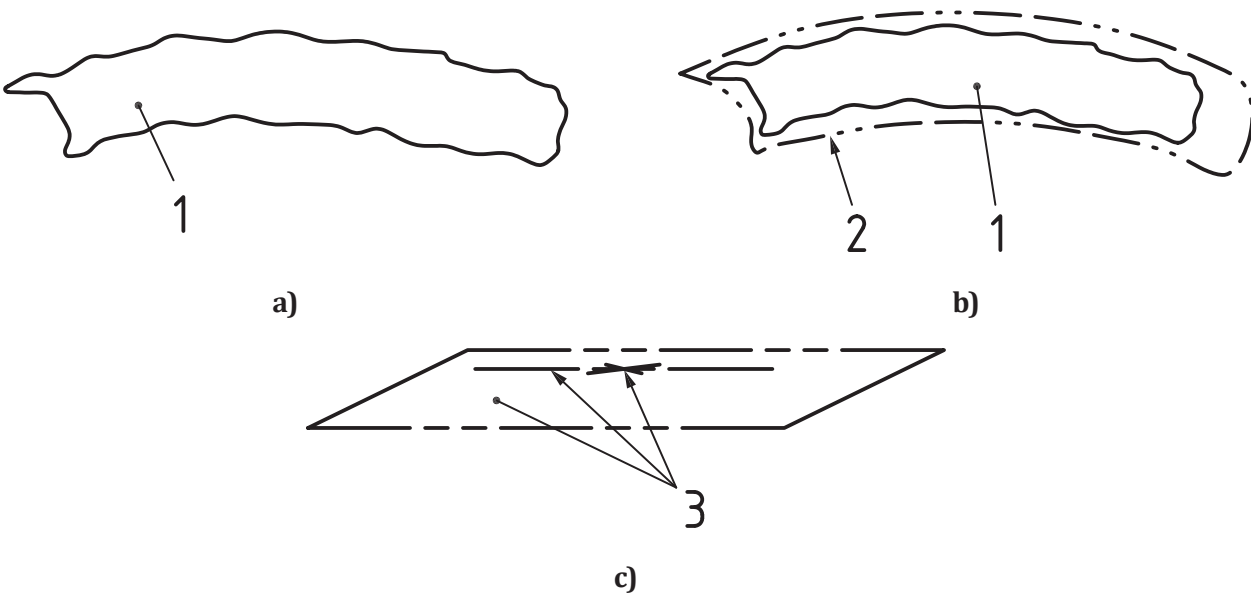


Figure C.15 — Complex surface — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surface is obtained after partition/extraction [see Figure C.16 a)].
- According to 6.3.2, the single datum is characterized by the situation features of the complex surface associated with the real integral feature without external constraints [see Figure C.16 b)]. The invariance class of the nominal surface is complex and the situation features are a plane, a straight line and a point [see Figure C.16 c)].



Key

- 1 datum feature: real integral feature
- 2 associated feature
- 3 single datum – situation feature of the associated feature: plane, straight line and point

Figure C.16 — Establishing a single datum from a complex surface

C.2.9 Intersecting planes

Figure C.17 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use the real integral surface resulting from the collection of two intersecting nominally planar surfaces, which constitutes a feature of size (a wedge), to establish a datum by considering its size fixed. The datum is used alone to orient and/or locate the tolerance zone relative to the situation feature of the associated feature of size. The situation features are a plane (the bisector plane) and a straight line (intersection of the two associated planes, with a fixed angle between them).

Drawing indication (reading input or writing input)

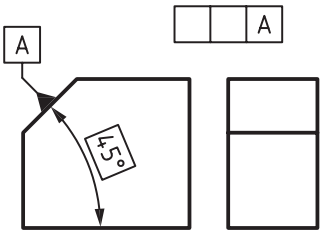
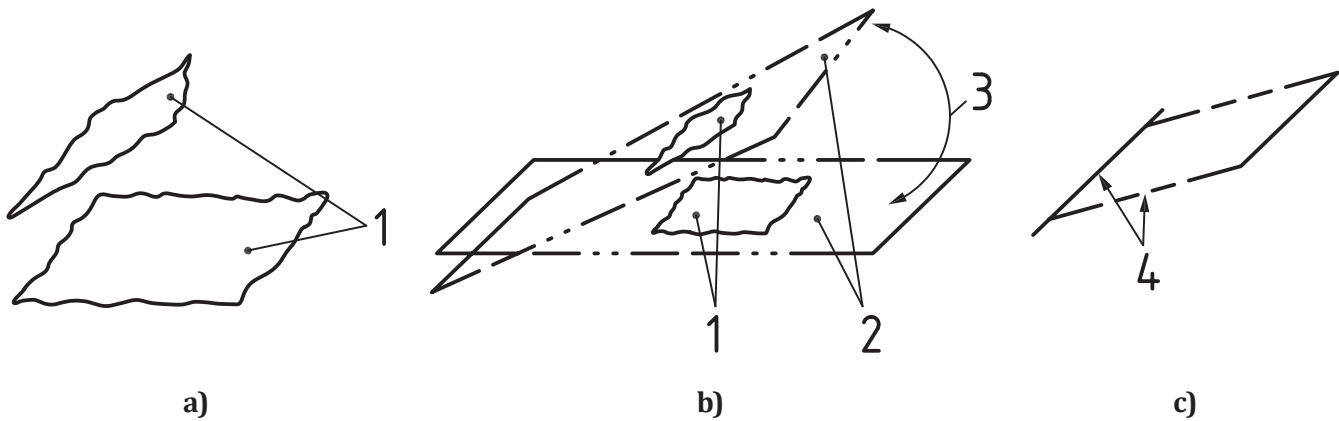


Figure C.17 — Intersecting planes — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The two real integral surfaces are obtained after partition/extraction and collection and constitute a feature of size with fixed angle [see Figure C.18 a)].
- According to 6.3.2, the single datum consists of the situation features of the collection of the two intersecting planes associated with the real integral features used for establishing the single datum. The association is global and an internal orientation constraint (fixed angle of 45°) exists [see Figure C.18 b)]. The invariance class of the collection of nominally planar surfaces is prismatic and the situation features are a plane (the bisector plane) and a straight line (the intersecting straight line) [see Figure C.18 c)].



Key

- 1 datum feature: real integral feature
- 2 associated features with orientation constraint (fixed angle 45°)
- 3 internal orientation constraint (angularity)
- 4 single datum – situation features of the associated feature: plane and straight line

Figure C.18 — Establishing a single datum from intersecting planes

C.2.10 Two parallel opposite planes (defined as a feature of size)

Figure C.19 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use the real integral surface resulting from the collection of two nominally parallel planar surfaces, which is a feature of size, to establish a datum by considering the size variable. The datum is used alone to orient and/or locate the tolerance zone relative to its situation feature which is a plane (the median plane of the two associated planes).

Drawing indication
(reading input or writing input)

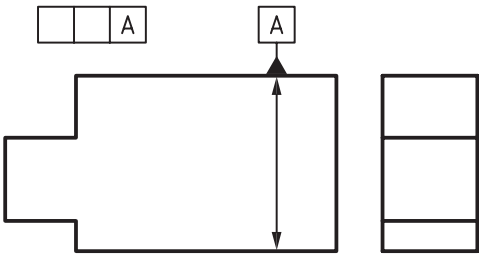
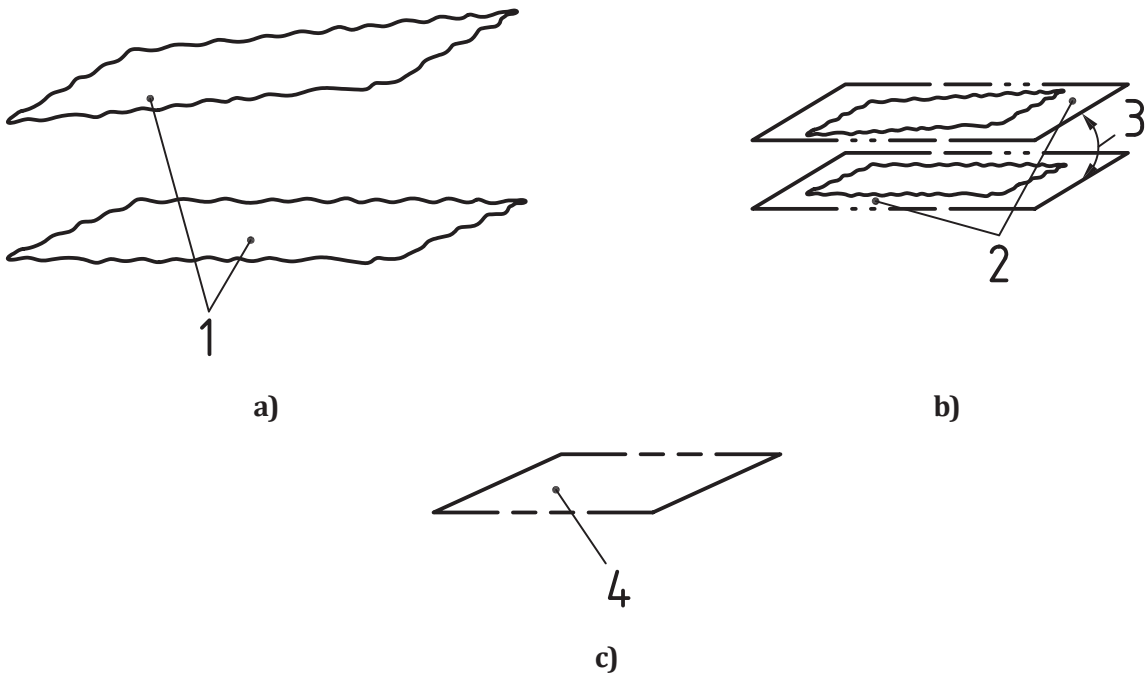


Figure C.19 — Two parallel planes — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The two real integral surfaces are obtained after partition/extraction and collection and constitute a feature of size with variable size [see Figure C.20 a)].
- According to 6.3.2, the single datum is characterized by the situation feature of the collection of two parallel planes associated with the real integral features used for establishing the single datum. The association is global and an internal orientation constraint of parallelism exists (the distance between the two planes is variable) [see Figure C.20 b)]. The invariance class of the collection of nominal surfaces is planar and the situation feature is a plane (the median plane of the two associated planes) [see Figure C.20 c)].



Key

- 1 datum feature: real integral feature
- 2 associated features with variable distance
- 3 internal orientation constraint only (parallelism) (the size is variable for association)
- 4 single datum – situation feature of the collection surface: plane

Figure C.20 — Establishing a single datum from a feature of size — Two parallel opposite planes

C.2.11 Three datum targets on a plane

Figure C.21 illustrates the drawing indication of the design intent.

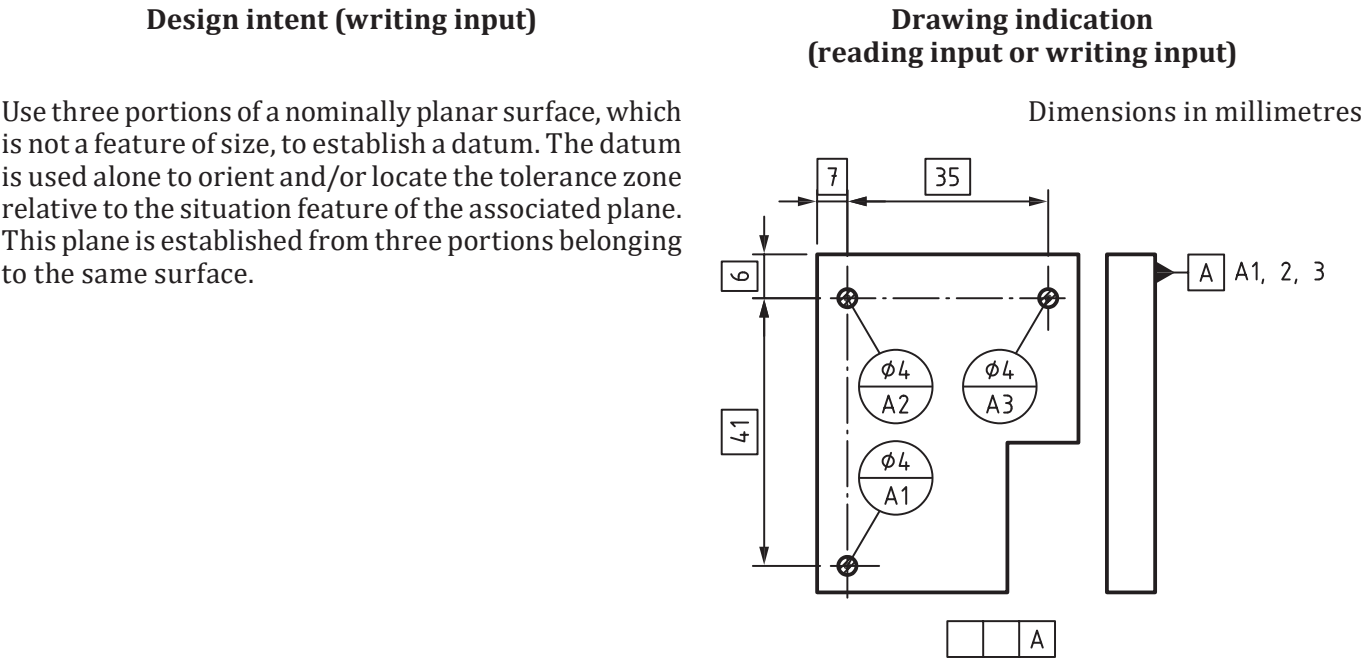
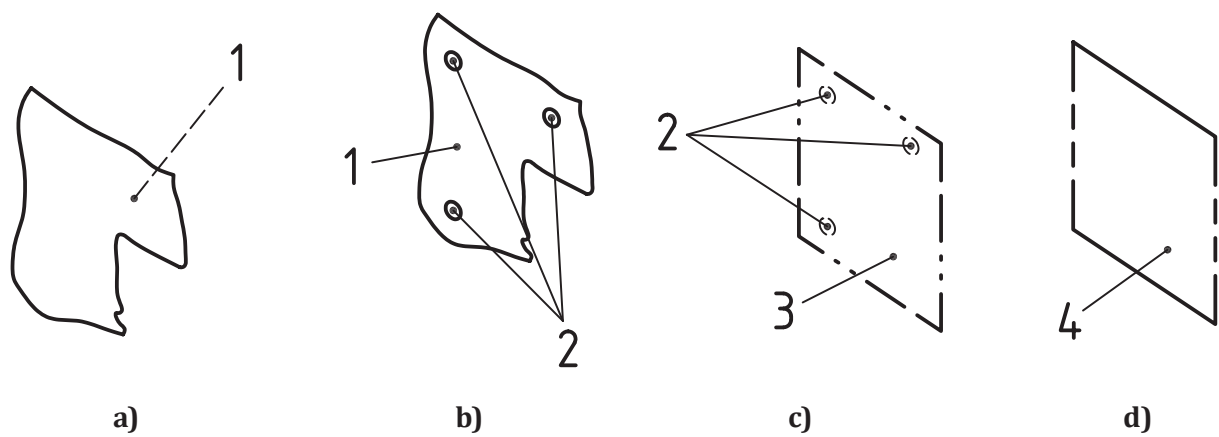


Figure C.21 — Three datum targets on a plane — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The datum target areas are obtained after partition/extraction by using the TEDs which define the location and the dimension of the areas from the datum feature [see Figures C.22 a) and C.22 b)].
- According to 6.3.2, the single datum is characterized by the situation feature of the plane associated with the real integral feature without external constraints [see Figure C.22 c)]. The invariance class of the nominal surface is planar and the situation feature is a plane [see Figure C.22 d)].



Key

- 1 datum feature: real integral feature
- 2 datum targets taken on the datum feature
- 3 associated feature to the three datum targets
- 4 single datum – situation feature of the collection surface: plane

Figure C.22 — Establishing a single datum from three datum targets on a planar surface

C.3 Examples of common datums

C.3.1 Two coplanar planes

[Figure C.23](#) illustrates the drawing indication of the design intent.

Design intent (writing input)

Use two integral, nominally planar surfaces constrained to be coplanar, which are not features of size, simultaneously to establish a datum. The datum is used to orient and/or locate the tolerance zone relative to a plane. This plane is the situation feature of the collection of the two associated planes.

**Drawing indication
(reading input or writing input)**

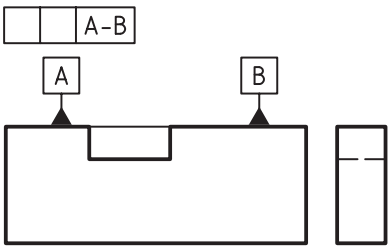
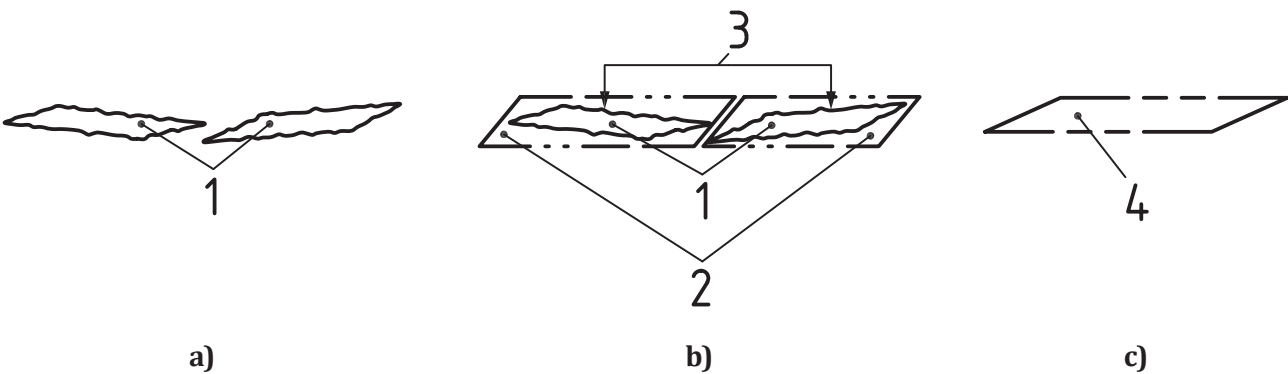


Figure C.23 — Two coplanar planes — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surfaces are obtained after partition/extraction and collection [see [Figure C.24 a](#)].
- According to [6.3.3](#), the common datum is characterized by the situation feature of the collection of two coplanar planes associated with the real integral features used for establishing the common datum. The association is global and with internal constraints: coplanar (i.e. distance zero and parallel) [see [Figure C.24 b](#)]. The invariance class of the collection of nominal surfaces is planar and the situation feature is a plane [see [Figure C.24 c](#)].



Key

- 1 datum features: real integral features
- 2 associated features (two associated planes)
- 3 orientation and location constraint (coplanarity) between associated features
- 4 single datum – situation feature of the collection surface: plane

Figure C.24 — Establishing a common datum from two coplanar planes

C.3.2 Two coaxial cylinders

Figure C.25 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use two integral, nominally cylindrical and coaxial, surfaces, which are features of size, simultaneously to establish a datum by considering their sizes variable and the orientation constraint (parallelism) and location constraint (coaxiality). The datum is used to orient and/or locate the tolerance zone relative to an axis. This axis is the situation feature of the collection of the two associated cylinders.

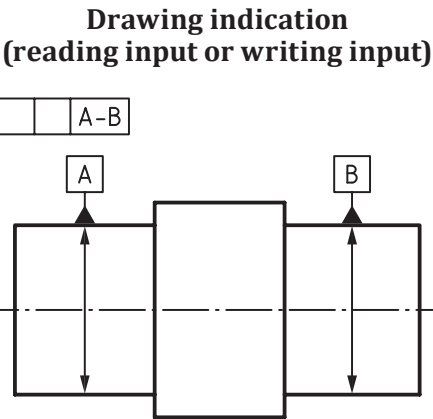
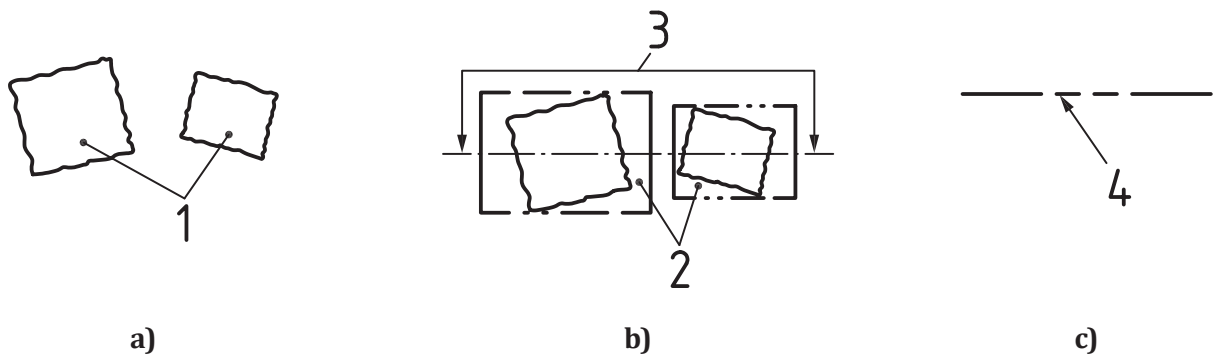


Figure C.25 — Two coaxial cylinders — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surfaces are obtained after partition/extraction and collection [see Figure C.26 a)].
- According to 6.3.3, the common datum is characterized by the situation feature of the collection of two coaxial cylinders associated with the real integral features used for establishing the common datum. The association is global and with internal constraints: distance zero and parallel (coaxial) [see Figure C.26 b)]. The invariance class of the collection of nominal surfaces is cylindrical and the situation feature is the common axis of the two cylinders [see Figure C.26 c)].



Key

- 1 datum feature: real integral feature
- 2 associated features with variable diameters
- 3 orientation constraint (parallelism) and location constraint (coaxiality)
- 4 single datum – situation feature of the associated feature: straight line (common axis)

Figure C.26 — Establishing a common datum from two coaxial cylinders

C.3.3 Plane and cylinder perpendicular to each other

Figure C.27 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use two integral surfaces, which are a nominally planar surface (which is not a feature of size) and a cylindrical surface (which is a feature of size) respectively, and which are nominally perpendicular to each other, simultaneously to establish a datum by considering the size of the cylinder variable and the orientation constraint (perpendicularity) between the axis of the associated cylinder and the associated plane (without defined order). The datum is used to orient and/or locate the tolerance zone relative to a straight line and one of its points. The axis and the point are the situation features of the collection of the two associated features.

Drawing indication
(reading input or writing input)

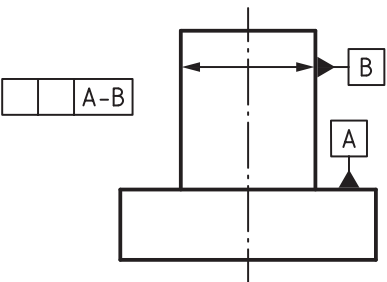
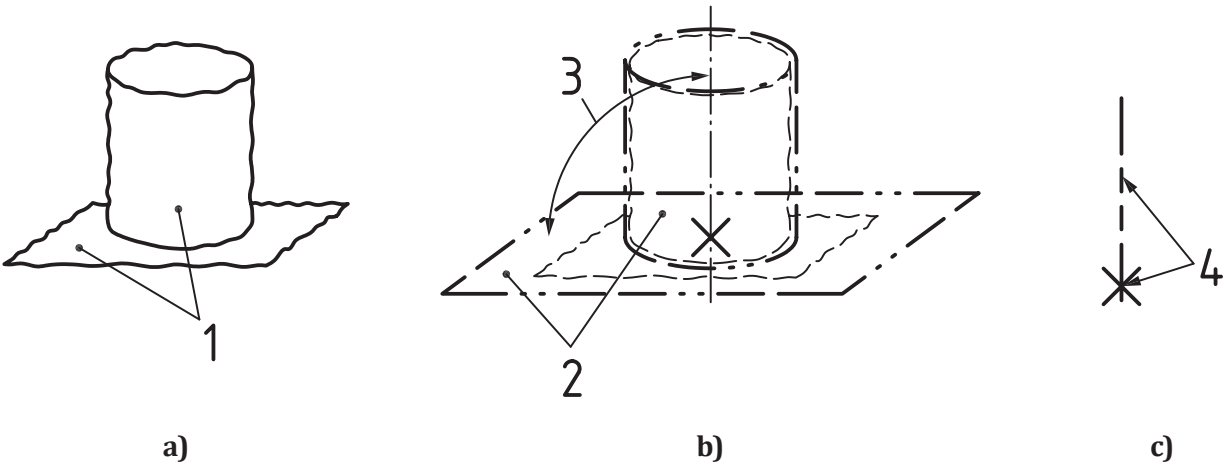


Figure C.27 — Plane and perpendicular cylinder — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surfaces are obtained after partition/extraction and collection [see Figure C.28 a)].
- According to 6.3.3, the common datum is characterized by the situation features of the collection of a plane and a cylinder perpendicular to each other associated with the real integral features used for establishing the common datum. The association is global and with internal constraints: an angle of 90° between the axis and the plane [see Figure C.28 b)]. The invariance class of the collection of nominal surfaces is revolute and the situation features are the axis of the cylinder and the point resulting from the intersection of the plane and the axis [see Figure C.28 c)].



Key

- 1 datum features: real integral features
- 2 associated features (an associated cylinder with variable diameter and an associated plane)
- 3 orientation constraint (perpendicularity)
- 4 single datum – situation feature of the associated feature: straight line and one point

Figure C.28 — Establishing a common datum from a plane and a perpendicular cylinder

C.3.4 Two parallel cylinders

Figure C.29 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use two integral, nominally cylindrical and parallel surfaces, which are features of size, simultaneously to establish a datum by considering the size of the cylinders variable and the orientation constraint (parallelism) and location constraint (l mm) between the two axes. The datum is used to orient and/or locate the tolerance zone relative to a plane and a straight line in the plane. The plane and the straight line are the situation features of the collection of the two associated features.

Drawing indication (reading input and writing input)

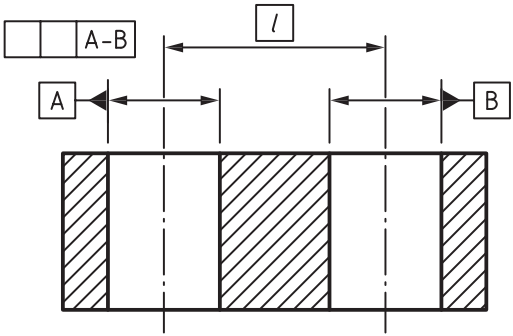
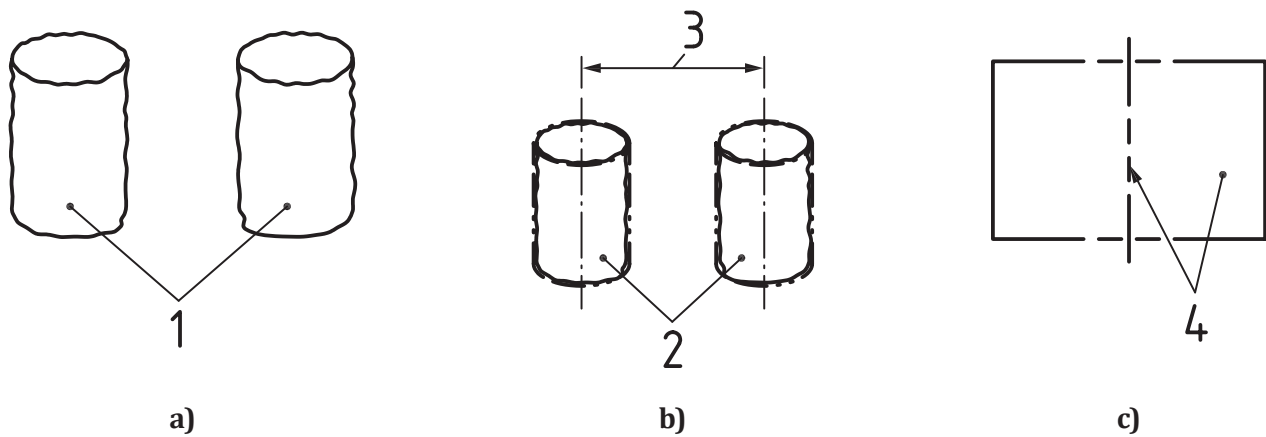


Figure C.29 — Two parallel cylinders — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surfaces are obtained after partition/extraction and collection [see Figure C.30 a)].
- According to 6.3.3, the common datum is characterized by the situation features of the collection of two parallel cylinders associated with the real integral features used for establishing the common datum. The association is global and with internal constraints: distance between the two axes and parallelism [see Figure C.30 b)]. The invariance class of the collection of nominal surfaces is prismatic and the situation features are the plane containing the two axes and the median straight line of the axes of the two associated cylinders [see Figure C.30 c)].



Key

- 1 datum features: real integral features
- 2 associated features with variable diameters
- 3 orientation constraint (parallelism) and location constraint (l distance)
- 4 common datum – situation feature of the collection surface: plane and straight line

Figure C.30 — Establishing a common datum from two parallel cylinders

C.3.5 Pattern of five cylinders

Figure C.31 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use five integral, nominally identical and cylindrical surfaces, which are features of size, with parallel axes together simultaneously to establish a datum by considering the size of cylinders variable and the orientation constraints (parallelism) and location constraints (d mm and $4 \times 72^\circ$). The datum is used to orient and/or locate the tolerance zone relative to a plane and a straight line in the plane. The plane and the straight line are the situation features of the collection of the five associated features.

Drawing indication (reading input or writing input)

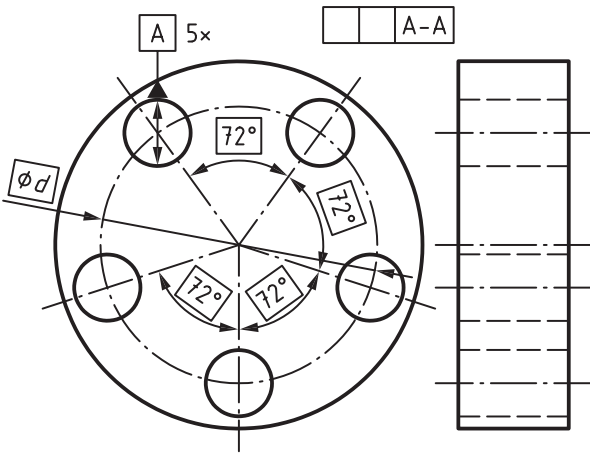


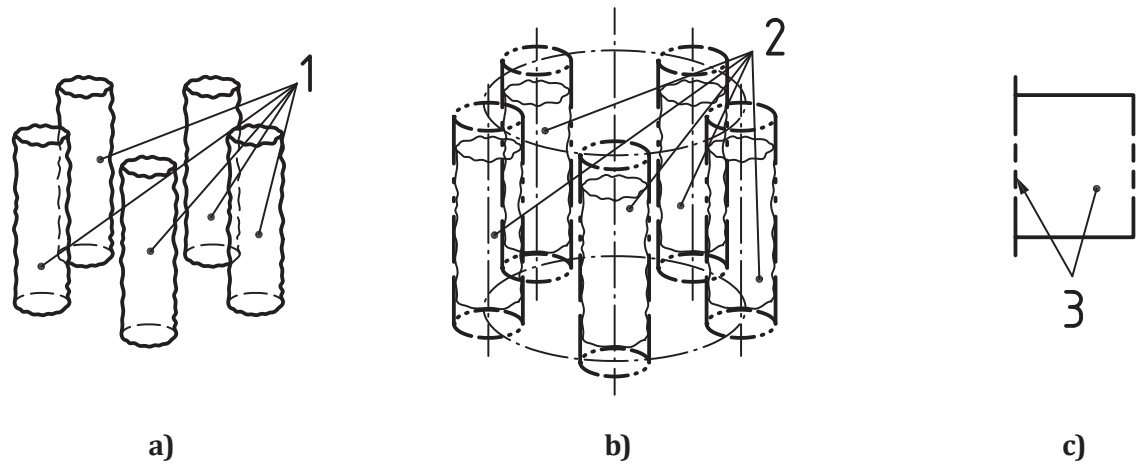
Figure C.31 — Pattern of five cylinders — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surfaces are obtained after partition/extraction and collection [see Figure C.32 a)].
- According to 6.3.3, the common datum is characterized by the situation features of the collection of five parallel cylinders associated with the real integral features used for establishing the common datum. The association is simultaneous and with internal constraints: parallelism, distance and angle between the five axes [see Figure C.32 b)]. The invariance class of the collection of nominal surfaces is prismatic

and the situation features are the median straight line of the axes of the five associated cylinders and a plane containing one axis and this median straight line [see [Figure C.32 c](#)].

NOTE The actual planar situation feature is specified by the tolerance frame and by TED(s), if applicable.



Key

- 1 datum features: real integral features
- 2 associated features: with variable diameters, orientation constraint (parallelism) and location constraint (given by the TEDs and angles of 72°)
- 3 common datum – situation features of the collection surface: plane and straight line

Figure C.32 — Establishing a common datum from a pattern of five cylinders

C.3.6 Two parallel planes

[Figure C.33](#) illustrates the drawing indication of the design intent.

Design intent (writing input)

Use two integral, nominally planar and parallel surfaces, which are not features of size, simultaneously to establish a datum with a constraint of distance between the two planes. The datum is used to orient and/or locate the tolerance zone relative to a plane. This plane is the situation feature of the collection of the two associated planes which are constrained to be parallel and separated by a value indicated as a TED.

Drawing indication
(reading input or writing input)

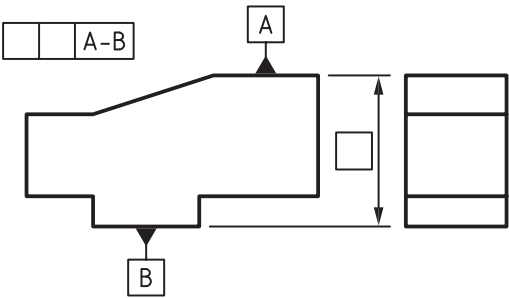
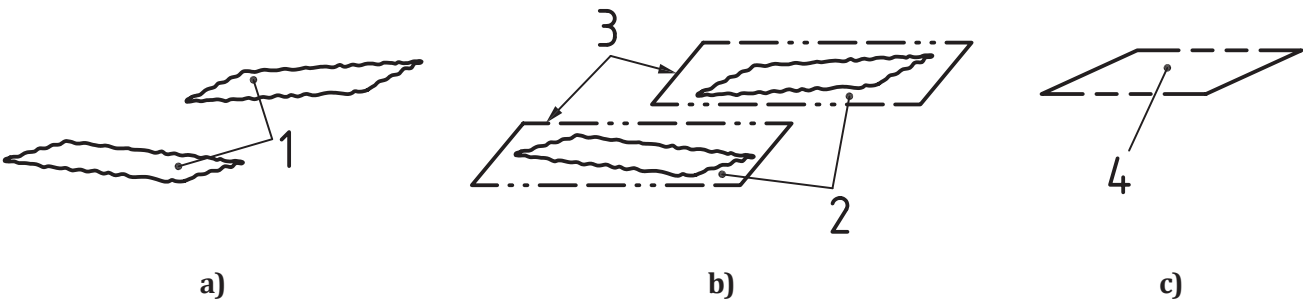


Figure C.33 — Two parallel planes — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The two real integral surfaces, which do not constitute a feature of size, are obtained after partition/ extraction and collection [see [Figure C.34 a](#)].
- According to [6.3.3](#), the common datum is characterized by the situation features of the collection of two parallel planes associated with the real integral features used for establishing the common datum. The association is global and with internal constraints: fixed distance defined by a TED and parallelism [see

Figure C.34 b)]. The invariance class of the collection of nominal surfaces is planar and the situation feature is a plane [see Figure C.34 c)].



Key

- 1 datum features: two real integral features
- 2 associated features (two associated planes)
- 3 orientation constraint (parallelism) and location constraint (distance defined by a TED)
- 4 common datum – situation feature of the collection surface: plane

Figure C.34 — Establishing a common datum from two parallel planes

C.4 Examples of datum systems

C.4.1 Three perpendicular planes

Figure C.35 illustrates the drawing indication of the design intent.

Design intent (writing input)

Use three integral, nominally planar and perpendicular surfaces, which are not features of size, to establish a datum system consisting of three single datums with orientation constraints (perpendicularity) between them. The datums are used together sequentially in a given order to orient and locate the tolerance zone relative to a plane, one of its straight lines and one of its points (equivalent to three planes). These are the situation features of the collection of the three associated planes.

**Drawing indication
(reading input or writing input)**

Figure C.35 — Three perpendicular planes — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surfaces are obtained after partition/extraction and collection (see [Figure C.36](#)).

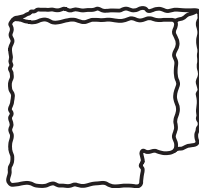
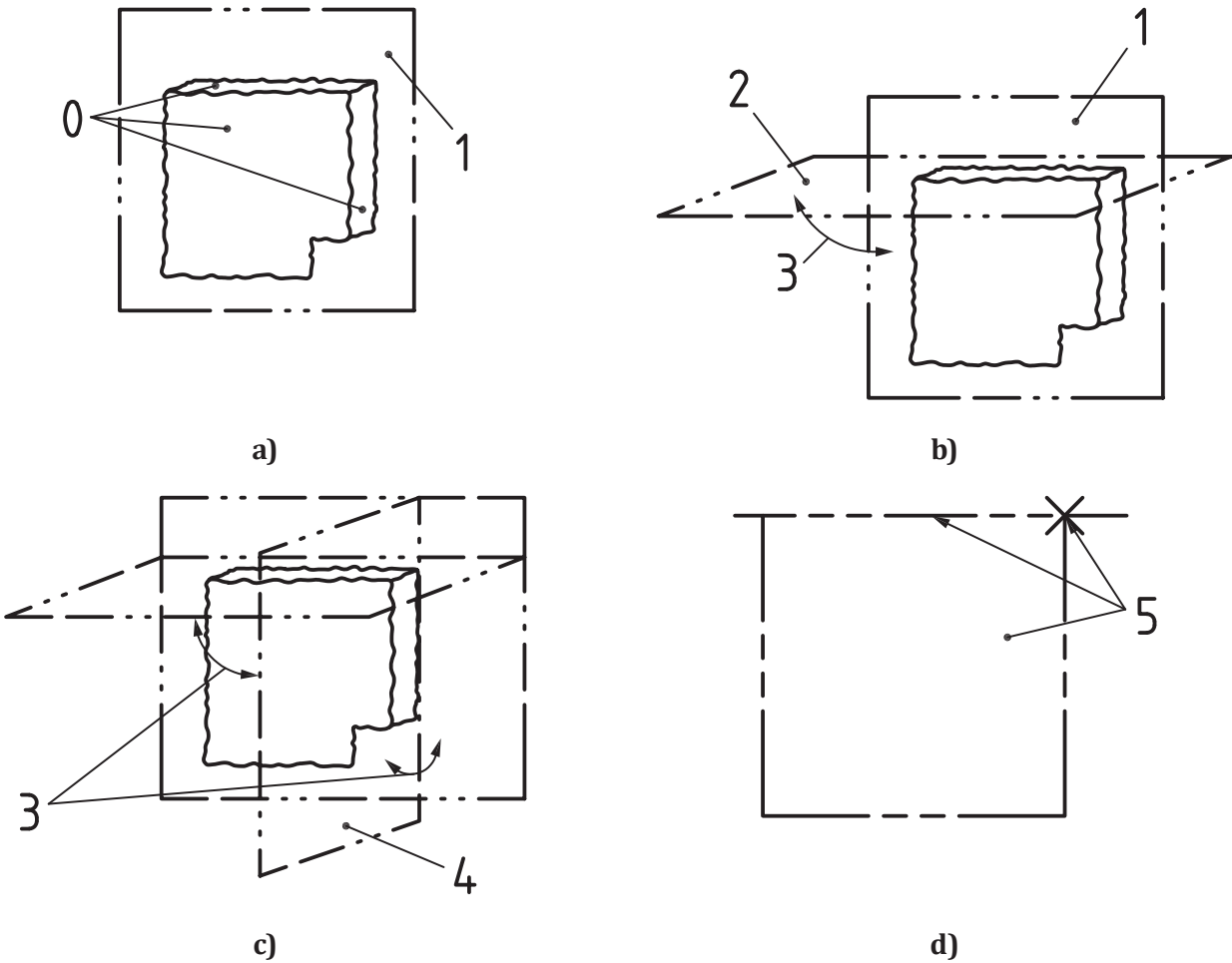


Figure C.36 — Three perpendicular planes — Set of three real integral surfaces

- According to [6.3.4](#), the datum system is characterized by the situation features of the collection of three planes associated with the real integral features with orientation constraints between them: angle of 90° . In the datum system, the primary datum is an associated plane [see [Figure C.37 a](#)]; the secondary datum is an associated plane that respects the orientation constraint from the primary single datum [see [Figure C.37 b](#)]; the tertiary datum is an associated plane which respects the orientation constraint firstly from the primary datum and secondly from the secondary datum [see [Figure C.37 c](#)].

The invariance class of the collection of nominal surfaces that make up the datum system is complex and the situation features are a plane (corresponding to the primary datum), a straight line (the intersection between this plane and the plane corresponding to the secondary datum) and a point (the intersection between the straight line of the secondary datum and the plane corresponding to the tertiary datum) [see [Figure C.37 d](#)].



Key

- 0 datum features: real integral features
- 1 associated plane (primary datum in this case) to the datum feature identified by the datum feature identifier A
- 2 associated plane (secondary datum in this case) identified by the datum feature identifier B, with orientation constraint from the primary datum
- 3 orientation constraint (perpendicularity)
- 4 associated plane (tertiary datum in this case) identified by the datum feature identifier C, with orientation constraints from the primary datum and secondary datum
- 5 datum system: plane (primary datum), straight line (intersection between the primary and secondary datum) and point (intersection of the three datums)

Figure C.37 — Establishing a datum system from three perpendicular planes

C.4.2 Cylinder and perpendicular plane

Figure C.38 illustrates the drawing indication of the design intent.

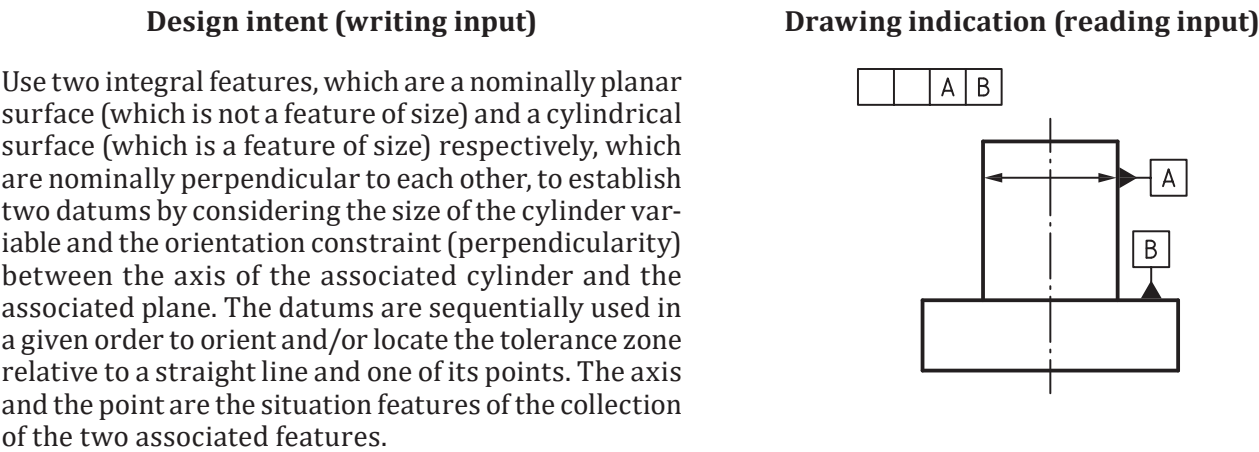


Figure C.38 — Cylinder and perpendicular plane — Input for reading (drawing indication) and for writing (design intent)

Explanation:

- The real integral surfaces are obtained after partition/extraction and collection (see Figure C.39).

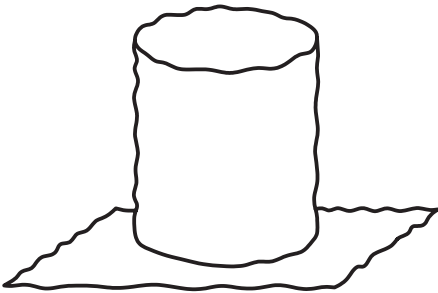
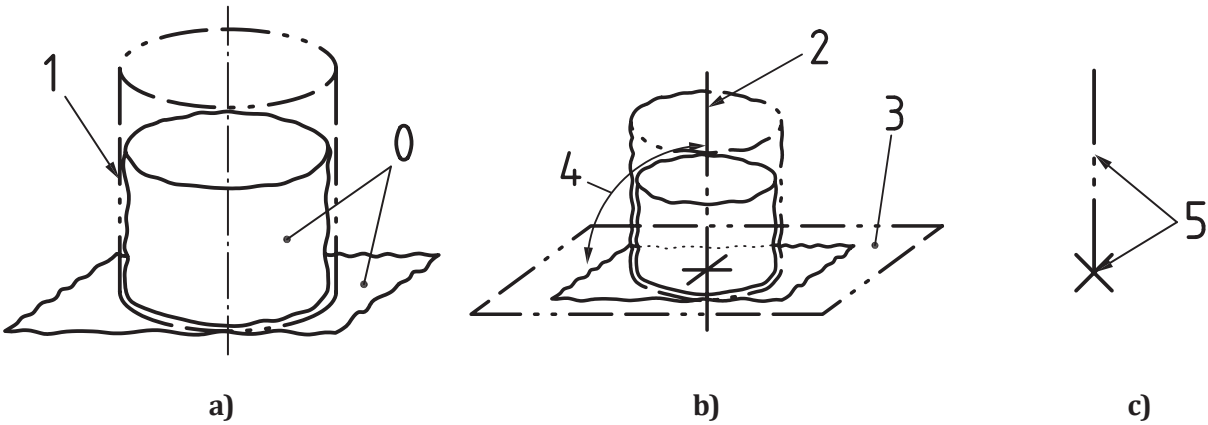


Figure C.39 — Cylinder and perpendicular plane — Set of two real integral surfaces

- According to 6.3.4, the datum system is characterized by the situation features of the collection of a plane and a cylinder perpendicular to each other, associated with the real integral features, with an orientation constraint between them: angle of 90°. In the datum system, the primary datum is an associated cylinder [see Figure C.40 a)]; the secondary datum is an associated plane which respects the orientation constraint from the primary single datum [see Figure C.40 b)].
- The invariance class of the collection of nominal surfaces is revolute and the situation features are the axis of the cylinder and the point resulting from the intersection between the axis and the plane [see Figure C.40 c)].



Key

- 0 datum features: real integral features
- 1 associated cylinder to the datum feature
- 2 primary datum (axis of the associated cylinder)
- 3 associated plane (secondary datum in this case) with orientation constraint from the primary datum
- 4 orientation constraint (perpendicularity)
- 5 datum system: straight line (primary datum) and point (intersection between the primary and secondary datum)

Figure C.40 — Establishing a datum system from a cylinder and a perpendicular plane

Annex D
(informative)

Former practices

D.1 Indication of a specific cross-section of a cylinder as a datum feature

It was former practice to indicate the specific cross-section of a cylinder as shown in [Figure D.1](#).

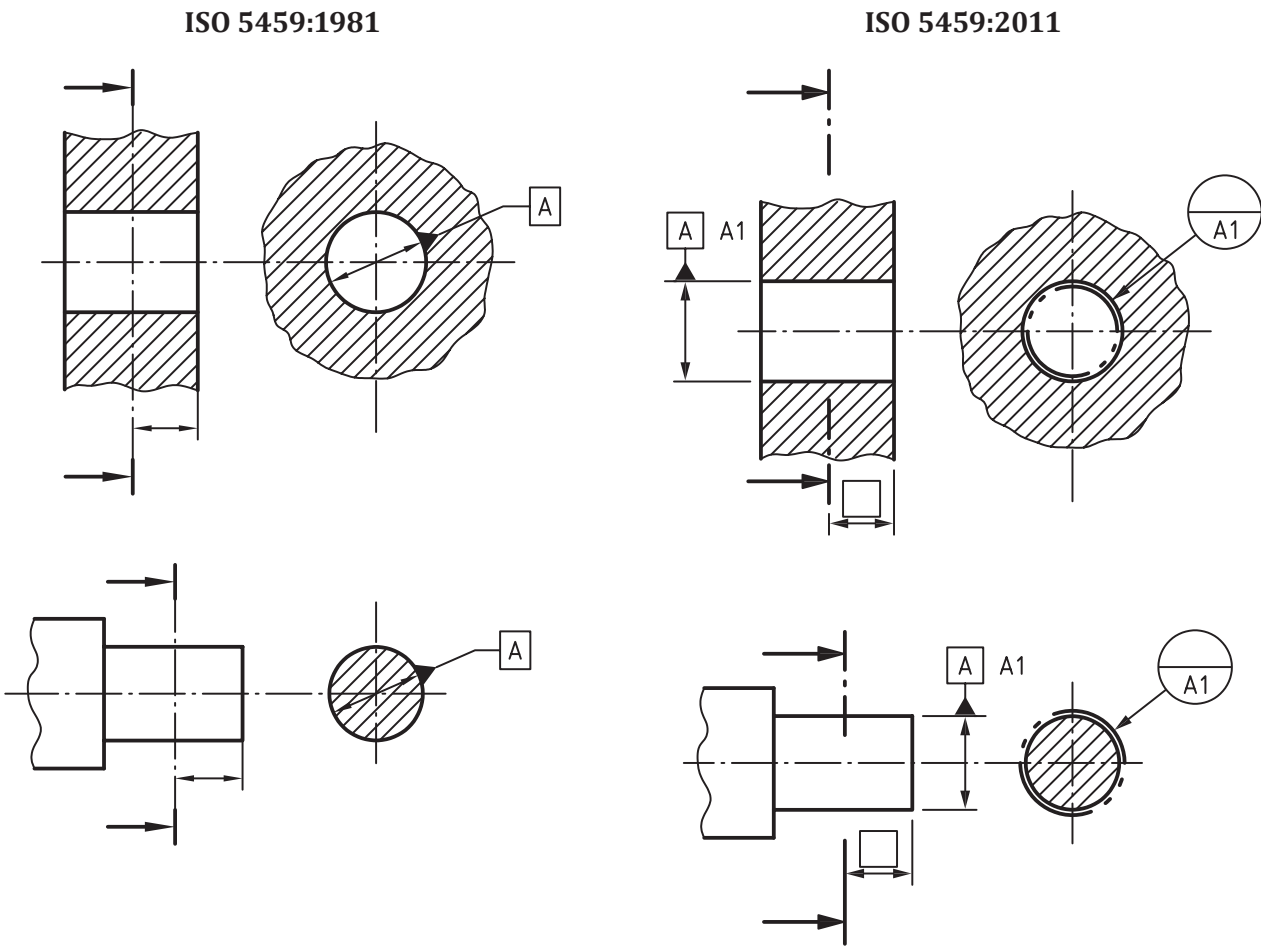


Figure D.1 — Indication of the centre of a cylinder

D.2 Datum target line indication

It was former practice, for target lines, to connect the two crosses with a continuous narrow line, as shown in [Figure D.2](#).



Figure D.2 — Connection of two crosses

D.3 Common datum indication

It was former practice to indicate a common datum established from a group of features with an indication of single datum, as shown in [Figure D.3](#).

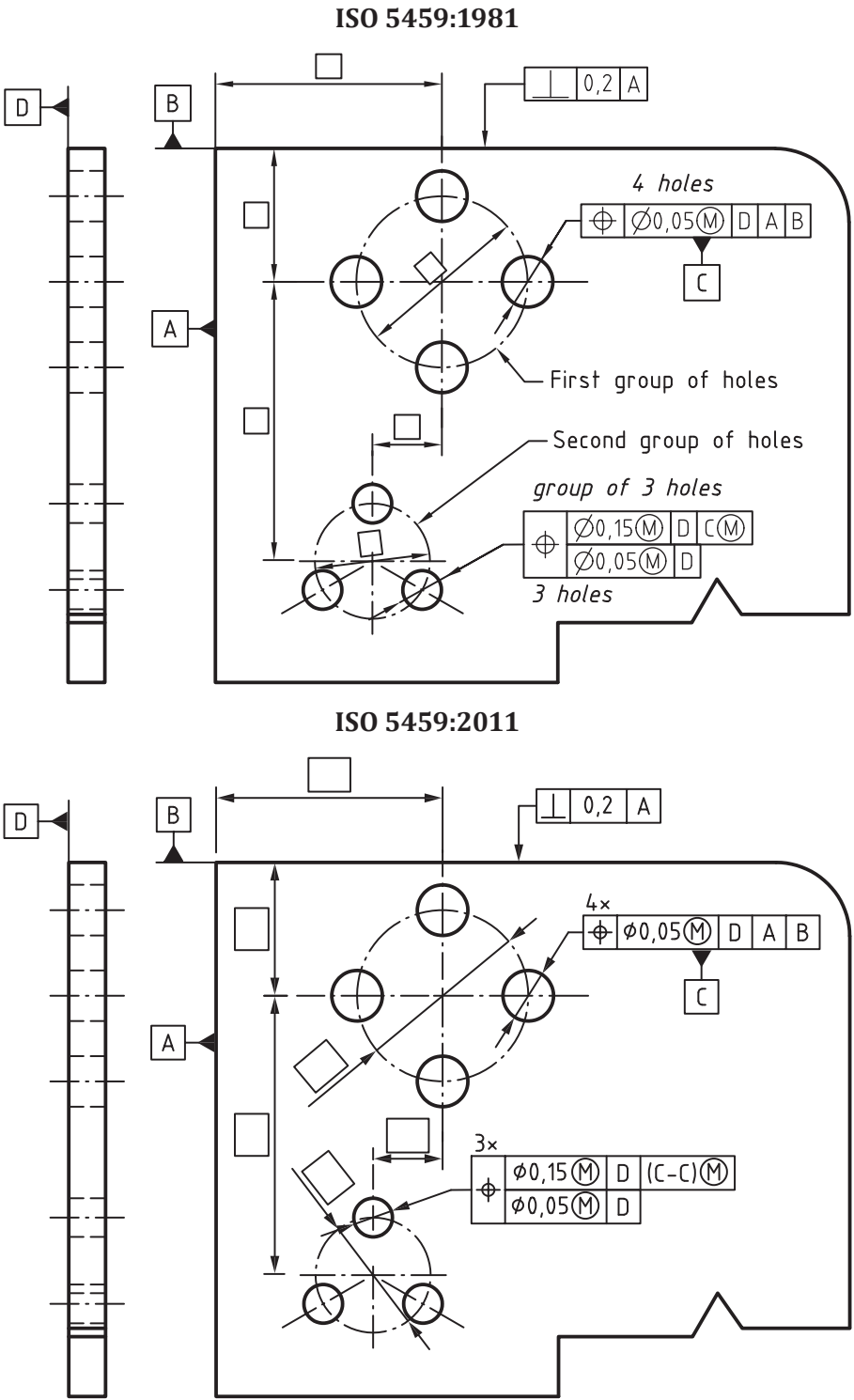
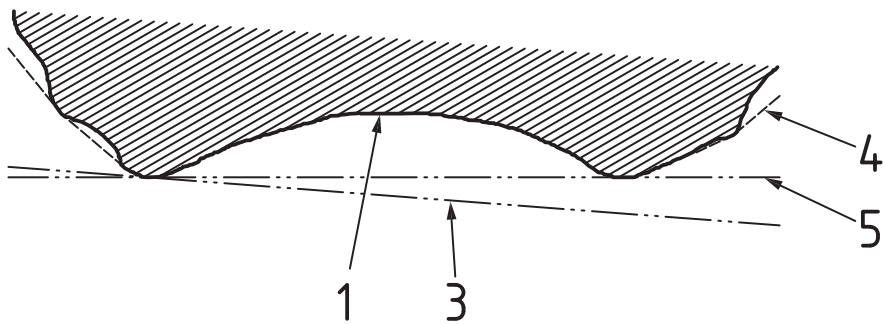


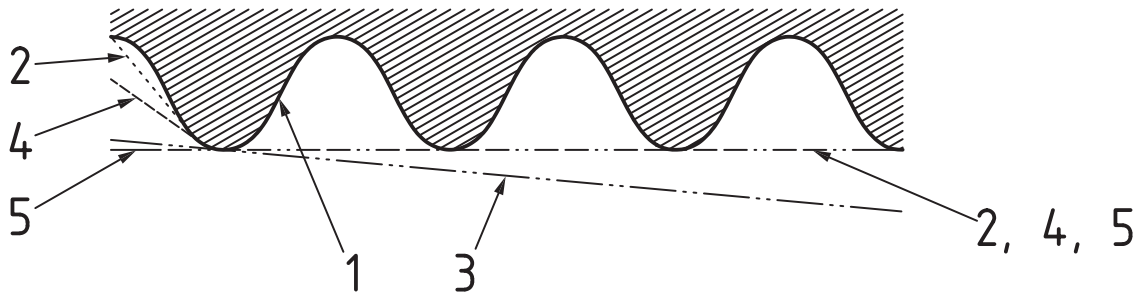
Figure D.3 — Indication of a common datum

D.4 CE, GE and new filter for a planar single datum case

To establish a single datum from a datum feature that is nominally a plane, the association criterion for datum feature used to establish a single datum (not for a common datum) has been modified from minimax with external material constraint (i.e. [CE]) to least squares with external material constraint (i.e. [GE]), which has a reduced variation due to the border effect(s) of the datum feature. [Figures D.4 a\)](#) and [D.4 b\)](#) illustrate the changing of default association criteria between this document and ISO 5459:2011 and the influence of border effects for the association criterion using the new filter introduced for a single plane. To reduce the impact of border effects like these examples show, the default association criterion has been changed.



a) Example showing the new default filtration, the default association according to ISO 5459:2011, and the new default association criterion



b) Example showing the new default filtration, the default association according to ISO 5459:2011, and the new default association criterion for a wavy surface

Key

- 1 datum feature (extracted integral feature)
- 2 filtered feature using a convex hull filter (not completely defined in ISO 5459:2011)
- 3 associated feature using the default association criteria for the plane ([CE]) as defined in ISO 5459:2011
- 4 filtered feature using the new filtration [EM] as presented in [Annex I](#)
- 5 associated feature using the new default association criterion for a plane ([GE]) as defined in this document

NOTE 1 Figure a) shows (in 2D) the plane corresponding to the result of the new default association criterion for a single plane (labelled 5) as making exact contact with the datum feature. This contact accurately represents the behaviour of the [GE] association criterion, which is a least squares (Gaussian) association objective function that is constrained to be external to the material. It is important not to confuse with the [G+] association criterion, which is an unconstrained least squares fit that is subsequently translated to be just external to the material.

NOTE 2 Figure b) shows (in 2D) the plane corresponding to result of the new default association criterion for a single plane (labelled 5) as overlapping with the filtered feature (labelled 4) for a significant portion. Although the two have slightly different orientations, it is negligible enough that representing them with the same line (as in the figure) is a reasonable depiction. It is important to differentiate this from the translated least squares (the [G+] associated plane, not shown) which exhibits a noticeable gap on the right side between the filtered feature and the [G+] associated plane.

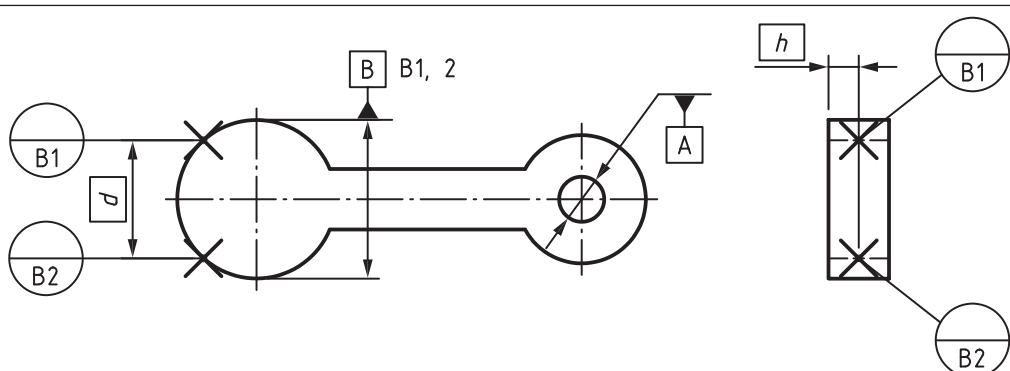
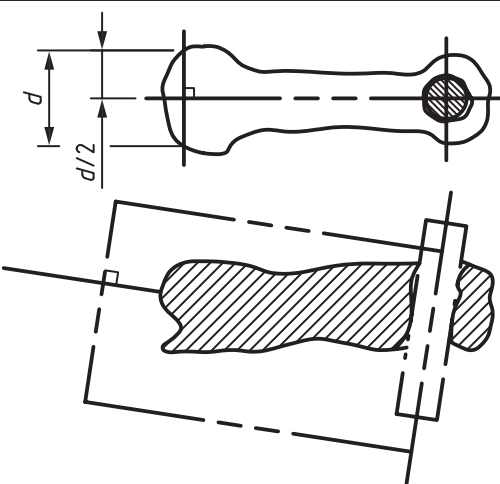
NOTE 3 It is important not to confuse [GE] and [G+]: the association criterion [G+] (not shown in this figure) is an unconstrained least squares fit that is subsequently translated to be just external to the material.

Figure D.4 — Two-dimensional view showing the impact of the default filtration and the changing of default association criteria between this document and ISO 5459:2011

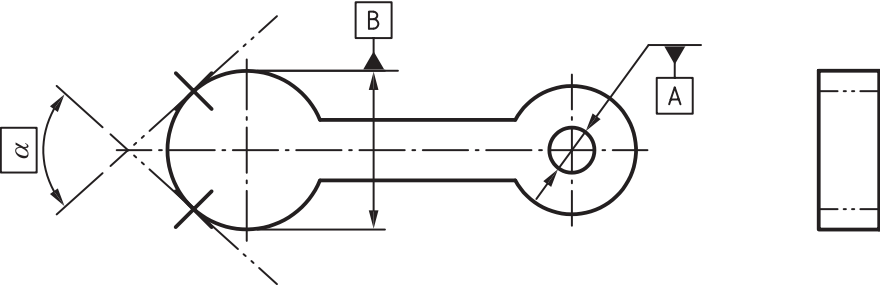
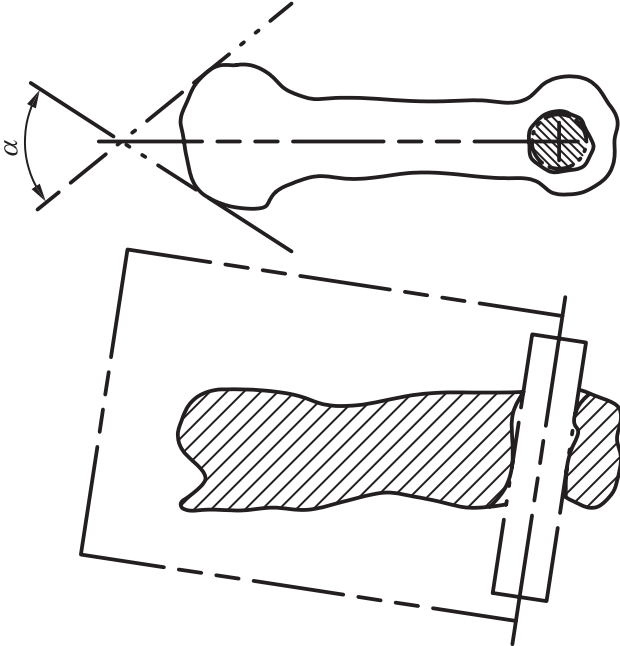
Annex E
(informative)

Examples of a datum system or a common datum established with contacting features

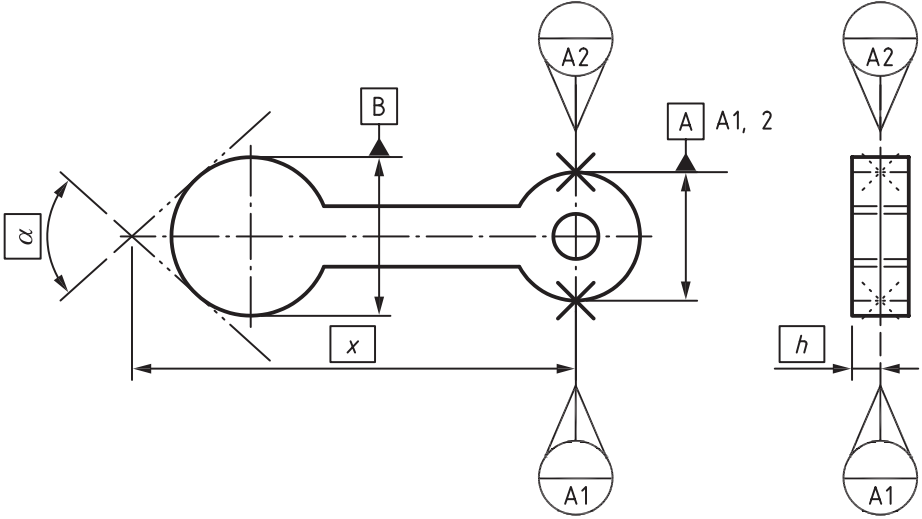
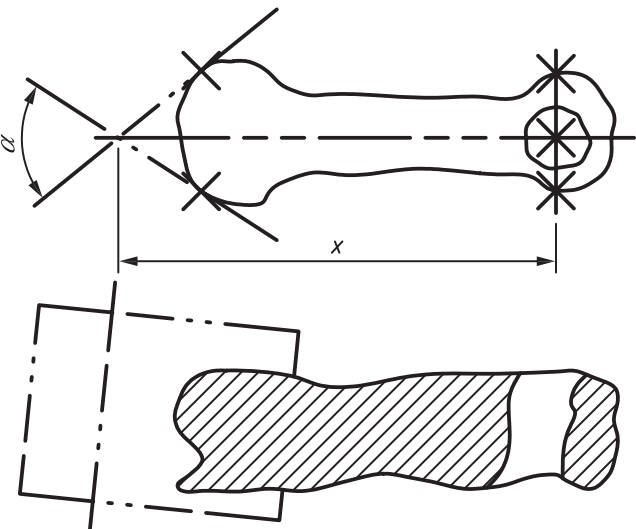
E.1 Example 1

Indication in the tolerance frame	A B[CF]
Example of indication on the drawing	
Meaning	The datum system should be established in the following manner. - First, associate a cylinder to the extracted surface, which is identified by the datum feature identifier “A”. The primary datum is the axis of the associated cylinder (defining an axis of rotation for a coordinate system). - Second, associate a set of two points, the distance between which is fixed as “d” with a TED. The manner in which the datum targets are dimensioned results in the intersection between the real surface and two straight lines, which are perpendicular to the primary datum axis and TED “d” apart (illustrated below). The secondary datum is a plane bisecting the distance between the datum target points and containing the primary datum axis (defining a stop in rotation for a coordinate system). The datum system is a set consisting of a straight line and a coincident plane.
Illustration	

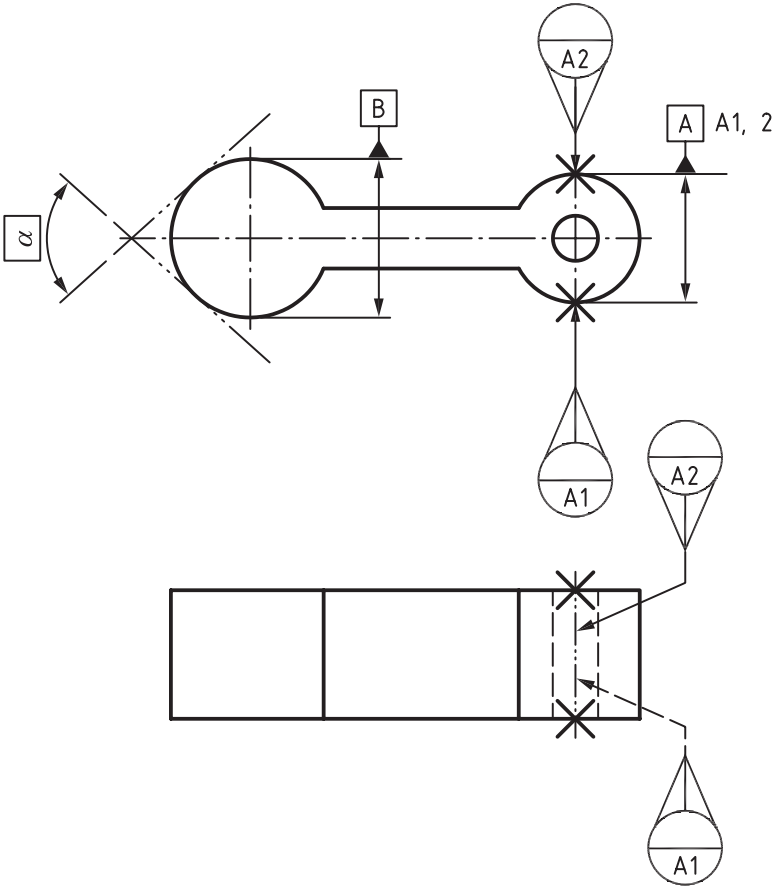
E.2 Example 2

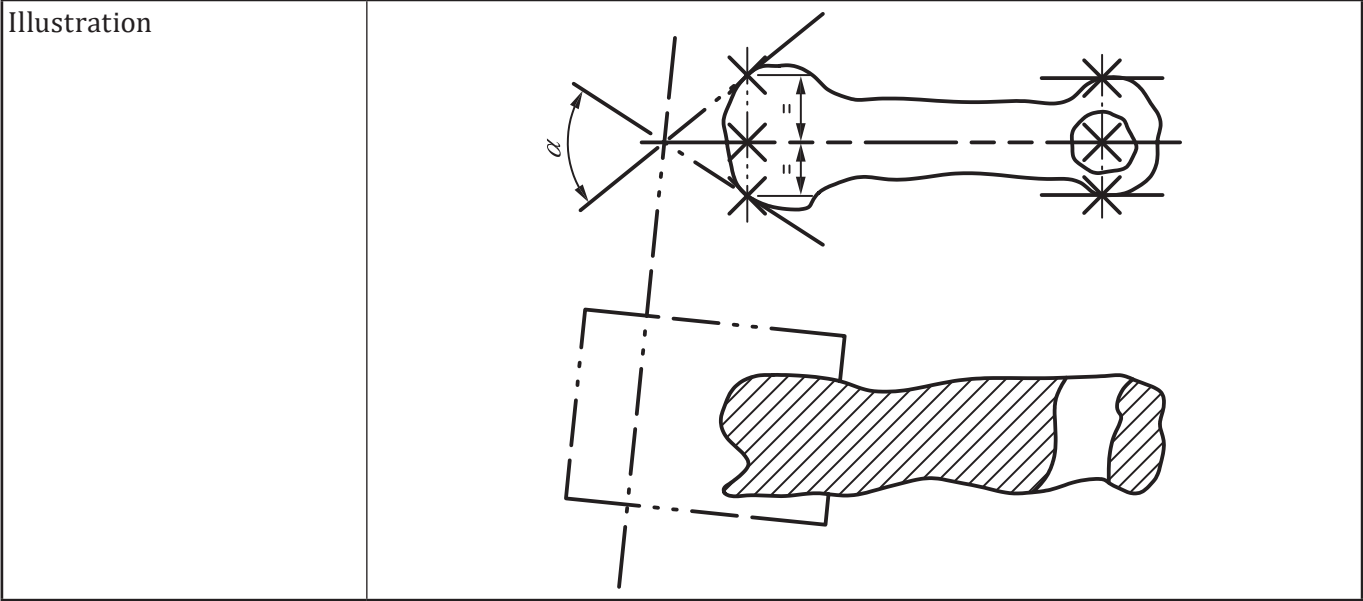
Indication in the tolerance frame	A B[CF]
Example of indication on the drawing	
Meaning	The datum system should be established in the following manner. — First, associate a cylinder with the datum feature, which is identified by the datum feature identifier “A”. The primary datum is the axis of the associated cylinder (defining an axis of rotation for a coordinate system). — Second, associate a set of two segments, the distance between which is variable. The manner in which the datum targets are dimensioned results in the intersection between the real surface and the two planes producing the wedge angle α. The line of intersection between the two planes of the wedge is parallel to the primary datum axis. The secondary datum is a bisector plane containing the primary datum axis (defining a stop in rotation for a coordinate system). The datum system is a set consisting of a straight line and a coincident plane.
Illustration	

E.3 Example 3

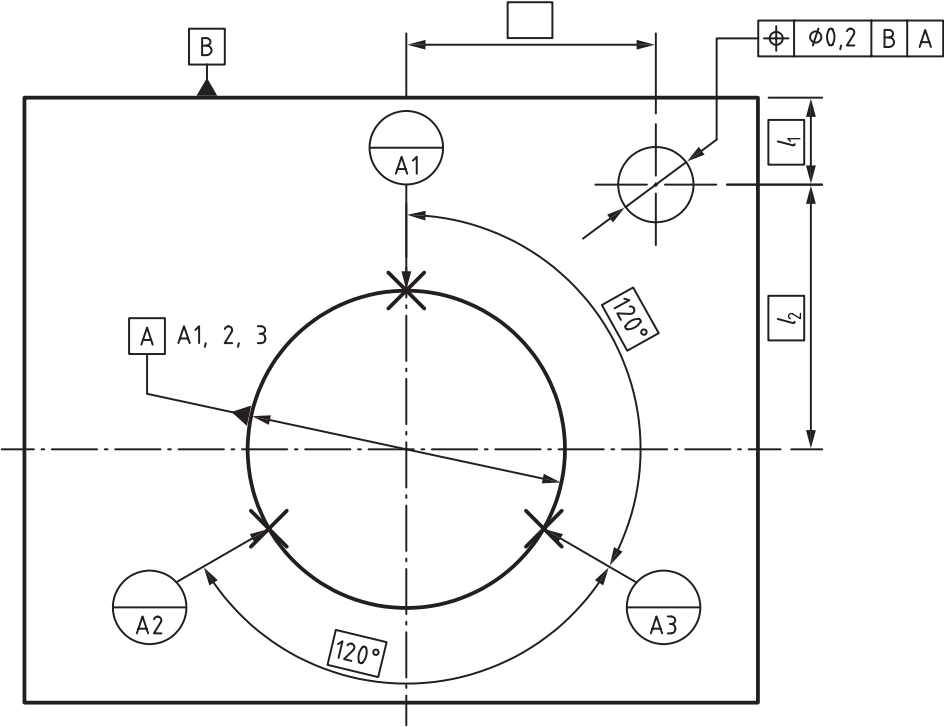
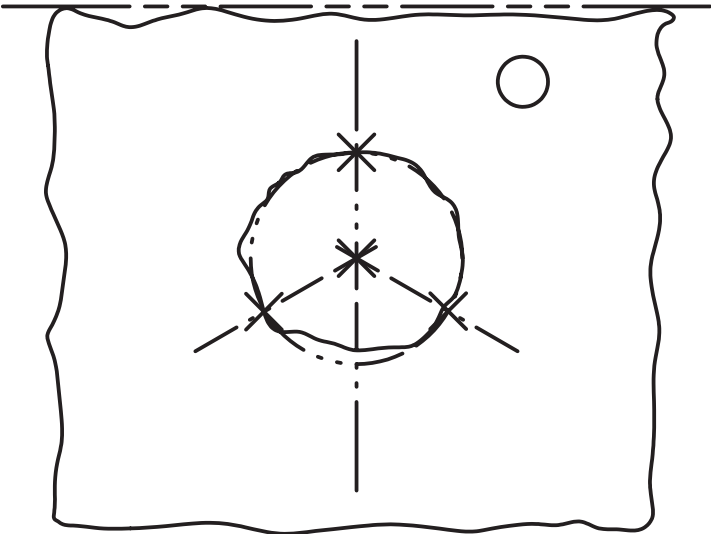
Indication in the tolerance frame	(A-B)[CF]
Example of indication on the drawing	
Meaning	The common datum should be established by associating two contacting features simultaneously with the real surface of the workpiece. The first contacting feature is a wedge with a fixed angle providing two datum targets, the distance between which is variable. The second contacting feature is a set of two datum targets, where the distance between the apex of the wedge and the points is fixed by a TED. (The first contacting feature generates nominally a datum line at the intersection of the wedge angle and leaves a degree of freedom in rotation, while the second contacting feature removes this remaining degree of freedom with a median feature.) The median plane of the wedge is coincident with the median point constructed from the two datum targets. The datum system is a set consisting of two perpendicular planes.
Illustration	

E.4 Example 4

Indication in the tolerance frame	(A-B)[DV][CF]
Example of indication on the drawing	
Meaning	The common datum should be established by associating two contacting features simultaneously with the real surface of the workpiece. The first contacting feature is a wedge with a fixed angle providing two datum targets, the distance between which is variable. The second contacting feature is a set of two parallel planes, defining as interface with the part two datum target lines. (The first contacting feature generates a datum line at the intersection of the wedge angle and leaves a degree of freedom in rotation, while the second contacting feature removes this remaining degree of freedom with a median feature.) The median plane of the wedge contains the median line constructed from the two datum targets. With the modifier [DV], the distance between the apex line of the wedge and median line is not fixed. The datum system is a set consisting of two perpendicular planes.



E.5 Example 5 — Example for a “three jaw chuck”

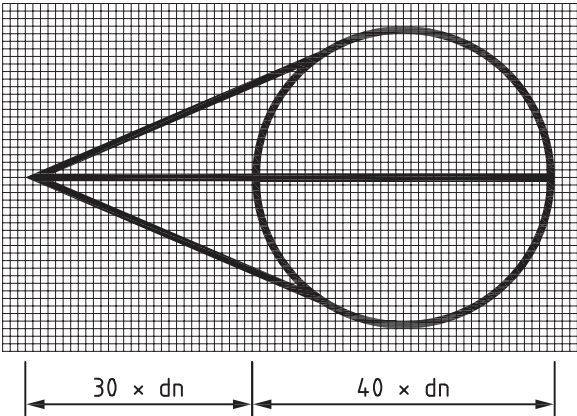
Indication in the tolerance frame	B A
Example of indication on the drawing	
Meaning	The datum system should be established by associating a plane, and associating a cylinder, established from three datum target lines. To define datum targets A1, A2 and A3, it is necessary to construct the intersections between three straight lines constrained in orientation by TEDs (120°) and coinciding at the same point. The datum system is a set consisting of a plane and a perpendicular straight line (axis of the associated cylinder constrained in orientation from the datum B).
Illustration	

Annex F
(normative)

Relations and dimensions of graphical symbols

In order to harmonize the sizes of the symbols specified in this document with those of the other inscriptions on the drawing (dimensions, letters, tolerances), the rules given in [Figure F.1](#) shall be observed. See also [Figures F.2](#), [F.3](#) and [F.4](#). These rules are in accordance with ISO 81714-1. The letters dn denote the narrow line width.

NOTE dn represents the thickness of a line as defined in ISO 7083, to conserve the proportionality between drawing indication by using a grid to draw a symbol as defined in ISO 81714-1.



NOTE A diameter line is indicated and depends on the orientation of the symbol.

Figure F.1 — Moveable target frame

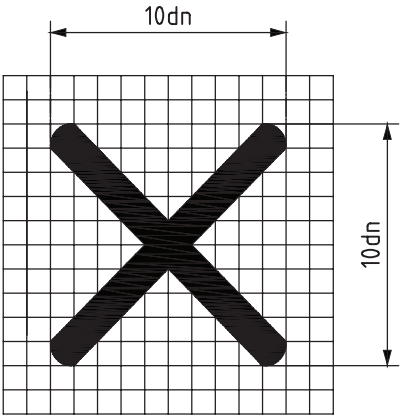
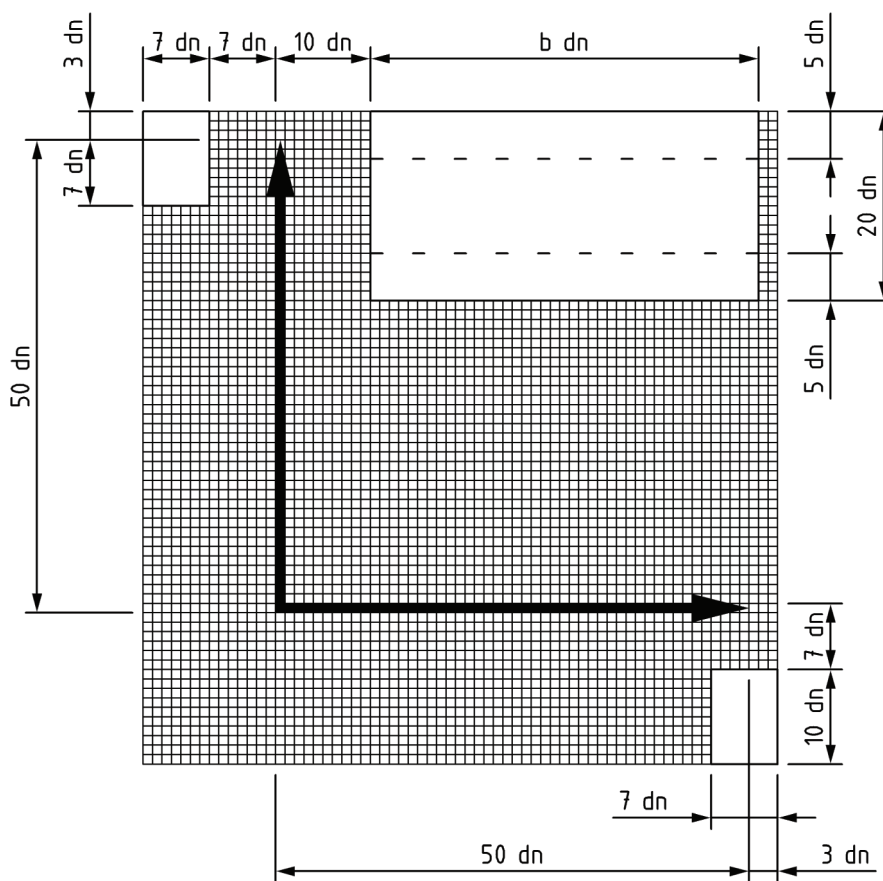


Figure F.2 — Wide line cross (e.g. for datum target point or situation feature which is a point)



NOTE The large central zone is potentially dedicated to the indication of the datum coordinate system identifier and depends on its length. The other areas are dedicated to the naming of the axis label. Their width depends on the number, a , of the characters.

Figure F.4 — Datum coordinate system indicator in one annotation plane on 3D model

Annex G (normative)

Establishment of a datum coordinate system from a datum system

G.1 General

When it is necessary to define a datum coordinate system, the datum coordinate system shall be established on a datum or a datum system that locks all six degrees of freedom.

G.2 Indication of a datum coordinate system

If it is necessary to define a datum coordinate system, it shall be indicated with a datum coordinate system identifier placed closed to a datum coordinate system indicator.

NOTE 1 The rules in this document for establishing a datum coordinate system are restricted to cases where the datum, or datum system, locks all six degrees of freedom. This is to ensure that a unique datum coordinate system can be established once the labelling and direction of the axes are indicated. Rules for establishing datum coordinate systems on datum systems where one or more degrees of freedom are not locked, are yet to be defined.

The datum coordinate system is a right-handed Cartesian coordinate system, unless otherwise indicated (see ISO 5456-1). Each axis shall be labelled. The positive direction shall be shown using an arrow on the termination of the axis, defining the positive direction.

This datum coordinate system shall be indicated in the technical product documentation in accordance with [Annex F](#). In a 2D specification, only two axes with their labels are necessary to define the datum coordinate system, see [Figure F.3](#). In a 3D specification, three axes with their labels are necessary to define the datum coordinate system, based on 2D annotation as presented in [Figure F.4](#).

NOTE 2 The datum coordinate system is defined from a datum system established from one or more datum features. There is no inherent relation between the coordinate system used to define the CAD model and datum coordinate systems. A coordinate system can be applied to a datum system to facilitate the process control.

G.3 Datum coordinate system indicator

If necessary, the datum coordinate system indicator shall be placed on the drawing at its nominal situation.

The location and orientation of a datum coordinate system are defined from the resulting datum system (i.e. from the situation features a point on a straight line in a plane), with explicit or implicit TEDs in accordance with its representation given by the datum coordinate system indicator:

- for the location, linear TEDs between the origin of the datum coordinate system and the situation feature of type point of the resulting datum system;
- for the orientation, angular TEDs from the situation features of type straight line and of type plane of the resulting datum system.

NOTE 1 A datum coordinate system can be established by:

- coinciding the origin of the coordinate system with the point of the resulting datum system;
- aligning one axis of the coordinate system with the straight line of the resulting datum system;
- aligning another axis of the coordinate system with the plane of the resulting datum system.

The labelling of the axes of the datum coordinate system and the directions are defined by the datum coordinate system indicator.

NOTE 2 The datum coordinate system is defined from a datum (single or common) or a datum system, established from one or more datum features. The coordinate system used in a CAD system to define the CAD model is interpretable. And for each real workpiece the coordinate system is unique, but it can be different for different real workpieces. A coordinate system can be applied to datum systems as a means of enabling a process control.

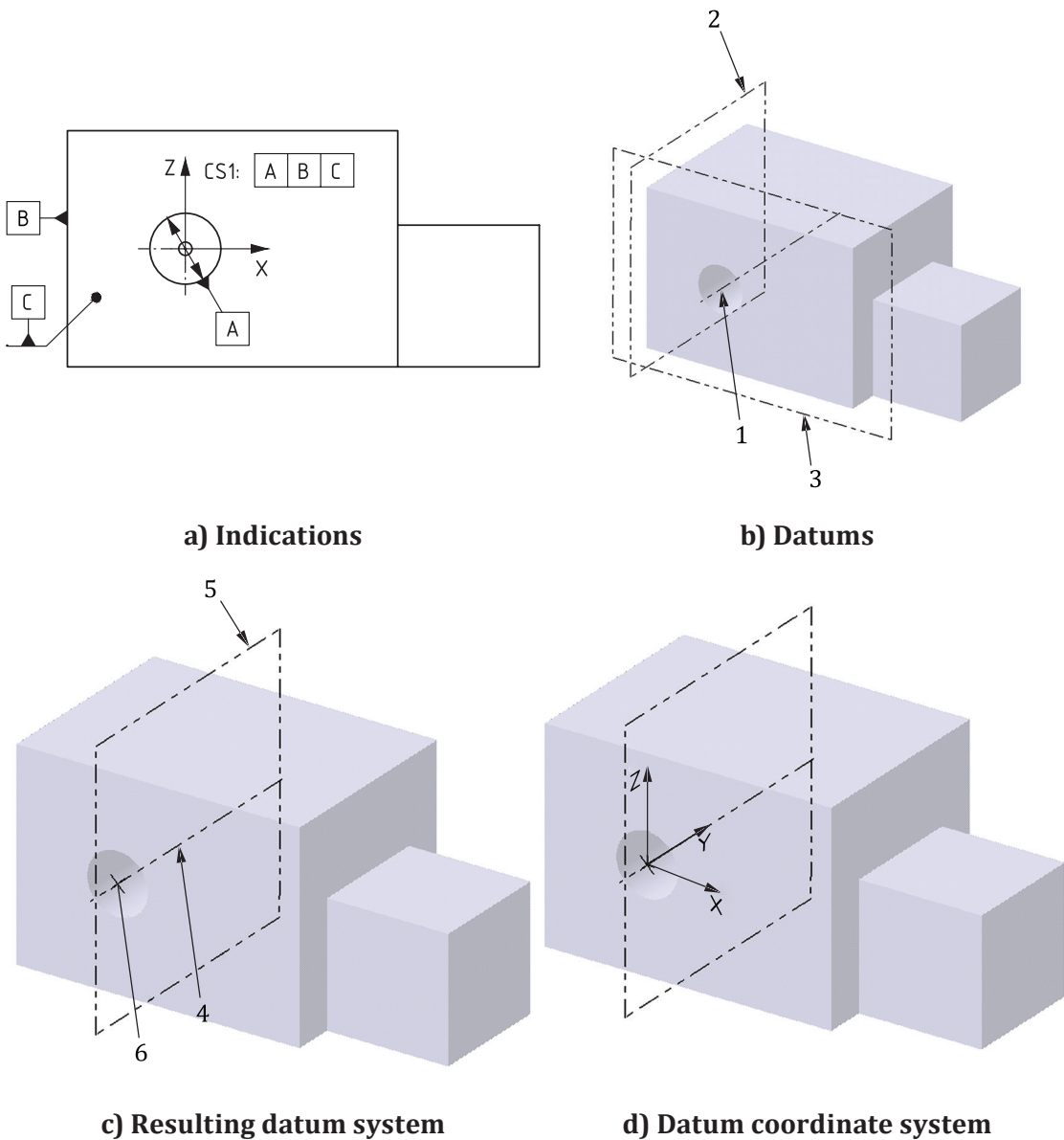
NOTE 3 An axis label of the datum coordinate system (e.g. "X", "Y", "Z") does not imply any connection to the hierarchy of the datum system, i.e. axis "X" is not necessarily the axis of the primary datum.

In the example shown in [Figure G.1](#):

- the primary datum consists of a straight line;
- the secondary datum consists of a plane;
- the tertiary datum consists of a plane.

The datum system consists of a set of a plane, a straight line (on the plane) and a point (on the line). The sequence of the datums determines the situation features:

- the straight line is defined by the primary datum;
- the plane includes the straight line and is orientated from the secondary datum;
- the point is the intersection between the straight line and the tertiary datum.



Key

- 1 datum A
- 2 datum B in the datum system, with orientation constraint (parallel) from datum A
- 3 datum C in the datum system, with orientation constraints from 1 and 2
- 4 situation feature straight line of the datum system (equal to 1)
- 5 situation feature plane of the datum system (contains 4 and is parallel to 2)
- 6 situation feature point of the datum system (on 4, intersection of 3 with 4)

Figure G.1 — Example of establishment of a datum coordinate system

G.4 Datum coordinate system identifier

The datum coordinate system identifier consists of the letters CS. If several datum coordinate systems are indicated on the drawing, then the letters CS shall be followed without space by a unique positive integer, e.g. CS1. It is placed preferably to the location allocated to it relatively to the datum coordinate system indicator, excepted when its indication can overlap other indication. In this case, it shall be linked with it through a leader line pointing on an axis of datum coordinate system indicator.

The datum coordinate system is defined by default as a right-handed Cartesian coordinate system.

The datum coordinate system identifier shall be completed by either:

- a direct indication of a datum system, or a datum (single or common), i.e. the datum coordinate system identifier, followed by a colon, followed by a datum section [see [Figure G.2 a\)](#)]; or
- an indirect indication of a datum section, i.e. the datum coordinate system identifier, followed by a colon, followed by an indirect indication of a datum section [[Figure G.2 b\)](#)].

NOTE See [7.4.2.11](#) for indirect indication of datum system.

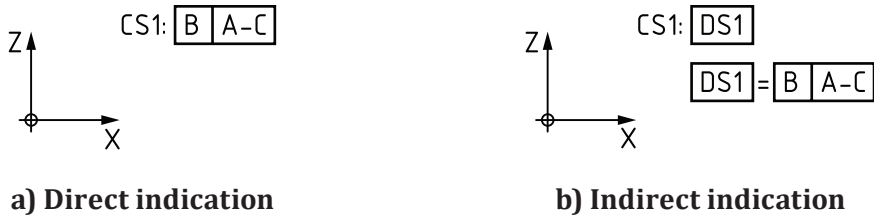


Figure G.2 — Indication of a datum coordinate system

Annex H
(informative)

Filter symbols and attached nesting index

Table H.1 gives the list of filter symbols and attached nesting index which can be used to define the filtration (see 8.2 and 8.3). The filtered feature is then used to establish the associated feature.

For more information on filters, see the ISO 16610 series.

Table H.1 — Filter symbols and attached nesting index

Symbol	Name	Nesting index
G	Gaussian	Cut off length Cut off UPR
S	Spline	Cut off length Cut off UPR
SW	Spline wavelet	Cut off length Cut off UPR
CW	Complex wavelet	Cut off length Cut off UPR
W	Wavelet	Cut off length Cut off UPR
RG	Robust Gaussian	Cut off length Cut off UPR
RS	Robust spline	Cut off length Cut off UPR
OB	Opening ball	Ball radius
OH	Opening horizontal segment	Segment length
OD	Opening disc	Disc radius
CB	Closing ball	Ball radius
CBD	Closing ball dedicated	Ball radius
CH	Closing horizontal segment	Segment length
CD	Closing disc	Disc radius
AB	Alternating series ball	Ball radius
AH	Alternating series horizontal segment	Segment length
AD	Alternating series disk	Disc radius
F	Fourier	Wavelength UPR number
H	Harmonic	Wavelength UPR number
EM	Envelop morphological	—

Annex I

(informative)

Issue of orientation and location constraints in datum systems

I.1 Construction of associated feature for defining a datum system

The rules and definitions used for defining a datum system (see 6.2.4 and 6.3.4) indicate that:

- the associated feature used to define a secondary datum has its orientation constrained to the primary datum, but not its location;
- the associated feature used to define a tertiary datum has its orientation constrained to the primary and secondary datums, but not its location.

This defines a datum system which can, but does not always, correctly represent the functional requirement (see Examples 1 to 3). For ease of notation in this annex, |A|B|C| denotes a datum section constituted by the ordered sequence of datum indicators defining the datum A, B and C.

I.2 Example 1

A datum system, introduced by the datum section |A|B|C|, can be defined based on the datum features shown on the drawing in Figure I.1.

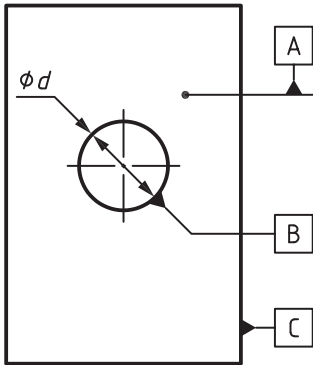


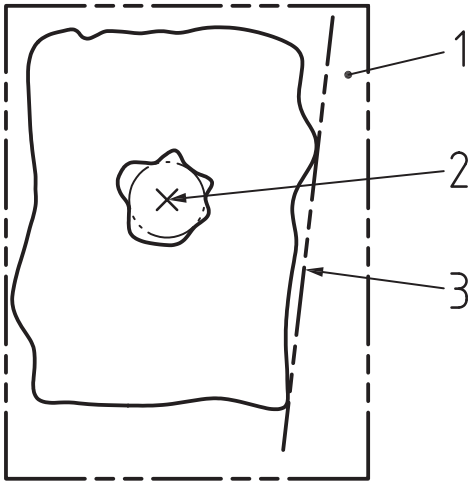
Figure I.1 — Example showing indication of datum features

According to the rules in 6.2.4 and 6.3.4, the associated feature used to establish the tertiary datum C, is:

- constrained in orientation perpendicular to the primary datum A;
- not constrained in location (the distance of the associated feature from the secondary datum B is variable).

The result is the outcome shown in Figure I.2.

NOTE In this example, considering location constraint to establish the tertiary datum C from the secondary datum B would not make sense in most cases.



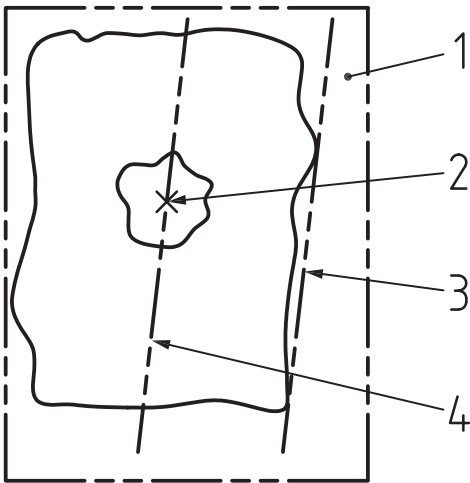
Key

- 1 situation feature of the associated feature to the primary datum feature A: a plane
- 2 situation feature of the associated feature to the secondary datum feature B: a straight line
- 3 situation feature of the associated feature to the tertiary datum feature C: a plane

Figure I.2 — Situation features of the features associated to the datums features defined to establish the datum system |A|B|C| on a possible real part

The resulting datum system (i.e. set of one or more situation features, see [Figure I.3](#)) consists of the following:

- a plane (corresponding to the primary datum A);
- a point (intersection between the primary datum A and the secondary datum B);
- a straight line (parallel to the tertiary datum C, included in the plane and through with the secondary datum B).



Key

- 1 plane (corresponding to the primary datum A)
- 2 point (intersection between the primary datum A and the secondary datum B)
- 3 straight line (resulting from the intersection of 1 and the tertiary datum C)
- 4 straight line (included in 1 and parallel to 3)

NOTE The resulting datum system is, in this case, the set of 1, 2 and 4.

Figure I.3 — Resulting datum system required in this case

I.3 Example 2

A datum system |A|B|C| can be defined based on the datum features shown in Figure I.4. To establish the secondary datum, the feature associated to datum feature B has an orientation constraint (perpendicularity) with the primary datum. To establish the tertiary datum, the feature associated to the datum feature C has an orientation constraint with the primary datum (perpendicularity) and the secondary datum (parallelism).

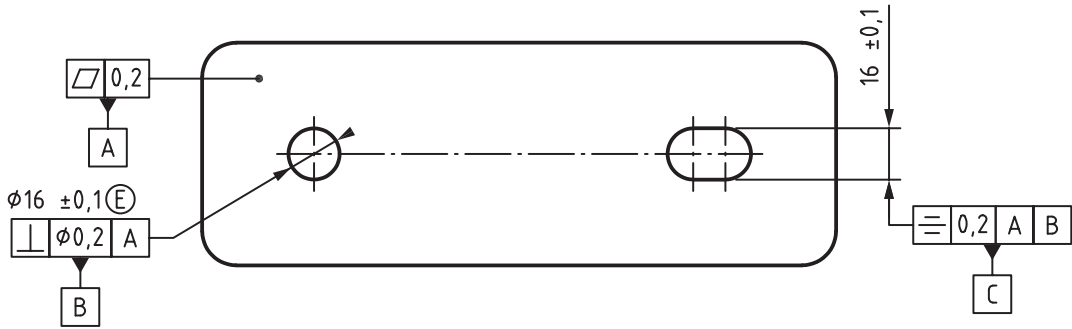
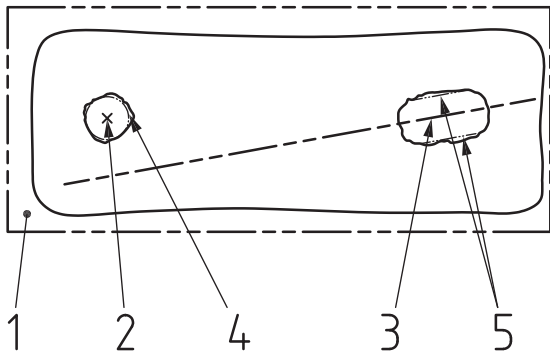


Figure I.4 — Example showing the indication for datum features

According to the rules defined in 6.2.4 and 6.3.4, the associated feature used to establish the tertiary datum, C, is:

- constrained in orientation perpendicular to the primary datum A;
- not constrained in location (the distance of the associated feature from datum B is variable).

The result is the outcome shown in Figure I.5.



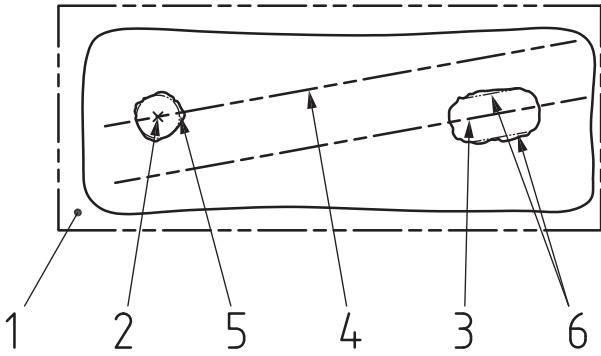
Key

- 1 situation feature of the associated feature to the primary datum feature A: a plane
- 2 situation feature of the associated feature to the secondary datum feature B: a straight line (perpendicular to 1)
- 3 situation feature of the associated feature to the tertiary datum feature C: a plane (perpendicular to 1 and parallel to 2)
- 4 associated feature to the datum feature (of size) B: the maximum inscribed cylinder
- 5 associated features to the datum feature (of size) C: the maximum inscribed two opposite planes with orientation constraint from 1

NOTE The straight line 2 is shown end-on (appears as a point) and the plane 3 is shown edge-on (appears as a straight line).

Figure I.5 — Situation features of the features associated to the datums features defined to establish for the datum system |A|B|C| on a possible real part

The resulting datum system is orientated to the tertiary datum feature, but not aligned with it (see [Figure I.6](#)).



Key

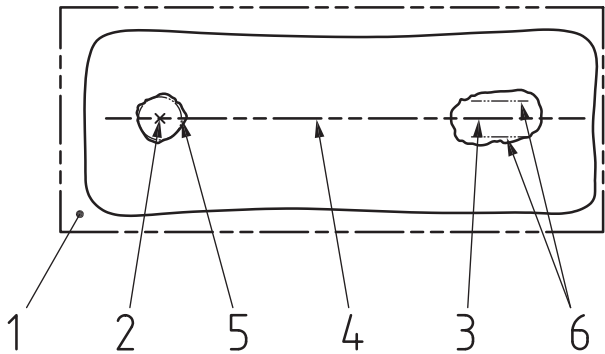
- 1 plane (corresponding to the primary datum A)
- 2 point (corresponding to intersection between the primary datum A and the secondary datum B)
- 3 plane (corresponding to the tertiary datum C)
- 4 straight line (included in 1, aligned with 2 and parallel to 3)
- 5 associated feature to the datum feature (of size) B: the maximum inscribed cylinder
- 6 associated features to the datum feature (of size) C: the maximum inscribed two opposite planes with orientation constraint from 1

NOTE The resulting datum system is the set of 1, 2 and 4.

Figure I.6 — Resulting datum system according to current rules

This would require the associated feature to datum feature C to be constrained by the location relationship (distance of zero millimetre) with the situation feature of the associated feature to the secondary datum B:

variations in the location of the datum feature C would no longer result in variations in the orientation of the resulting datum system. It would make no difference to the definition of the secondary datum of the resulting datum system, because the associated feature to datum feature has only an orientation relationship with the primary datum.



Key

- 1 plane (corresponding to the primary datum A)
- 2 point (corresponding to intersection between the primary datum A and the secondary datum B)
- 3 plane [situation feature of the feature of size associated to the datum feature C, established with orientation constraint from the primary datum A and with location constraint from the secondary datum B (through with)]
- 4 straight line (included in 1, aligned with 2 and parallel to 3)
- 5 associated feature to the datum feature (of size) B: the maximum inscribed cylinder
- 6 associated features to the datum feature (of size) C: the maximum inscribed two opposite planes with orientation constraint and location constraint from 1

NOTE The resulting datum system is, in this case, the set of 1, 2 and 4.

Figure I.7 — Resulting datum system that is required functionally in some cases

In order to achieve the result shown in [Figure I.7](#), a note in the technical product documentation can be used to change from the current default rule. This specific drawing indication should indicate which specific associated features used to define the secondary or tertiary datums should have their locations, as well as their orientations, fixed. This would require the associated feature for datum C to be constrained by the location relationship (distance of zero) with datum B. Variations in the location of the tertiary datum feature would no longer result in variations in the orientation of the resulting datum system. It would make no difference to the definition of datum B, because the datum feature only has an orientation relationship with the primary datum.

Depending on the functional requirement, other approaches can also be considered, such as using a contacting feature.

I.4 Example 3

A datum system |A|B| can be defined based on the datum features shown in [Figure I.8](#). To establish the secondary datum the feature associated to the datum feature B has an orientation constraint (parallelism) with the primary datum A.

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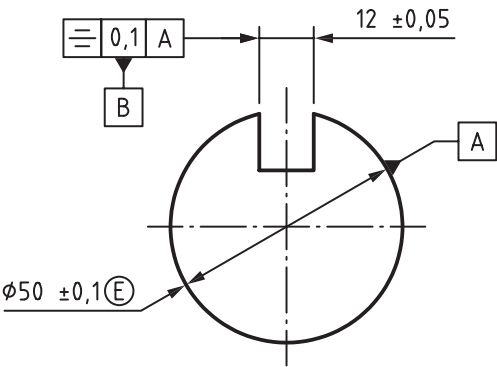
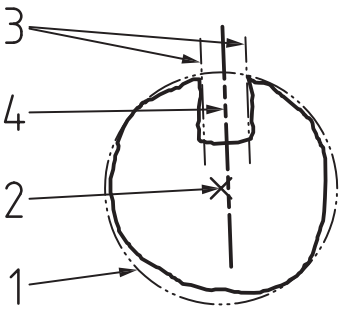


Figure I.8 — Example showing specification for datum features A and B

According to the rules in 6.2.4 and 6.3.4, the associated feature used to establish the secondary datum, datum B, is:

- constrained in orientation parallel to the primary datum A;
- not constrained in location (the distance of the associated feature from the primary datum A is variable).

The result is the outcome shown in Figure I.9.



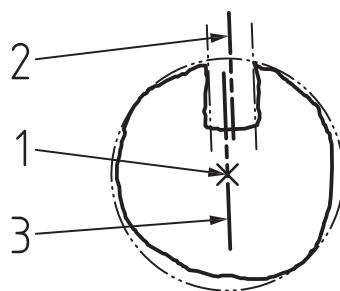
Key

- 1 associated feature to the primary datum feature A
- 2 situation feature of 1: a straight line
- 3 associated feature to the secondary datum feature (of size) B
- 4 situation feature of 3: a plane

Figure I.9 — Situation features of the associated features to the datums features defined to establish the datum system [A|B] on a possible real part

The resulting datum system is orientated to the secondary datum, but not aligned with it (see Figure I.10).

NOTE The line is shown end-on and the plane edge-on in Figure I.10.

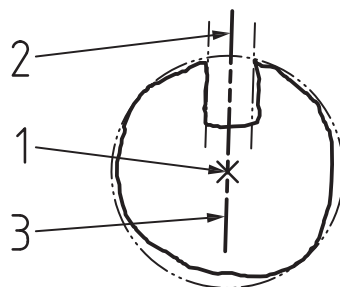
**Key**

- 1 straight line (corresponding to the primary datum A)
- 2 plane (corresponding to the secondary datum B)
- 3 plane (parallel to 2 and aligned to 1)

NOTE The resulting datum system is the set of 1 and 3.

Figure I.10 — Resulting datum system according to current rules

In some cases, the functional requirement would be to have the resulting datum system aligned to the secondary datum feature as shown in [Figure I.11](#).

**Key**

- 1 straight line (corresponding to the primary datum A)
- 2 plane (situation feature of the feature associated to the datum feature B, established with parallel orientation constraint and location constraint, 0 mm apart, from the primary datum A)
- 3 plane (coincident with plane 2)

NOTE The resulting datum system is the set of 1 and 3.

Figure I.11 — Resulting datum system that is required functionally in some cases

In order to achieve the result shown in [Figure I.11](#), a note in the technical product documentation can be used to change from the current default rule and to indicate which specific associated features used to define the secondary or tertiary datums should have their locations, as well as their orientations, fixed.

This would require the associated feature for datum B to be constrained by the location relationship (implied distance of zero) with datum A.

Depending on the functional requirement, other approaches can also be considered, such as using a contacting feature.

Annex J (normative)

Filtration of a datum feature which is nominally a plane

J.1 General

This annex defines the default filtration when the nominal datum feature is a plane used to establish a single datum. For this geometry, the default filtration is the [EM] filtration (see [J.2.2](#)).

The reason for introducing the default filtration in conjunction with the new default association criterion for a single surface, nominally planar, to establish a single datum plane is to improve the rules stated in ISO 5459:2011 for datum features that are nominally a single plane (see [Clause D.4](#)). Specifically, the aims are:

- to lower the level of specification ambiguity by presenting a detailed description of filtration and association;
- to provide a resulting datum closer to the functional intent; this has been obtained mostly by ensuring that the resulting datum is close to the contact plane that can be expected when the real part rests on the real datum feature;
- to increase the stability of the resulting datum; stability is obtained when a small change of the real feature leads to a small change in the resulting datum.

These aims are better obtained by applying first the filtration [EM] and then the association [GE], which is why this combination is now the default. Designers should use caution when deviating from this recommended combination of filter and association.

The concepts and methods used in this annex use the results of recent research papers (see References [\[11\]](#) and [\[12\]](#)). It is envisioned that research will continue on this subject to provide more general methods allowing rules for other geometries to be included.

Other sources of ambiguity, such as the lack of specification for the extraction operation, are not considered in this annex.

J.2 Filtration

J.2.1 General

This annex defines a filter [EM] using a functional perspective for a surface, which is nominally a plane. It is the default filter for single datums that are nominally a planar surface. The purpose of the [EM] filter is to remove voids from the surface that cannot be accessed by a perfect form mating datum surface. This default filter improves the final datum by removing the influence of those voids.

This functional description is mathematically modelled using an ideal mating surface that can be rotated and translated to capture all the reasonable mating interactions. Conceptually speaking, the union of all these mating interactions is the filtered surface. Mathematically, two additional concepts need to be added to control the filtered surface:

- first, a limited extent needs to be defined so the filtered surface does not extend to infinity;
- second, the word “reasonable” needs a precise mathematical definition.

The filtration described in this annex applies to the set of points of the extracted integral feature, which is nominally a single plane. The extraction is not described in this annex. This procedure of filtration can be

equally used for an infinite number of points representing a continuous surface or a finite number of points representing a discrete surface. In both cases, the filtered feature is a continuous surface.

This annex also provides a symbol to specify the use of the filter described hereafter on a plane nominal datum feature and a symbol to specify the use of the filter described hereafter together with the association described in this annex on a plane nominal datum feature.

J.2.2 Mathematical concepts and symbols

In order to present the mathematical concepts involved in the application of a filter to the extracted integral surface, the symbols given in Table J.1 are used.

Table J.1 — Symbols

Symbol	Description
E	set of points of the extracted integral feature
A	a plane, associated with E with a least squares (Gaussian) criterion, without material constraint
$\hat{n}$	surface normal vector of A , such that it points external from the material
δ	total range of deviation between E and A
E_p	2D region obtained by projecting the points of E to A (perpendicular to A) and then computing the convex hull of the projected points
D	largest distance between any two points in E_p and called the diameter of E_p
S	set of half-spaces, where each half-space satisfies the conditions below
H	half-space in S
$\hat{h}$	surface normal vector of H , such that it points into the half-space
U	union of all half-spaces H in S
F	filtered feature, which is the portion of the boundary of U whose projection to A perpendicular to A lies within the region E_p

NOTE 1 The region E_p helps define the limited extent presented in J.2.1.

The conditions to be satisfied by each half-space H in S are the following:

- Its surface normal $\hat{h}$ shall form an angle with $\hat{n}$ that is no more than $4\delta / D$ radians. In extreme cases of unusually large form deviation over a small extent where this angle exceeds 0,1 radian, which is not recommended, then the angle shall be limited to 0,1 radian.
- It shall not contain any points of E (besides tangential contact), ensuring it is external to the material.

These conditions constitute the precise mathematical definition of “reasonable” as discussed in J.2.1.

NOTE 2 Other filters can be specified to adjust the specification according to the functional intent for the establishment of a datum from a plane nominal datum feature (see Annex H).

It is the designer’s responsibility to check the compatibility of the specified modifiers.

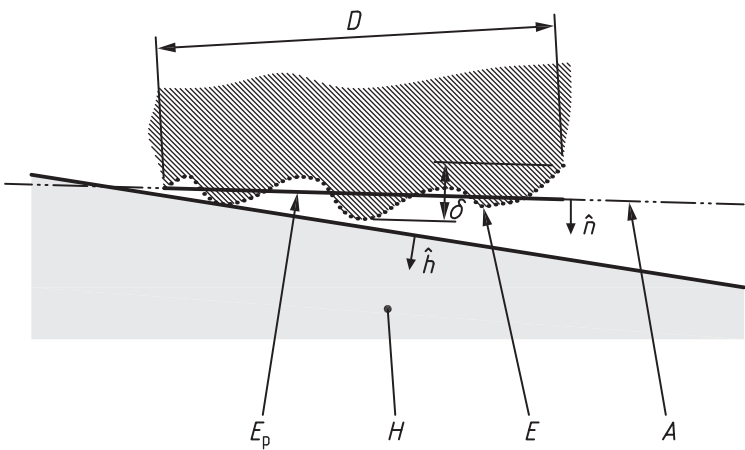
NOTE 3 The symbol [EM] has no meaning for any datum geometry besides single datum planes.

J.3 Examples

The [EM] filter reduces to a closing ball filter with infinite-diameter structuring element (i.e. a convex hull-producing filter) over most of the filtered surface. The edges are handled differently, for two reasons:

- the closing ball filter makes no distinction between inside the material and outside the material;
- the closing ball filter has no concept for what a “reasonable” mating interaction means.

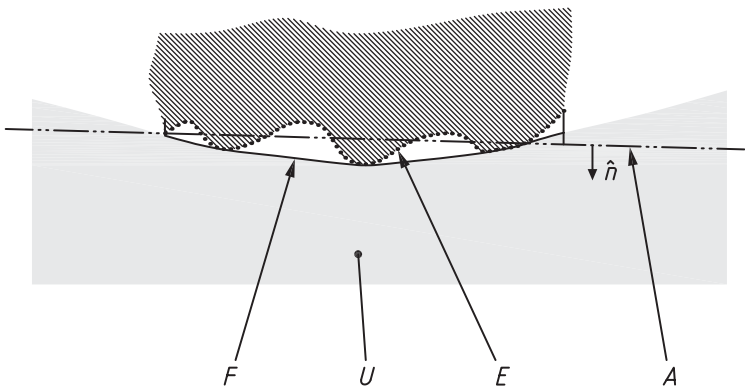
Figure J.1 shows an illustration of a typical half-space H in S . Figure J.2 illustrates what a typical filtered feature F looks like.



Key

- E real integral surface or set of its extracted points
- A associated plane with least squares objective function without material constraint
- $\hat{n}$ normal vector to A
- δ range of deviations established between A and E
- H one of the half-spaces in the set S (structuring element of filtration with infinite radius)
- $\hat{h}$ normal vector of the planar boundary of H
- E_p the 2D region that is obtained by projecting the points of E to A (perpendicular to A) and then computing the convex hull of the projected points
- D maximum distance between two points of E_p (as defined in J.2.2)

Figure J.1 — Illustration of a typical half-space H in the set S shown in two dimensions



Key

- E real integral surface or set of its extracted points
- A associated plane with least squares objective function without material constraint
- $\hat{n}$ normal vector to A
- U union of all half-spaces H (as presented in Figure J.1)
- F filtered feature: boundary of U

NOTE The angles between any local normal vector of F and $\hat{n}$ are under condition (see J.2.2).

Figure J.2 — Illustration of the filtered feature, F , and of the union of the half-space, U , shown in two dimensions

Annex K
(informative)

Relation to the GPS matrix model

K.1 General

The ISO GPS matrix model given in ISO 14638 gives an overview of the ISO GPS system of which this document is a part. The fundamental rules of ISO GPS given in ISO 8015 apply to this document and the default decision rules given in ISO 14253-1 apply to specifications made in accordance with this document, unless otherwise indicated.

K.2 Information about the document and its use

This document describes the terminology, the rules, the explanations and the manner in which datums and datum systems are indicated in technical product documentation.

K.3 Position in the GPS matrix model

This document is a general ISO GPS standard which influences chain links A, B and C of the chains of standards on size, distance, form, orientation, location, run-out and (profile, areal) surface texture in the GPS matrix model, as shown in [Table K.1](#). The rules and principles given in this document apply to all segments of the ISO GPS matrix which are indicated with a filled dot (•).

Table K.1 — Position in the GPS matrix model

	Chain links						
	A	B	C	D	E	F	G
	Symbols and indications	Feature requirements	Feature properties	Conformance and non-conformance	Measurement	Measurement equipment	Calibration
Size	•	•	•				
Distance	•	•	•				
Form	•	•	•				
Orientation	•	•	•				
Location	•	•	•				
Run-out	•	•	•				
Profile surface texture	•	•	•				
Areal surface texture	•	•	•				
Surface imperfections							

K.4 Related International Standards

The related International Standards are those of the chains of standards indicated in [Table K.1](#).

Bibliography

- [1] ISO 2768-1, *General tolerances — Part 1: Tolerances for linear and angular dimensions without individual tolerance indications*
- [2] ISO 5456-1, *Technical drawings — Projection methods — Part 1: Synopsis*
- [3] ISO 5459:1981⁴⁾, *Technical drawings — Geometrical tolerancing — Datums and datum-systems for geometrical tolerances*
- [4] ISO 5459:2011⁵⁾, *Geometrical product specifications (GPS) — Geometrical tolerancing — Datums and datum systems*
- [5] ISO 7083, *Technical product documentation — Symbols used in technical product documentation — Proportions and dimensions*
- [6] ISO 8015, *Geometrical product specifications (GPS) — Fundamentals — Concepts, principles and rules*
- [7] ISO 14253-1, *Geometrical product specifications (GPS) — Inspection by measurement of workpieces and measuring equipment — Part 1: Decision rules for verifying conformity or nonconformity with specifications*
- [8] ISO 14638, *Geometrical product specifications (GPS) — Matrix model*
- [9] ISO 16610 (all parts), *Geometrical product specifications (GPS) — Filtration*
- [10] ISO 81714-1, *Design of graphical symbols for use in the technical documentation of products — Part 1: Basic rules*
- [11] SHAKARJI, C.M., SRINIVASAN, V. Theory and algorithm for planar datum establishment using constrained total least-squares. *Procedia CIRP*. 2016, 43 pp.232–237
- [12] SHAKARJI, C.M., SRINIVASAN, V. Toward a new mathematical definition of datums in standards to support advanced manufacturing. In: *International Manufacturing Science and Engineering Conference* (Vol. 51371, p. V003T02A016). American Society of Mechanical Engineers, 2018
- [13] SHAKARJI, C.M., SRINIVASAN, V. Toward a New Mathematical Definition of Datums in Standards to Support Advanced Manufacturing. *Journal of Computing and Information Science in Engineering*. 2023, 23(2) p.021013

4) Cancelled and replaced by ISO 5459:2011.

5) Cancelled and replaced by this document.

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